Post-Collection Analysis Guide — nsight-systems

282–358 minutes

NVIDIA Nsight Systems Analysis guide.

Overview#

Once you have profiled using Nsight Systems there are many options for analyzing the collected data, as well as to output it in various formats. These options are available from the CLI or the GUI.

Statistical Analysis#

Statistical Reports Shipped With Nsight Systems#

The Nsight Systems development team created and maintains a set of report scripts for some of the commonly requested statistical reports. These scripts will be updated to adapt to any changes in SQLite schema or internal data structures.

These scripts are located in the Nsight Systems package in the Target-<architecture>/reports directory. The following standard reports are available:

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw,

mangled kernel name data is not available, the default demangled names will be used.

Note

All time values given in nanoseconds by default. If you wish to output the results using a different time unit, use the `--timeunit` option when running the recipe.

cuda_api_gpu_sum[:nvtx-name][:base:mangled] – CUDA Summary (API/Kernels/MemOps)#

Arguments

- nvtx-name : Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will cause summary to be over the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather than the templated name.

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this kernel
- Instances : Number of executions of this kernel
- Avg : Average execution time of this kernel
- Med : Median execution time of this kernel

- Min : Smallest execution time of this kernel
- Max : Largest execution time of this kernel
- StdDev : Standard deviation of execution time of this kernel
- Category : Category of the operation
- Operation : Name of the kernel

This report provides a summary of CUDA API calls, kernels and memory operations, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that API call’s, kernel’s, or memory operation’s percent of the execution time of the APIs, kernels and memory operations listed, and not a percentage of the application wall or CPU execution time.

This report combines data from the `cuda_api_sum`, `cuda_gpu_kern_sum`, and `cuda_gpu_mem_size_sum` reports. It is very similar to profile section of `nvprof --dependency-analysis`.

cuda_api_sum – CUDA API Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this function
- Num Calls : Number of calls to this function
- Avg : Average execution time of this function
- Med : Median execution time of this function
- Min : Smallest execution time of this function
- Max : Largest execution time of this function

- StdDev : Standard deviation of the time of this function
- Name : Name of the function

This report provides a summary of CUDA API functions and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that function’s percent of the execution time of the functions listed, and not a percentage of the application wall or CPU execution time.

cuda_api_trace – CUDA API Trace#

Arguments - None

Output: All time values default to nanoseconds

- Start : Timestamp when API call was made
- Duration : Length of API calls
- Name : API function name
- Result : Return value of API call
- CorrID : Correlation used to map to other CUDA calls
- Pid : Process ID that made the call
- Tid : Thread ID that made the call
- T-Pri : Run priority of call thread
- Thread Name : Name of thread that called API function

This report provides a trace record of CUDA API function calls and their execution times.

cuda_gpu_kern_gb_sum[:nvtx-name][:basel:mangled] – CUDA GPU Kernel/Grid/Block Summary#

Arguments

- nvtx-name - Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will cause summary to be over the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather than the templated name.

  Note

  The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

  Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this kernel
- Instances : Number of calls to this kernel
- Avg : Average execution time of this kernel
- Med : Median execution time of this kernel
- Min : Smallest execution time of this kernel
- Max : Largest execution time of this kernel
- StdDev : Standard deviation of the time of this kernel
- GridXYZ : Grid dimensions for kernel launch call
- BlockXYZ : Block dimensions for kernel launch call
- Name : Name of the kernel

This report provides a summary of CUDA kernels and their execution times. Kernels are sorted by grid dimensions, block dimensions, and kernel name. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that kernel’s percent of the execution time of the kernels listed, and not a percentage of the application wall or CPU execution time.

cuda_gpu_kern_sum[:nvtx-name][:basel:mangled] – CUDA GPU Kernel Summary#

Note

In recent versions of Nsight Systems, this report was expanded to include and sort by CUDA grid and block dimensions. This change was made to accommodate developers doing a certain type of optimization work. Unfortunately, this change caused an unexpected burden for developers doing a different type of optimization work. In order to service both use-cases, this report has been returned to the original form, without grid or block information. A new report, called `cuda_gpu_kern_gb_sum`, has been created that retains the grid and block information.

Arguments

- nvtx-name - Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will cause summary to be over the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather than the templated name.

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this kernel
- Instances : Number of calls to this kernel
- Avg : Average execution time of this kernel
- Med : Median execution time of this kernel
- Min : Smallest execution time of this kernel
- Max : Largest execution time of this kernel
- StdDev : Standard deviation of the time of this kernel
- Name : Name of the kernel

This report provides a summary of CUDA kernels and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that kernel’s percent of the execution time of the kernels listed, and not a percentage of the application wall or CPU execution time.

cuda_gpu_mem_size_sum – CUDA GPU MemOps Summary (by Size)#

Arguments - None

Output:

- Total : Total memory utilized by this operation
- Count : Number of executions of this operation
- Avg : Average memory size of this operation
- Med : Median memory size of this operation
- Min : Smallest memory size of this operation

- Max : Largest memory size of this operation
- StdDev : Standard deviation of the memory size of this operation
- Operation : Name of the operation

This report provides a summary of GPU memory operations and the amount of memory they utilize.

cuda_gpu_mem_time_sum – CUDA GPU MemOps Summary (by Time)#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this operation
- Count : Number of operations to this type
- Avg : Average execution time of this operation
- Med : Median execution time of this operation
- Min : Smallest execution time of this operation
- Max : Largest execution time of this operation
- StdDev : Standard deviation of execution time of this operation
- Operation : Name of the memory operation

This report provides a summary of GPU memory operations and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that operation’s percent of the execution time of the operations listed, and not a percentage of the application wall or CPU execution time.

cuda_gpu_sum[:nvtx-name][:basel:mangled] – CUDA GPU Summary (Kernels/MemOps)#

Arguments

- nvtx-name - Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will cause summary to be over the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather than the templated name.

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this kernel
- Instances : Number of executions of this kernel
- Avg : Average execution time of this kernel
- Med : Median execution time of this kernel
- Min : Smallest execution time of this kernel
- Max : Largest execution time of this kernel
- StdDev : Standard deviation of execution time of this kernel
- Category : Category of the operation
- Operation : Name of the kernel

This report provides a summary of CUDA kernels and memory operations, and their execution times. Note that the "Time" column is calculated using a summation of the "Total Time" column, and represents that kernel's or memory operation's percent of the execution time of the kernels and memory operations listed, and not a percentage of the application wall or CPU execution time.

This report combines data from the `cuda_gpu_kern_sum` and `cuda_gpu_mem_time_sum` reports. This report is very similar to output of the command `nvprof --print-gpu-summary`.

cuda_gpu_trace[:nvtx-name][:base|:mangled] – CUDA GPU Trace#

Arguments

- nvtx-name - Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will display the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will display the raw mangled name of the kernel, rather than the templated name.

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- Start : Timestamp of start time
- Duration : Length of event
- CorrId : Correlation ID
- GrdX, GrdY, GrdZ : Grid values

- BlkX, BlkY, BlkZ : Block values
- Reg/Trd : Registers per thread
- StcSMem : Size of Static Shared Memory
- DymSMem : Size of Dynamic Shared Memory
- Bytes : Size of memory operation
- Throughput : Memory throughput
- SrcMemKd : Memcpy source memory kind or memset memory kind
- DstMemKd : Memcpy destination memory kind
- Device : GPU device name and ID
- Ctx : Context ID
- GreenCtx: Green context ID
- Strm : Stream ID
- Name : Trace event name

This report displays a trace of CUDA kernels and memory operations. Items are sorted by start time.

cuda_kern_exec_sum[:nvtx-name][:base:mangled] – CUDA Kernel Launch & Exec Time Summary#

Arguments

- nvtx-name - Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will cause summary to be over the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather

than the templated name.

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- PID : Process ID that made kernel launch call
- TID : Thread ID that made kernel launch call
- DevId : CUDA Device ID that executed kernel (which GPU)
- Count : Number of kernel records
- QCount : Number of kernel records with positive queue time

Average, Median, Minimum, Maximum, and Standard Deviation for:

- TAvg, TMed, TMin, TMax, TStdDev : Total time
- AAvg, AMed, AMin, AMax, AStdDev : API time
- QAvg, QMed, QMin, QMax, QStdDev : Queue time
- KAvg, KMed, KMin, KMax, KStdDev : Kernel time
- API Name : Name of CUDA API call used to launch kernel
- Kernel Name : Name of CUDA Kernel

This report provides a summary of the launch and execution times of CUDA kernels. The launch and execution is broken down into three phases: “API time,” the execution time of the CUDA API call on the CPU used to launch the kernel; “Queue time,” the time between the launch call and the kernel execution; and “Kernel time,” the kernel execution time on the GPU. The “total time” is not a just sum of the other times, as the phases sometimes overlap. Rather, the total time runs from the start of the API call to end of the API call or the end of the kernel execution,

whichever is later.

The reported queue time is measured from the end of the API call to the start of the kernel execution. The actual queue time is slightly longer, as the kernel is enqueue somewhere in the middle of the API call, and not in the final nanosecond of function execution. Due to this delay, it is possible for kernel execution to start before the CUDA launch call returns. In these cases, no queue time will be reported. Only kernel launches with positive queue times are included in the queue average, minimum, maximum, and standard deviation calculations. The “QCount” column indicates how many launches had positive queue times (and how many launches were involved in calculating the queue time statistics). Subtracting “QCount” from “Count” will indicate how many kernels had no queue time.

Be aware that having a queue time is not inherently bad. Queue times indicate that the GPU was busy running other tasks when the new kernel was scheduled for launch. If every kernel launch is immediate, without any queue time, that _may_ indicate an idle GPU with poor utilization. In terms of performance optimization, it should not necessarily be a goal to eliminate queue time.

cuda_kern_exec_trace[:nvtx-name][:basel:mangled] – CUDA Kernel Launch & Exec Time Trace#

Arguments

- nvtx-name - Optional argument, if given, will prefix the kernel name with the name of the innermost enclosing NVTX range.
- base - Optional argument, if given, will cause summary to be over the base name of the kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather than the templated name.

Note: the ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- API Start : Start timestamp of CUDA API launch call
- API Dur : Duration of CUDA API launch call
- Queue Start : Start timestamp of queue wait time, if it exists
- Queue Dur : Duration of queue wait time, if it exists
- Kernel Start : Start timestamp of CUDA kernel
- Kernel Dur : Duration of CUDA kernel
- Total Dur : Duration from API start to kernel end
- PID : Process ID that made kernel launch call
- TID : Thread ID that made kernel launch call
- DevId : CUDA Device ID that executed kernel (which GPU)
- API Function : Name of CUDA API call used to launch kernel
- GridXYZ : Grid dimensions for kernel launch call
- BlockXYZ : Block dimensions for kernel launch call
- Kernel Name : Name of CUDA Kernel

This report provides a trace of the launch and execution time of each CUDA kernel. The launch and execution is broken down into three phases: “API time,” the execution time of the CUDA API call on the CPU used to launch the kernel; “Queue time,” the time between the launch call and the kernel execution; and “Kernel time,” the kernel execution time on the GPU. The “total time” is not a just sum of the other times, as the phases sometimes overlap. Rather, the total time runs from the start of the API call to end of the API call or the end of the kernel execution, whichever is later.

The reported queue time is measured from the end of the API call to the start of the kernel execution. The actual queue time is slightly longer, as the kernel is enqueue somewhere in the middle of the API call, and not in the final nanosecond of function execution. Due to this delay, it is possible for kernel execution to start before the CUDA

launch call returns. In these cases, no queue times will be reported.

Be aware that having a queue time is not inherently bad. Queue times indicate that the GPU was busy running other tasks when the new kernel was scheduled for launch. If every kernel launch is immediate, without any queue time, that _may_ indicate an idle GPU with poor utilization. In terms of performance optimization, it should not necessarily be a goal to eliminate queue time.

dx11_pix_sum – DX11 PIX Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this rage
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of D3D11 PIX CPU debug markers, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that range’s percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

dx12_gpu_marker_sum – DX12 GPU Command List PIX Ranges Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of DX12 PIX GPU command list debug markers, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that range’s percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

dx12_pix_sum – DX12 PIX Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range

- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of D3D12 PIX CPU debug markers, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that range’s percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

mpi_event_sum – MPI Event Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this event
- Instances : Number of instances of this event
- Avg : Average execution time of this event
- Med : Median execution time of this event
- Min : Smallest execution time of this event
- Max : Largest execution time of this event
- StdDev : Standard deviation of execution time of this event
- Source: Original source class of event data

- Name : Name of MPI event

  This report provides a summary of all recorded MPI events. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that event’s percent of the total execution time of the listed events, and not a percentage of the application wall or CPU execution time.

  mpi_event_trace – MPI Event Trace#

  Arguments - None

  Output: All time values default to nanoseconds

- Start : Start timestamp of event
- End : End timestamp of event
- Duration : Duration of event
- Event : Name of event type
- Pid : Process Id that generated the event
- Tid : Thread Id that generated the event
- Tag : MPI message tag
- Rank : MPI Rank that generated event
- PeerRank : Other MPI rank of send or receive type events
- RootRank : Root MPI rank for broadcast type events
- Size : Size of message for uni-directional operations (send & recv)
- CollSendSize : Size of sent message for collective operations
- CollRecvSize : Size of received message for collective operations

  This report provides a trace record of all recorded MPI events.

Note that MPI_Sendrecv events with different rank, tag, or size values are broken up into two separate report rows, one reporting the send, and one reporting the receive. If only one row exists, the rank, tag, and size can assumed to be the same.

mpi_msg_size_sum – MPI Message Size Summary#

Arguments - None

Output: Message size values are in bytes

- Total Message Volume : Aggregated message size from all instances of this API function
- Instances : Number of instances of this API function
- Avg : Average message size of this API function
- Med : Median message size of this API function
- Min : Smallest message size of this API function
- Max : Largest message size of this API function
- StdDev : Standard deviation of message size for this API function
- Source : Message source (p2p, coll_send, coll_recv)
- Name : Name of the MPI API function

This report provides a message size summary of all collective and point-to-point MPI calls.

Note that for MPI collectives the report presents the sent message with Source equal to `coll_send` and the received message with Source equal to `coll_recv`.

network_congestion[:ticks_threshold=<ticks_per_ms>] – Network Devices Congestion#

Arguments

- ticks_threshold=<ticks_per_ms> - Threshold in ticks/ms above which we report congestion. Default is 10000.

  Output: All time values default to nanoseconds

- Start : Start timestamp of congestion interval
- End : End timestamp of congestion interval
- Duration : Duration of congestion interval
- Send wait rate: Rate of congestion during the interval
- GUID : The device GUID
- Name : The device name

  This report displays congestion events with a high send wait rate. By default, only events with a send wait rate above 10000 ticks/ms are shown, but a custom threshold value can be set.

  Each event defines a period of time when the device experienced some level of congestion. The level of congestion is defined by the send wait rate, given in time ticks per millisecond (ticks/ms). The specific duration of a tick is device specific, but can be assumed to be nanoseconds in scale. Congestion is measured by counting the number of ticks during which the port had data to transmit, but no data was sent because of insufficient credits or because of lack of arbitration. The presented value of send wait rate is the amount of ticks counted during an event, normalized over the event's duration. Higher send wait rate values indicate more congestion.

  Because the specific duration of a tick is device dependent, analysis should focus on the relative send wait rates of events generated by the same device. Comparing absolute send wait rates across devices is only meaningful if the time tick duration is known to be similar.

  For IB Switch metrics, we do not present the device name, only the GUID.

nvtx_gpu_proj_sum – NVTX GPU Projection Summary#

Arguments - None

Output: All time values default to nanoseconds

- Range : Name of the NVTX range
- Style : Range style; Start/End or Push/Pop
- Total Proj Time: Total projected time used by all instances of this range name
- Total Range Time: Total original NVTX range time used by all instances of this range name
- Range Instances : Number of instances of this range
- Proj Avg : Average projected time for this range
- Proj Med : Median projected time for this range
- Proj Min : Minimum projected time for this range
- Proj Max : Maximum projected time for this range
- Proj StdDev : Standard deviation of projected times for this range
- Total GPU Ops : Total number of GPU ops
- Avg GPU Ops : Average number of GPU ops
- Avg Range Lvl : Average range stack depth
- Avg Num Child : Average number of children ranges

This report provides a summary of NVTX time ranges projected from the CPU to the GPU. Each NVTX range contains one or more GPU operations. A GPU operation is considered to be “contained” by the NVTX range if the CUDA API call used to launch the operation is within the NVTX range. Only ranges that start and end on the same thread are taken into account.

The projected range will have the start timestamp of the start of the first enclosed GPU operation and the end timestamp of the end of the last enclosed GPU operation. This report then summarizes all the range instances by name and style. Note that in cases when one NVTX range might enclose another, the time of the child(ren) range(s) is not subtracted from the parent range. This is because the projected times may not strictly overlap like the original

NVTX range times do. As such, the total projected time of all ranges might exceed the total sampling duration.

nvtx_gpu_proj_trace – NVTX GPU Projection Trace#

Arguments - None

Output: All time values default to nanoseconds

- Name : Name of the NVTX range
- Projected Start : Projected range start timestamp
- Projected Duration : Projected range duration
- Orig Start : Original NVTX range start timestamp
- Orig Duration : Original NVTX range duration
- Style : Range style; Start/End or Push/Pop
- PID : Process ID
- TID : Thread ID
- NumGPUOps : Number of enclosed GPU operations
- Lvl : Stack level, starts at 0
- NumChild : Number of children ranges
- RangeId : Arbitrary ID for range
- ParentId : Range ID of the enclosing range
- RangeStack : Range IDs that make up the push/pop stack

This report provides a trace of NVTX time ranges projected from the CPU onto the GPU. Each NVTX range contains one or more GPU operations. A GPU operation is considered to be “contained” by an NVTX range if the CUDA API call used to launch the operation is within the NVTX range. Only ranges that start and end on the same thread are

taken into account.

The projected range will have the start timestamp of the first enclosed GPU operation and the end timestamp of the last enclosed GPU operation, as well as the stack state and relationship to other NVTX ranges.

nvtx_kern_sum[:base:mangled] – NVTX Range Kernel Summary#

Arguments

- base - Optional argument, if given, will cause summary to be over the base name of the CUDA kernel, rather than the templated name.
- mangled - Optional argument, if given, will cause summary to be over the raw mangled name of the kernel, rather than the templated name.

Note

The ability to display mangled names is a recent addition to the report file format, and requires that the profile data be captured with a recent version of Nsight Systems. Re-exporting an existing report file is not sufficient. If the raw, mangled kernel name data is not available, the default demangled names will be used.

Output: All time values default to nanoseconds

- NVTX Range : Name of the range
- Style : Range style; Start/End or Push/Pop
- PID : Process ID for this set of ranges and kernels
- TID : Thread ID for this set of ranges and kernels
- NVTX Inst : Number of NVTX range instances
- Kern Inst : Number of CUDA kernel instances
- Total Time : Total time used by all kernel instances of this range

- Avg : Average execution time of the kernel
- Med : Median execution time of the kernel
- Min : Smallest execution time of the kernel
- Max : Largest execution time of the kernel
- StdDev : Standard deviation of the execution time of the kernel
- Kernel Name : Name of the kernel

This report provides a summary of CUDA kernels, grouped by NVTX ranges. To compute this summary, each kernel is matched to one or more containing NVTX range in the same process and thread ID. A kernel is considered to be "contained" by an NVTX range if the CUDA API call used to launch the kernel is within the NVTX range. The actual execution of the kernel may last longer than the NVTX range. A specific kernel instance may be associated with more than one NVTX range if the ranges overlap. For example, if a kernel is launched inside a stack of push/pop ranges, the kernel is considered to be "contained" by all of the ranges on the stack, not just the deepest range. This becomes very confusing if NVTX ranges appear inside other NVTX ranges of the same name.

Once each kernel is associated to one or more NVTX range(s), the list of ranges and kernels grouped by range name, kernel name, and PID/TID. A summary of the kernel instances and their execution times is then computed. The "NVTX Inst" column indicates how many NVTX range instances contained this kernel, while the "Kern Inst" column indicates the number of kernel instances in the summary line.

nvtx_pushpop_sum – NVTX Push/Pop Range Summary#

Arguments - None

Output: All time values given in nanoseconds

- Time : Percentage of "Total Time"
- Total Time : Total time used by all instances of this range

- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of NV Tools Extensions Push/Pop Ranges and their execution times. Note that the "Time" column is calculated using a summation of the "Total Time" column, and represents that range's percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

nvtx_pushpop_trace – NVTX Push/Pop Range Trace#

Arguments - None

Output: All time values default to nanoseconds

- Start : Range start timestamp
- End : Range end timestamp
- Duration : Range duration
- DurChild : Duration of all child ranges
- DurNonChild : Duration of this range minus child ranges
- Name : Name of the NVTX range
- PID : Process ID
- TID : Thread ID

- Lvl : Stack level, starts at 0
- NumChild : Number of children ranges
- RangeId : Arbitrary ID for range
- ParentId : Range ID of the enclosing range
- RangeStack : Range IDs that make up the push/pop stack
- NameTree : Range name prefixed with level indicator

This report provides a trace of NV Tools Extensions Push/Pop Ranges, their execution time, stack state, and relationship to other push/pop ranges.

nvtx_startend_sum – NVTX Start/End Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of NV Tools Extensions Start/End Ranges and their execution times. Note that the "Time" column is calculated using a summation of the "Total Time" column, and represents that range's percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

nvtx_sum – NVTX Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of "Total Time"
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Style : Range style; Start/End or Push/Pop
- Range : Name of the range

This report provides a summary of NV Tools Extensions Start/End and Push/Pop Ranges, and their execution times. Note that the "Time" column is calculated using a summation of the "Total Time" column, and represents that range's percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

nvvideo_api_sum – NvVideo API Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this function
- Num Calls : Number of calls to this function
- Avg : Average execution time of this function
- Med : Median execution time of this function
- Min : Smallest execution time of this function
- Max : Largest execution time of this function
- StdDev : Standard deviation of the time of this function
- Event Type : Which API this function belongs to
- Name : Name of the function

This report provides a summary of NvVideo API functions and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that function’s percent of the execution time of the functions listed, and not a percentage of the application wall or CPU execution time.

openacc_sum – OpenACC Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of event type
- Count : Number of event type

- Avg : Average execution time of event type
- Med : Median execution time of event type
- Min : Smallest execution time of event type
- Max : Largest execution time of event type
- StdDev : Standard deviation of execution time of event type
- Name : Name of the event

This report provides a summary of OpenACC events and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that event type’s percent of the execution time of the events listed, and not a percentage of the application wall or CPU execution time.

opengl_khr_gpu_range_sum – OpenGL KHR_debug GPU Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of OpenGL KHR_debug GPU PUSH/POP debug Ranges, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that range’s percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

opengl_khr_range_sum – OpenGL KHR_debug Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of OpenGL KHR_debug CPU PUSH/POP debug Ranges, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that range’s percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

openmp_sum – OpenMP Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of event type
- Count : Number of event type
- Avg : Average execution time of event type
- Med : Median execution time of event type
- Min : Smallest execution time of event type
- Max : Largest execution time of event type
- StdDev : Standard deviation of execution time of event type
- Name : Name of the event

This report provides a summary of OpenMP events and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that event type’s percent of the execution time of the events listed, and not a percentage of the application wall or CPU execution time.

osrt_sum – OS Runtime Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this function
- Num Calls : Number of calls to this function
- Avg : Average execution time of this function

- Med : Median execution time of this function
- Min : Smallest execution time of this function
- Max : Largest execution time of this function
- StdDev : Standard deviation of execution time of this function
- Name : Name of the function

This report provides a summary of operating system functions and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that function’s percent of the execution time of the functions listed, and not a percentage of the application wall or CPU execution time.

syscall_sum – Syscall Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this syscall
- Num Calls : Number of calls to this syscall
- Avg : Average execution time of this syscall
- Med : Median execution time of this syscall
- Min : Smallest execution time of this syscall
- Max : Largest execution time of this syscall
- StdDev : Standard deviation of execution time of this syscall
- Name : Name of the syscall

This report provides a summary of syscalls and their execution times. Note that the “Time” column is calculated

using a summation of the “Total Time” column, and represents that syscall’s percent of the execution time of the syscalls listed, and not a percentage of the application wall or CPU execution time.

um_cpu_page_faults_sum – Unified Memory CPU Page Faults Summary#

Arguments - None

Output:

> CPU Page Faults : Number of CPU page faults that occurred CPU Instruction Address : Address of the CPU instruction that caused the CPU page faults
>
> This report provides a summary of CPU page faults for unified memory.

um_sum[:rows=<limit>] – Unified Memory Analysis Summary#

Arguments

- rows=<limit> - Maximum number of rows returned by the query. Default is 10.

Output:

- Virtual Address : Virtual base address of the page(s) being transferred
- HtoD Migration Size : Bytes transferred from Host to Device
- DtoH Migration Size : Bytes transferred from Device to Host
- CPU Page Faults : Number of CPU page faults that occurred for the virtual base address
- GPU Page Faults : Number of GPU page faults that occurred for the virtual base address
- Migration Throughput : Bytes transferred per second

This report provides a summary of data migrations for unified memory.

um_total_sum – Unified Memory Totals Summary#

Arguments - None

Output:

- Total HtoD Migration Size : Total bytes transferred from host to device
- Total DtoH Migration Size : Total bytes transferred from device to host
- Total CPU Page Faults : Total number of CPU page faults that occurred
- Total GPU Page Faults : Total number of GPU page faults that occurred
- Minimum Virtual Address : Minimum value of the virtual address range for the pages transferred
- Maximum Virtual Address : Maximum value of the virtual address range for the pages transferred

This report provides a summary of all the page faults for unified memory.

vulkan_api_sum – Vulkan API Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all executions of this function
- Num Calls: Number of calls to this function
- Avg : Average execution time of this function
- Med : Median execution time of this function
- Min : Smallest execution time of this function
- Max : Largest execution time of this function

- StdDev : Standard deviation of the time of this function
- Name : Name of the function

  This report provides a summary of Vulkan API functions and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that function’s percent of the execution time of the functions listed, and not a percentage of the application wall or CPU execution time.

  vulkan_api_trace – Vulkan API Trace#

  Arguments - None

  Output: All time values default to nanoseconds

- Start : Timestamp when API call was made
- Duration : Length of API calls
- Name : API function name
- Event Class : Vulkan trace event type
- Context : Trace context ID
- CorrID : Correlation used to map to other Vulkan calls
- Pid : Process ID that made the call
- Tid : Thread ID that made the call
- T-Pri : Run priority of call thread
- Thread Name : Name of thread that called API function

  This report provides a trace record of Vulkan API function calls and their execution times.

  vulkan_gpu_marker_sum – Vulkan GPU Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range
- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of Vulkan GPU debug markers, and their execution times. Note that the “Time” column is calculated using a summation of the “Total Time” column, and represents that range’s percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

vulkan_marker_sum – Vulkan Range Summary#

Arguments - None

Output: All time values default to nanoseconds

- Time : Percentage of “Total Time”
- Total Time : Total time used by all instances of this range
- Instances : Number of instances of this range
- Avg : Average execution time of this range

- Med : Median execution time of this range
- Min : Smallest execution time of this range
- Max : Largest execution time of this range
- StdDev : Standard deviation of execution time of this range
- Range : Name of the range

This report provides a summary of Vulkan debug markers on the CPU, and their execution times. Note that the "Time" column is calculated using a summation of the "Total Time" column, and represents that range's percent of the execution time of the ranges listed, and not a percentage of the application wall or CPU execution time.

wddm_queue_sum – WDDM Queue Utilization Summary#

Arguments - None

Output: All time values default to nanoseconds

- Utilization : Percent of time when queue was not empty
- Instances : Number of events
- Avg : Average event duration
- Med : Median event duration
- Min : Minimum event duration
- Max : Maximum event duration
- StdDev : Standard deviation of event durations
- Name : Event name
- Q Type : Queue type ID
- Q Name : Queue type name

- PID : Process ID associated with event
- GPU ID : GPU index
- Context : WDDM context of queue
- Engine : Engine type ID
- Node Ord : WDDM node ordinal ID

This report provides a summary of the WDDM queue utilization. The utilization is calculated by comparing the amount of time when the queue had one or more active events to total duration, as defined by the minimum and maximum event time for a given Process ID (regardless of the queue context).

Report Formatters Shipped With Nsight Systems#

The following formats are available in Nsight Systems

Column#

Usage:

column[:nohdr][:nolimit][:nofmt][:<width>[:<width>]...]

Arguments

- `nohdr` : Do not display the header.
- `nolimit` : Remove 100 character limit from auto-width columns Note: This can result in extremely wide columns.
- `nofmt` : Do not reformat numbers.
- `<width>...` : Define the explicit width of one or more columns. If the value `.` is given, the column will auto-adjust. If a width of 0 is given, the column will not be displayed.

The column formatter presents data in vertical text columns. It is primarily designed to be a human-readable format

for displaying data on a console display.

Text data will be left-justified, while numeric data will be right-justified. If the data overflows the available column width, it will be marked with a "…" character, to indicate the data values were clipped. Clipping always occurs on the right-hand side, even for numeric data.

Numbers will be reformatted to make easier to visually scan and understand. This includes adding thousands-separators. This process requires that the string representation of the number is converted into its native representation (integer or floating point) and then converted back into a string representation to print. This conversion process attempts to preserve elements of number presentation, such as the number of decimal places, or the use of scientific notation, but the conversion is not always perfect (the number should always be the same, but the presentation may not be). To disable the reformatting process, use the argument `nofmt`.

If no explicit width is given, the columns auto-adjust their width based off the header size and the first 100 lines of data. This auto-adjustment is limited to a maximum width of 100 characters. To allow larger auto-width columns, pass the initial argument nolimit. If the first 100 lines do not calculate the correct column width, it is suggested that explicit column widths be provided.

Table#

Usage:

table[:nohdr][:nolimit][:nofmt][:<width>[:<width>]...]

Arguments

- `nohdr` : Do not display the header.
- `nolimit` : Remove 100 character limit from auto-width columns Note: This can result in extremely wide columns.
- `nofmt` : Do not reformat numbers.
- `<width>...` : Define the explicit width of one or more columns. If the value `.` is given, the column will auto-adjust. If a width of 0 is given, the column will not be displayed.

The table formatter presents data in vertical text columns inside text boxes. Other than the lines between columns, it is identical to the column formatter.

CSV#

Usage:

csv[:nohdr]

Arguments

- `nohdr` : Do not display the header.

The csv formatter outputs data as comma-separated values. This format is commonly used for import into other data applications, such as spread-sheets and databases.

There are many different standards for CSV files. Most differences are in how escapes are handled, meaning data values that contain a comma or space.

This CSV formatter will escape commas by surrounding the whole value in double-quotes.

TSV#

Usage:

tsv[:nohdr][:esc]

Arguments

- `nohdr` : Do not display the header.
- `esc` : escape tab characters, rather than removing them.

The TSV formatter outputs data as tab-separated values. This format is sometimes used for import into other data applications, such as spreadsheets and databases.

Most TSV import/export systems disallow the tab character in data values. The formatter will normally replace any tab characters with a single space. If the esc argument has been provided, any tab characters will be replaced with the literal characters “t”.

JSON#

Usage:

json

Arguments: no arguments

The JSON formatter outputs data as an array of JSON objects. Each object represents one line of data, and uses the column names as field labels. All objects have the same fields. The formatter attempts to recognize numeric values, as well as JSON keywords, and converts them. Empty values are passed as an empty string (and not nil, or as a missing field).

At this time the formatter does not escape quotes, so if a data value includes double-quotation marks, it will corrupt the JSON file.

HDoc#

hdoc[:title=<title>][:css=<URL>]

Arguments:

- `title` : string for HTML document title.
- `css` : URL of CSS document to include.

The HDoc formatter generates a complete, verifiable (mostly), standalone HTML document. It is designed to be opened in a web browser, or included in a larger document via an `<iframe>`.

HTable#

Usage:

htable

Arguments: no arguments

The HTable formatter outputs a raw HTML `<table>` without any of the surrounding HTML document. It is designed to be included into a larger HTML document. Although most web browsers will open and display the document, it is better to use the HDoc format for this type of use.

Expert Systems Analysis#

The Nsight Systems expert system is a feature aimed at automatic detection of performance optimization opportunities in an application's profile. It uses a set of predefined rules to determine if the application has known bad patterns.

Using Expert System from the CLI#

usage:

```
nsys [global-options] analyze [options]
    [nsys-rep-or-sqlite-file]
```

If a .nsys-rep file is given as the input file and there is no .sqlite file with the same name in the same directory, it will be generated.

Note

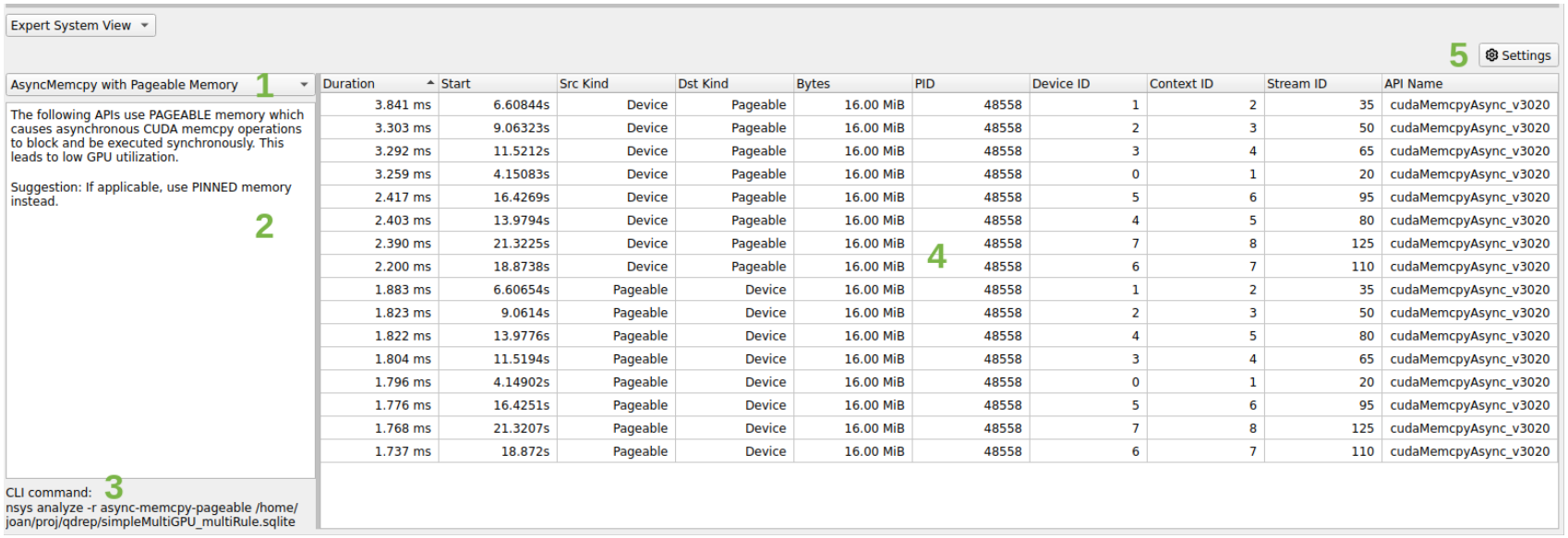
The Expert System view in the GUI will give you the equivalent command line.

Using Expert System from the GUI#

The Expert System View can be found in the same drop-down as the Events View. If there is no .sqlite file with the same name as the .nsys-rep file in the same directory, it will be generated.

The Expert System View has the following components:

1. Drop-down to select the rule to be run.
2. Rule description and advice summary.
3. CLI command that will give the same result.
4. Table containing results of running the rule.
5. Settings button that allows users to specify the rule's arguments.



Duration	Start	Src Kind	Dst Kind	Bytes	PID	Device ID	Context ID	Stream ID	API Name
3.841 ms	6.60844s	Device	Pageable	16.00 MiB	48558	1	2	35	cudaMemcpyAsync_v3020
3.303 ms	9.06323s	Device	Pageable	16.00 MiB	48558	2	3	50	cudaMemcpyAsync_v3020
3.292 ms	11.5212s	Device	Pageable	16.00 MiB	48558	3	4	65	cudaMemcpyAsync_v3020
3.259 ms	4.15083s	Device	Pageable	16.00 MiB	48558	0	1	20	cudaMemcpyAsync_v3020
2.417 ms	16.4269s	Device	Pageable	16.00 MiB	48558	5	6	95	cudaMemcpyAsync_v3020
2.403 ms	13.9794s	Device	Pageable	16.00 MiB	48558	4	5	80	cudaMemcpyAsync_v3020
2.390 ms	21.3225s	Device	Pageable	16.00 MiB	48558	7	8	125	cudaMemcpyAsync_v3020
2.200 ms	18.8738s	Device	Pageable	16.00 MiB	48558	6	7	110	cudaMemcpyAsync_v3020
1.883 ms	6.60654s	Pageable	Device	16.00 MiB	48558	1	2	35	cudaMemcpyAsync_v3020
1.823 ms	9.0614s	Pageable	Device	16.00 MiB	48558	2	3	50	cudaMemcpyAsync_v3020
1.822 ms	13.9776s	Pageable	Device	16.00 MiB	48558	4	5	80	cudaMemcpyAsync_v3020
1.804 ms	11.5194s	Pageable	Device	16.00 MiB	48558	3	4	65	cudaMemcpyAsync_v3020
1.796 ms	4.14902s	Pageable	Device	16.00 MiB	48558	0	1	20	cudaMemcpyAsync_v3020
1.776 ms	16.4251s	Pageable	Device	16.00 MiB	48558	5	6	95	cudaMemcpyAsync_v3020
1.768 ms	21.3207s	Pageable	Device	16.00 MiB	48558	7	8	125	cudaMemcpyAsync_v3020
1.737 ms	18.872s	Pageable	Device	16.00 MiB	48558	6	7	110	cudaMemcpyAsync_v3020

A context menu is available to correlate the table entry with the timeline. The options are the same as the Events View:

- Zoom to Selected on Timeline (ctrl+double-click)

The highlighting is not supported for rules that do not return an event but rather an arbitrary time range (e.g., GPU utilization rules).

The CLI and GUI share the same rule scripts and messages. There might be some formatting differences between the output table in GUI and CLI.

Expert System Rules#

Rules are scripts that run on the SQLite DB output from Nsight Systems to find common improvable usage patterns.

Each rule has an advice summary with explanation of the problem found and suggestions to address it. Only the top 50 results are displayed by default.

There are currently six rules in the expert system. They are described below. Additional rules will be made available in a future version of Nsight Systems.

CUDA Synchronous Operation Rules#

Asynchronous memcpy with pageable memory

This rule identifies asynchronous memory transfers that end up becoming synchronous if the memory is pageable. This rule is not applicable for Nsight Systems Embedded Platforms Edition

Suggestion: If applicable, use pinned memory instead

Synchronous Memcpy

This rule identifies synchronous memory transfers that block the host.

Suggestion: Use cudaMemcpy*Async APIs instead.

Synchronous Memset

This rule identifies synchronous memset operations that block the host.

Suggestion: Use cudaMemset*Async APIs instead.

Synchronization APIs

This rule identifies synchronization APIs that block the host until all issued CUDA calls are complete.

Suggestions: Avoid excessive use of synchronization. Use asynchronous CUDA event calls, such as cudaStreamWaitEvent and cudaEventSynchronize, to prevent host synchronization.

GPU Low Utilization Rules#

Nsight Systems determines GPU utilization based on API trace data in the collection. Current rules consider CUDA, Vulkan, DX12, and OpenGL API use of the GPU.

GPU Starvation

This rule identifies time ranges where a GPU is idle for longer than 500ms. The threshold is adjustable.

Suggestions: Use CPU sampling data, OS Runtime blocked state backtraces, and/or OS Runtime APIs related to thread synchronization to understand if a sluggish or blocked CPU is causing the gaps. Add NVTX annotations to CPU code to understand the reason behind the gaps.

Notes: For each process, each GPU is examined, and gaps are found within the time range that starts with the beginning of the first GPU operation on that device and ends with the end of the last GPU operation on that device. GPU gaps that cannot be addressed by the user are excluded. This includes:

- Profiling overhead in the middle of a GPU gap.

- The initial gap in the report that is seen before the first GPU operation.
- The final gap that is seen after the last GPU operation.

GPU Low Utilization

This rule identifies time regions with low utilization.

Suggestions: Use CPU sampling data, OS Runtime blocked state backtraces, and/or OS Runtime APIs related to thread synchronization to understand if a sluggish or blocked CPU is causing the gaps. Add NVTX annotations to CPU code to understand the reason behind the gaps.

Notes: For each process, each GPU is examined, and gaps are found within the time range that starts with the beginning of the first GPU operation on that device and ends with the end of the last GPU operation on that device. This time range is then divided into equal chunks, and the GPU utilization is calculated for each chunk. The utilization includes all GPU operations as well as profiling overheads that the user cannot address.

The utilization refers to the “time” utilization and not the “resource” utilization. This rule attempts to find time gaps when the GPU is or isn’t being used, but does not take into account how many GPU resources are being used. Therefore, a single running memcpy is considered the same amount of “utilization” as a huge kernel that takes over all the cores. If multiple operations run concurrently in the same chunk, their utilization will be added up and may exceed 100%.

Chunks with an in-use percentage less than the threshold value are displayed. If consecutive chunks have a low in-use percentage, the individual chunks are coalesced into a single display record, keeping the weighted average of percentages. This is why returned chunks may have different durations.

Advanced Report Analysis#

Nsight Systems Advanced Report Analysis is functionality to better support complex statistical analysis across multiple result files. Possible use cases for this functionality include:

- Multi-Node Analysis - When you run Nsight Systems across a cluster, it typically generates one result file per rank on the cluster. While you can load multiple result files into the GUI for visualization, this analysis system allows you to

run statistical analysis across all of the result files.

- Multi-Pass Analysis - Some features in Nsight Systems cannot be run together due to overhead or hardware considerations. For example, there are frequently more CPU performance counters available than your CPU has registers. Using this analysis, you could run multiple runs with different sets of counters and then analyze the results together.
- Multi-Run Analysis - Sometimes you want to compare two runs that were not taken at the same time together. Perhaps you ran the tool on two different hardware configurations and want to see what changed. Perhaps you are doing regression testing or performance improvement analysis and want to check your status. Comparing those result files statistically can show patterns.
- Complex/multi-phase analysis - Sometimes you may want to perform a complicated, or multi-phase analysis on one or more results files. The helper functionality available in the Advanced Analysis system can simplify common steps.
- Complex data output - Sometimes you want to be able to build complex visualizations from your analysis, rather than just tabular data from bare statistics.

Analysis Steps

Note

Prior to using advanced analysis, please make sure that you have installed all required dependencies. See **Installing Advanced Analysis System** in the **Installation Guide** for more information.

1. Generate the reports - Generate the reports as you always have, in fact, you can use reports that you have generated previously.
2. Set up - Choose the recipe (See Available Recipes, below), give it any required parameters, and run.
3. Launch Analysis - Nsight Systems will run the analysis, using your local system or Dask, as you have selected.
4. Output - the output is a directory containing an .nsys-analysis file, which can then be opened within the Nsight Systems GUI.
5. View the data - depending on your recipe, you can have any number of visualizations, from simple tabular

information to Jupyter notebooks which can be opened inside the GUI.

Available Advanced Analysis Recipes#

All advanced analysis recipes are run using the `recipe` CLI command switch.

usage:

nsys recipe [args] <recipe-name> [recipe args]

Nsight Systems provides several initial analysis recipes, mostly based around making our existing statistics and expert systems rules run multi-report.

These recipes can be found at `<target-linux-x64>/python/packages/nsys-recipe/recipes`. Please note that all recipes are in the form of python scripts. You may alter the given recipes or write your own to meet your needs. Refer to **Tutorial: Create a User-Defined Recipe** for an example of how to do this. However, be advised that the APIs may change for the next few versions. Additional recipes will be added on an ongoing basis.

For more information about a specific recipe, including recipe parameters, please use `nsys recipe [recipe name] --help`.

List of recipes

Each recipe will be tagged with one or more keywords to help understand its purpose.

Keywords	Description
Expert System	The recipe originated from the Expert System. A script with the same name is also available via `nsys analyze`, but its behavior and implementation may differ.
Stats System	The recipe originated from the Stats System. A script with the same name is also available via `nsys stats`, but its behavior and implementation may differ.

Keywords	Description
Trace	The recipe provides a trace record of individual events that are observable in the GUI timeline.
Summary	The recipe provides a summarized view of events, often representing aggregated data.
Pace	The recipe provides a detailed analysis of how a specific event progresses across the application.
Heatmap	The recipe provides a heatmap that visualizes patterns across the application.

- cuda_api_sumCUDA API Summary

  This recipe provides a summary of CUDA API functions and their execution times.

  Keywords: CUDA, Summary, Stats System

- cuda_api_syncCUDA Synchronization APIs

  This recipe identifies synchronization APIs that block the host until the issued CUDA calls are complete.

  Keywords: CUDA, Synchronization, Trace, Expert System

- cuda_gpu_kern_histCUDA GPU Kernel Duration Histogram

  This recipe represents the probability of the duration of a CUDA kernel among all its instances or all kernels in the program.

  Keywords: CUDA, Kernel, Histogram, Duration

- cuda_gpu_kern_paceCUDA GPU Kernel Pacing

  This recipe investigates the progress and consistency of a particular CUDA kernel throughout the application.

  Keywords: CUDA, Kernel, Pace

- cuda_gpu_kern_sumCUDA GPU Kernel Summary

  This recipe provides a summary of CUDA kernels and their execution times.

Keywords: CUDA, Kernel, Summary, Stats System

- cuda_gpu_mem_size_sumCUDA GPU MemOps Summary (by Size)
  This recipe provides a summary of GPU memory operations and the amount of memory they utilize.

  Keywords: CUDA, Memory, Summary, Stats System

- cuda_gpu_mem_time_sumCUDA GPU MemOps Summary (by Time)
  This recipe provides a summary of GPU memory operations and their execution times.

  Keywords: CUDA, Memory, Summary, Stats System

- cuda_gpu_time_util_mapCUDA GPU Time Utilization Heatmap
  This recipe calculates the percentage of time that CUDA kernels were running.

  Keywords: CUDA, Kernel, Heatmap

- cuda_memcpy_asyncCUDA Async Memcpy with Pageable Memory
  This recipe identifies asynchronous memory transfers that end up becoming synchronous if the memory is pageable.

  Keywords: CUDA, Memcpy, Trace, Expert System

- cuda_memcpy_syncCUDA Synchronous Memcpy
  This recipe identifies memory transfers that are synchronous.

  Keywords: CUDA, Memcpy, Trace, Expert System

- cuda_memset_syncCUDA Synchronous Memset
  This recipe identifies synchronous memset operations with pinned host memory or Unified Memory region.

  Keywords: CUDA, Memset, Trace, Expert System

- diffStatistics Diff
  This script compares outputs from two runs of the same statistical recipe.

  Keywords: Diff, Summary

- dx12_mem_opsDX12 Memory Operations

  This recipe flags problematic memory operations with warnings.

  Keywords: DX12, Memory, Trace, Expert System

- file_access_sumOS Runtime File Access Summary

  This recipe provides a summary of file access functions, including high-level overview of file access patterns across the system.

  Keywords: OSRT, Summary

- gfx_hotspotGraphics Hotspot Analysis

  This recipe generates a report of CPU hotspots for graphics applications.

  The output format for this recipe is different than other recipes. See gfx_hotspot Recipe.

  Keywords: DX12, Vulkan, Summary, Trace

- gpu_gapsGPU Gaps

  This recipe identifies time regions where a GPU is idle for longer than a set threshold.

  Keywords: CUDA, Utilization, Expert System

- gpu_metric_util_mapGPU Metric Utilization Heatmap

  This recipe calculates the percentage of SM Active, SM Issue, and Tensor Active metrics.

  Keywords: GPU Metrics, Heatmap

- gpu_time_utilGPU Time Utilization

  This recipe identifies time regions with low GPU utilization.

  Keywords: CUDA, Utilization, Expert System

- mpi_gpu_time_util_mapMPI and GPU Time Utilization Heatmap

  This recipe calculates the percentage of time that CUDA kernels were running and MPI communication was active, as well as their overlap.

Keywords: MPI, CUDA, Kernel, Utilization, Heatmap

- mpi_sumMPI Summary

  This recipe provides a summary of MPI functions and their execution times.

  Keywords: MPI, Summary

- nccl_gpu_overlap_traceNCCL GPU Overlap Trace

  This recipe calculates the percentage of overlap for communication and compute kernels. Communication kernels are identified by the 'nccl' prefix.

  Keywords: NCCL, CUDA, Kernel, Overlap, Trace

- nccl_gpu_proj_sumNCCL GPU Projection Summary

  This recipe provides a summary of NCCL functions projected from the CPU onto the GPU, and their execution times.

  Keywords: NCCL, CUDA, GPU Projection, Summary

- nccl_gpu_time_util_mapNCCL GPU Time Utilization Heatmap

  This recipe calculates the percentage of time that communication and compute kernels were running, as well as their overlap. Communication kernels are identified by the 'nccl' prefix.

  Keywords: NCCL, CUDA, Kernel, Utilization, Overlap, Heatmap

- nccl_sumNCCL Summary

  This recipe provides a summary of NCCL functions and their execution times.

  Keywords: NCCL, Summary

- network_map_awsAWS Metrics Heatmap

  This recipe displays heatmaps of AWS EFA metrics.

  Keywords: Network, AWS, EFA, Heatmap

- network_sumNetwork Traffic Summary

  This recipe provides a summary of the network traffic over NICs and InfiniBand Switches.

Keywords: Network, Summary

- network_traffic_mapNetwork Devices Traffic Heatmap

  This recipe displays heatmaps of sent traffic, received traffic, and congestion events for network devices.

  Keywords: Network, Heatmap

- nvtx_cpu_topdownCPU Topdown methodology metrics correlated to NVTX ranges

  This recipe calculates CPU Topdown methodology metrics for NVTX push/pop ranges based on collected PMU core events for Grace CPU.

  For details and use cases of this recipe, see nvtx_cpu_topdown Recipe.

  Keywords: NVTX, CPU Topdown, Metrics, Summary

- nvlink_sumNVLink Network Bandwidth Summary

  This recipe provides a summary of the NVLink network bandwidth.

  Keywords: NVLink, Summary

- nvtx_gpu_proj_paceNVTX GPU Projection Pacing

  This recipe investigates the progress and consistency of a particular NVTX range projected from the CPU onto the GPU, throughout the application.

  Keywords: NVTX, GPU Projection, Pace

- nvtx_gpu_proj_sumNVTX GPU Projection Summary

  This recipe provides a summary of NVTX time ranges projected from the CPU onto the GPU, and their execution times.

  Keywords: NVTX, GPU Projection, Summary, Stats System

- nvtx_gpu_proj_traceNVTX GPU Projection Trace

  This recipe provides a trace of NVTX time ranges projected from the CPU onto the GPU.

  Keywords: NVTX, GPU Projection, Trace, Stats System

- nvtx_paceNVTX Pacing

  This recipe investigates the progress and consistency of a particular NVTX range throughout the application.

  Keywords: NVTX, Pace

- nvtx_sumNVTX Range Summary

  This recipe provides a summary of NVTX Start/End and Push/Pop Ranges, and their execution times.

  Keywords: NVTX, Summary, Stats System

- osrt_sumOS Runtime Summary

  This recipe provides a summary of C library functions and their execution times.

  Keywords: OSRT, Summary, Stats System

- storage_util_mapStorage Metrics Heatmap

  This recipe displays heatmaps of storage devices metrics.

  Keywords: Storage, Heatmap

- ucx_gpu_time_util_mapUCX and GPU Time Utilization Heatmap

  This recipe calculates the percentage of time that CUDA kernels were running and UCX communication was active, as well as their overlap.

  Keywords: UCX, CUDA, Kernel, Heatmap

Recipe Output Examples#

A successful recipe run outputs a directory containing different files. This section gives some common examples of these output types.

Table

Trace or summary data will be stored in data storage formats such as CSV, Parquet, or Arrow. Typically, you can also access the same data within the output Jupyter notebook.

- Summary table:

Name	Count	Mean	Std (approx)	Min	Q1 (approx)	Median (approx)	Q3 (approx)	Max	Sum
cuEventCreate	1024	403.0	209.9	280	290.0	300.0	400.0	2040	412703
cuEventDestroy_v2	1024	444.9	425.6	230	300.0	335.0	430.0	12920	455543
cuEventQuery	176771	503.6	503.0	330	350.0	390.0	470.0	149101	89028223
cuEventRecord	8080	564.6	276.4	460	490.0	535.0	610.0	8030	4562037
cuIpcOpenMemHandle_v2	16	4723043.1	4316371.8	519724	1219136.0	2144399.2	17721903.2	22144656	75568690
cuMemcpyDtoDAsync_v2	8080	4159.1	1074.4	3340	3870.0	3975.0	4190.0	32560	33605478
cuStreamCreate	128	28568.4	103817.3	1430	1520.0	2015.0	3420.0	1928021	3656753
cuStreamDestroy_v2	128	6302.6	3507.5	4110	4535.0	5230.0	6700.0	28200	806736
cudaDeviceSynchronize_v3020	16	344109.9	470296.5	9390	89878.0	342900.2	773942.8	1027757	5505759
cudaEventCreateWithFlags_v3020	24	2009.6	2085.2	660	720.0	870.0	2890.0	5010	48231
cudaEventDestroy_v3020	24	2057.1	2448.2	610	630.0	665.0	4860.0	9040	49371
cudaEventRecord_v3020	24000	919.1	269.0	790	820.0	880.0	1020.0	39680	22058415

- Trace table:

Text	Start	End	PID	Stack Level	Children Count	Range ID	Parent ID	Range Stack
MPI_Warmup	2526325781	2757915766	10125	0	0	2	<NA>	[2]
Jacobi solve	2758082389	4752071048	10125	0	1000	3	<NA>	[3]
Jacobi iteration 0	2758082389	2758972273	10125	1	0	4	3	[3, 4]
Jacobi iteration 1	2760602440	2761412259	10125	1	0	5	3	[3, 5]
Jacobi iteration 2	2762986714	2763793238	10125	1	0	6	3	[3, 6]
Jacobi iteration 3	2765383629	2766194152	10125	1	0	7	3	[3, 7]
Jacobi iteration 4	2767774816	2768586043	10125	1	0	8	3	[3, 8]
Jacobi iteration 5	2770165011	2770975342	10125	1	0	9	3	[3, 9]
Jacobi iteration 6	2772557029	2773366305	10125	1	0	10	3	[3, 10]
Jacobi iteration 7	2774986360	2775794228	10125	1	0	11	3	[3, 11]

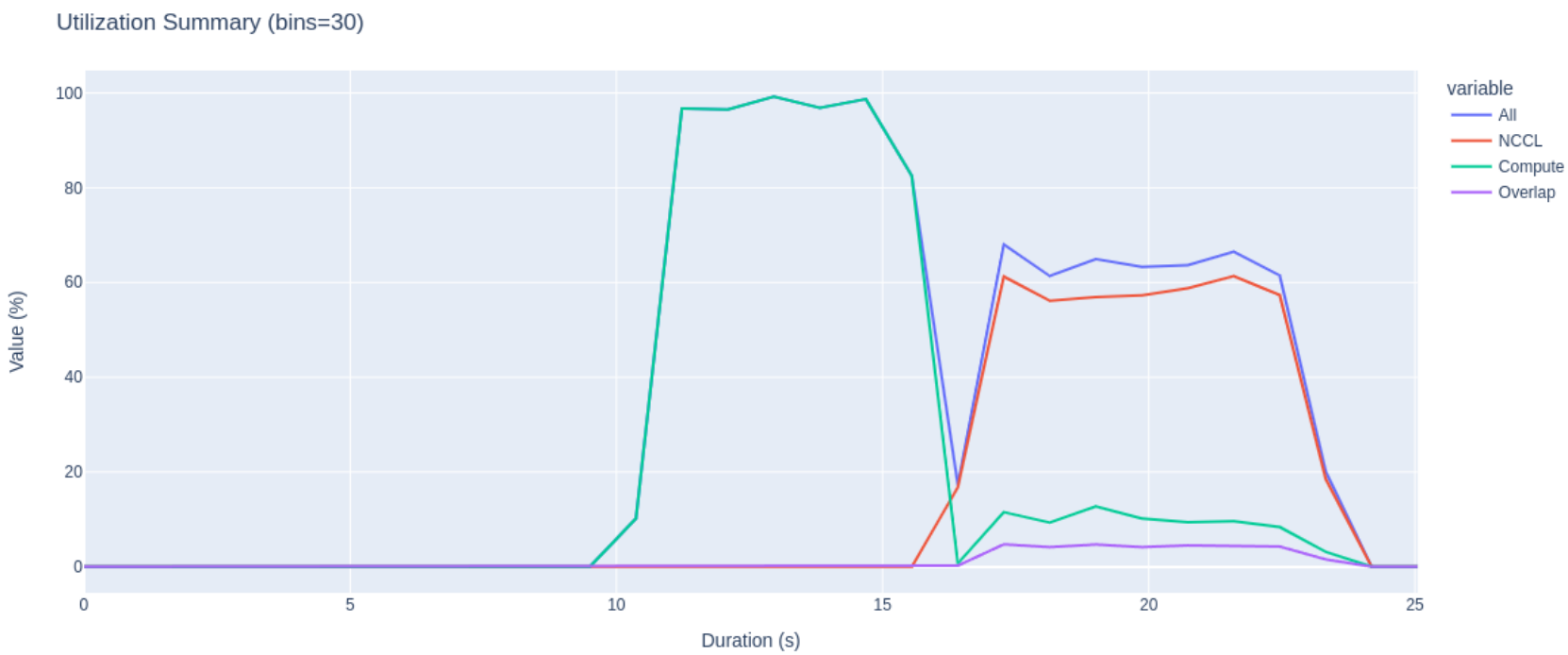
- Overlap table:

Name	Start	End	PID	DeviceID	Communication Overlap	Compute Overlap
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16675162477	16675231053	12354	0	0.0	0.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16675423245	16675508813	12354	0	0.0	100.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16676118892	16687832701	12354	0	0.0	5.2
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16687864125	16687919613	12354	0	0.0	100.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16688670524	16688800508	12354	0	0.0	100.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16697464305	16705822278	12354	0	0.0	0.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16705851910	16705877734	12354	0	0.0	100.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16706697605	16706881445	12354	0	0.0	100.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16707525988	16707638244	12354	0	0.0	100.0
ncclKernel_SendRecv_RING_SIMPLE_Sum_int8_t	16708227523	16708307459	12354	0	0.0	100.0

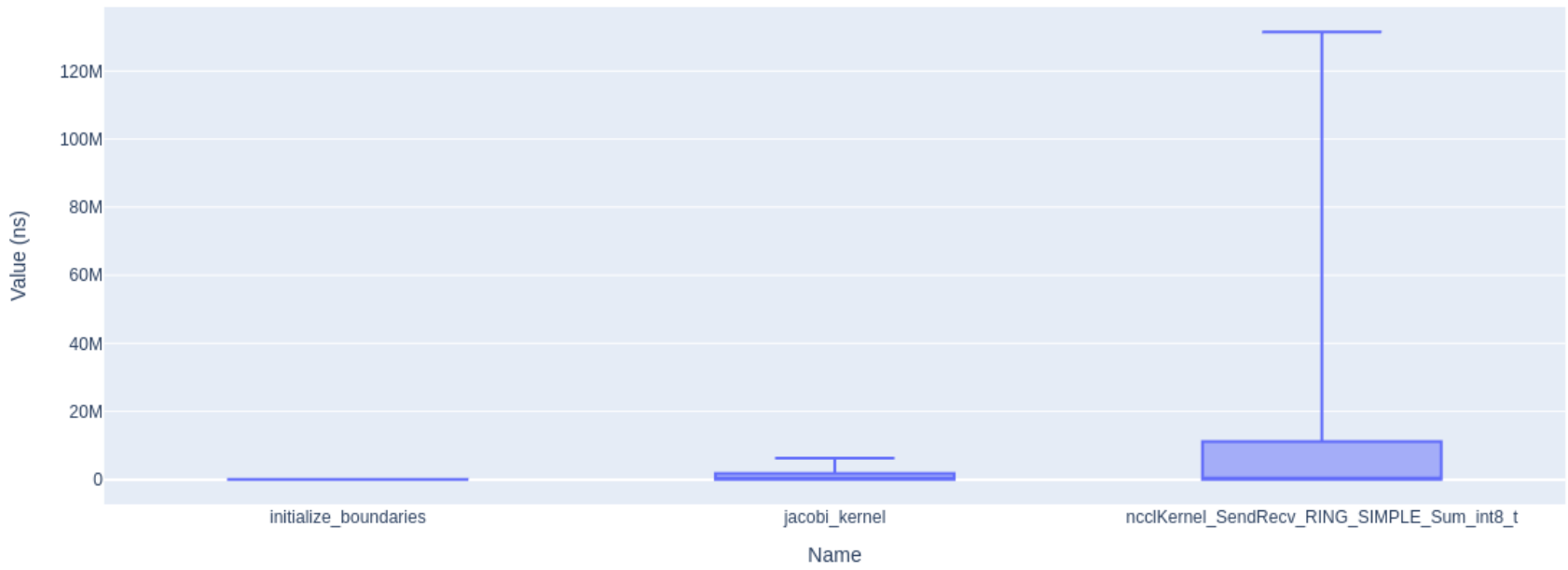
Visualization

Some recipes include data visualization in the output Jupyter notebooks. These graphs use Plotly, which provides interactivity.

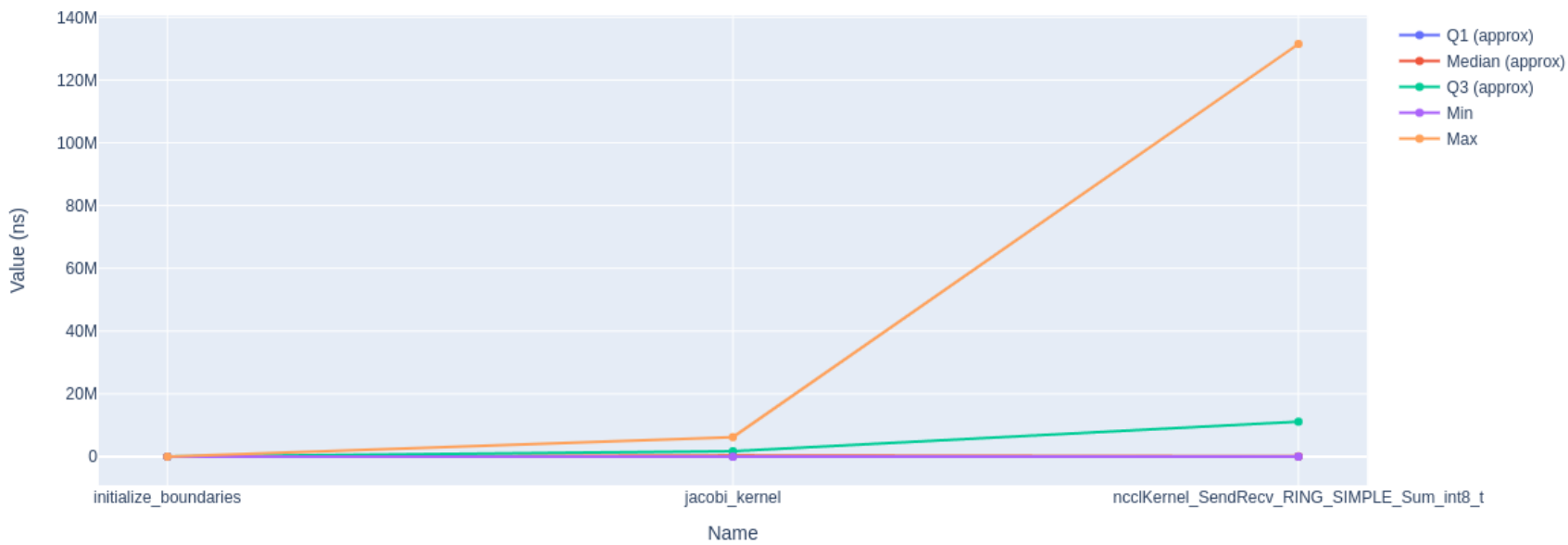
- Summary graph:


Utilization Summary (bins=30)
Value (%)
100
80
60
40
20
0
0
5
10
15
20
25
Duration (s)
variable
All
NCCL
Compute
Overlap

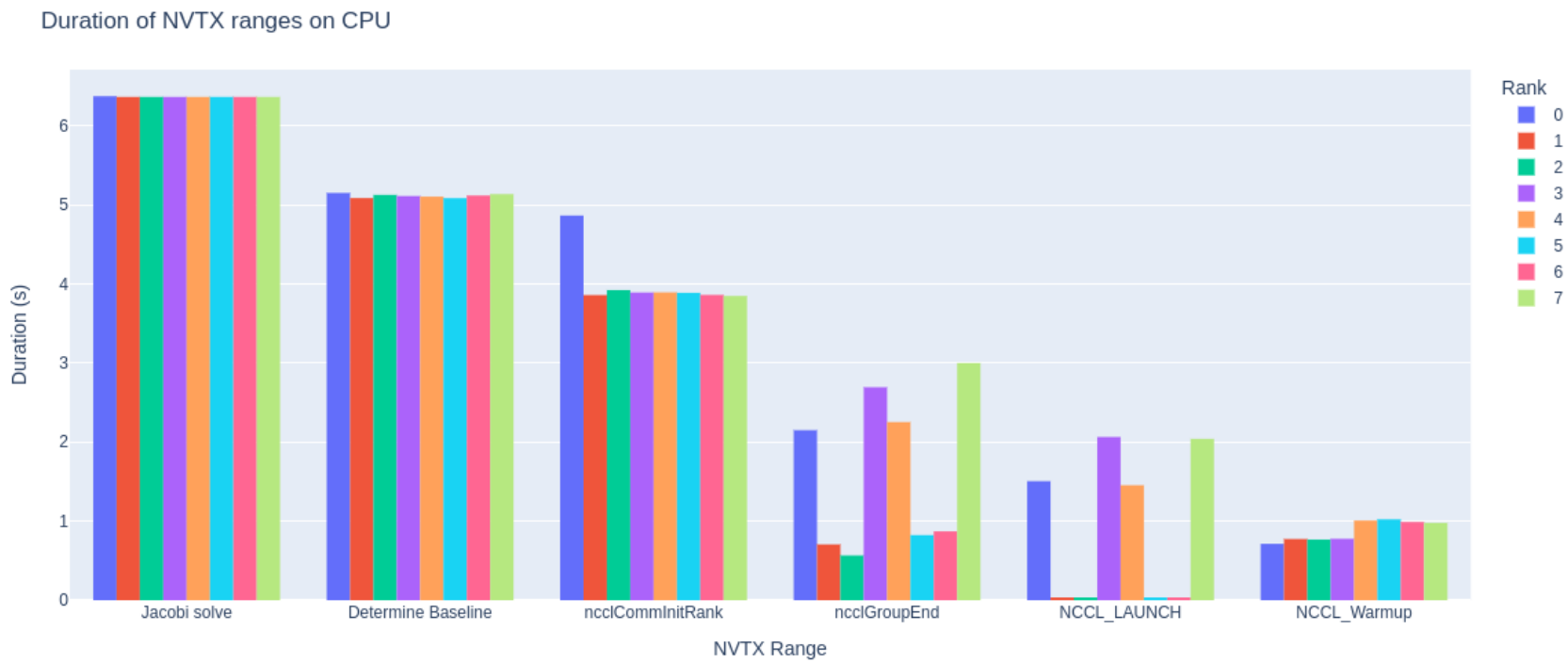
- Box plot:



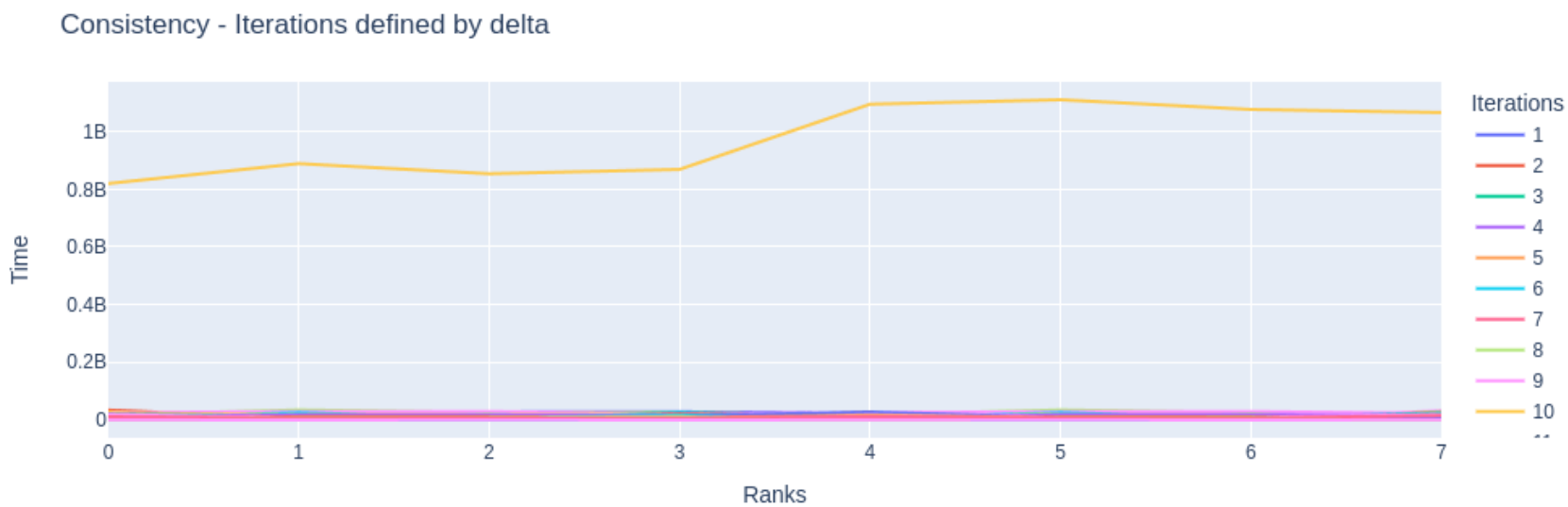
- Line graph:



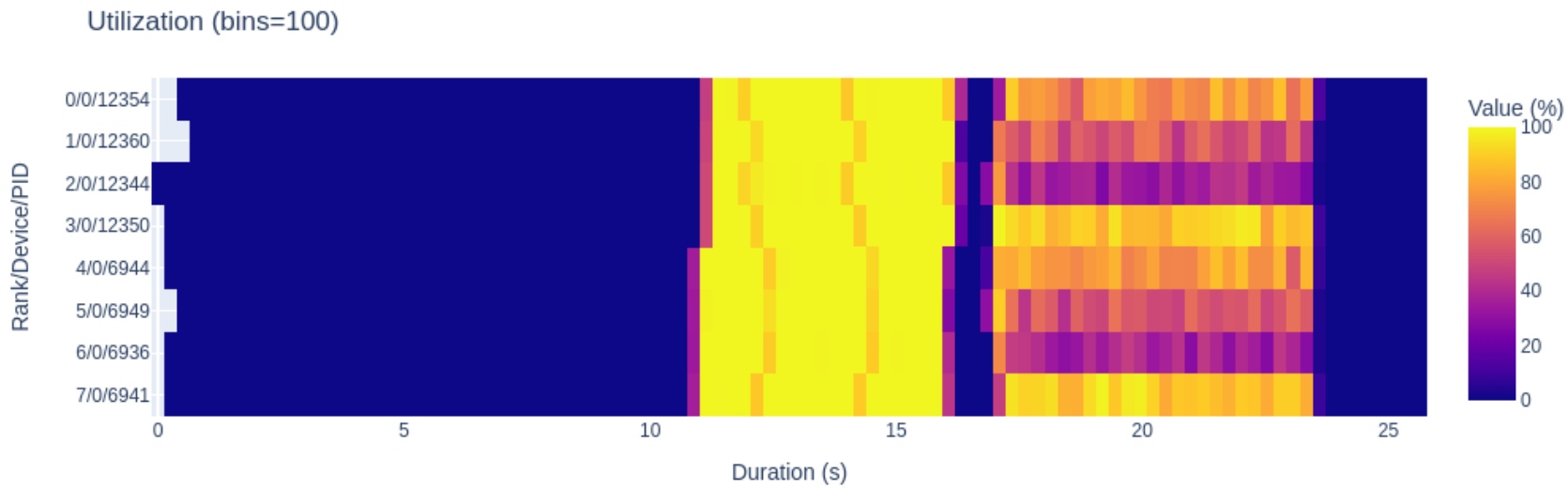
- Top N graph:


Duration of NVTX ranges on CPU
Duration (s)
0
1
2
3
4
5
6
Jacobi solve
Determine Baseline
ncclCommInitRank
ncclGroupEnd
NCCL_LAUNCH
NCCL_Warmup
NVTX Range
Rank
0
1
2
3
4
5
6
7

- Pace graph:

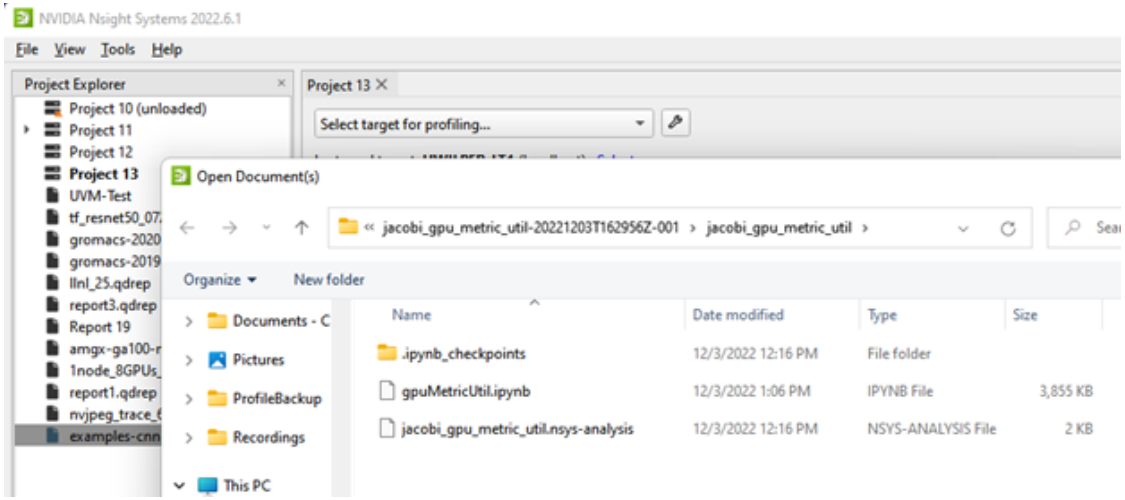

Consistency - Iterations defined by delta
Time
1B
0.8B
0.6B
0.4B
0.2B
0
0
1
2
3
4
5
6
7
Ranks
Iterations
1
2
3
4
5
6
7
8
9
10

- Heatmap:

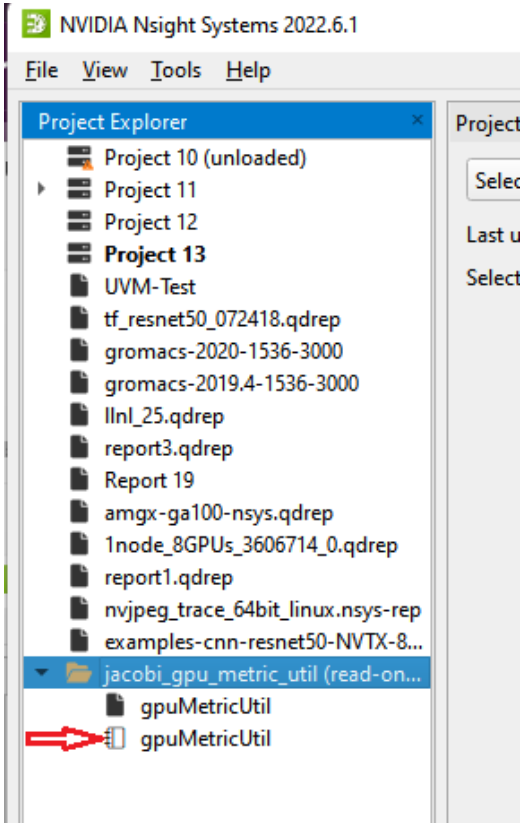

Utilization (bins=100)
Rank/Device/PID
0/0/12354
1/0/12360
2/0/12344
3/0/12350
4/0/6944
5/0/6949
6/0/6936
7/0/6941
0
5
10
15
20
25
Duration (s)
Value (%)
100
80
60
40
20
0

Opening in Jupyter Notebook#

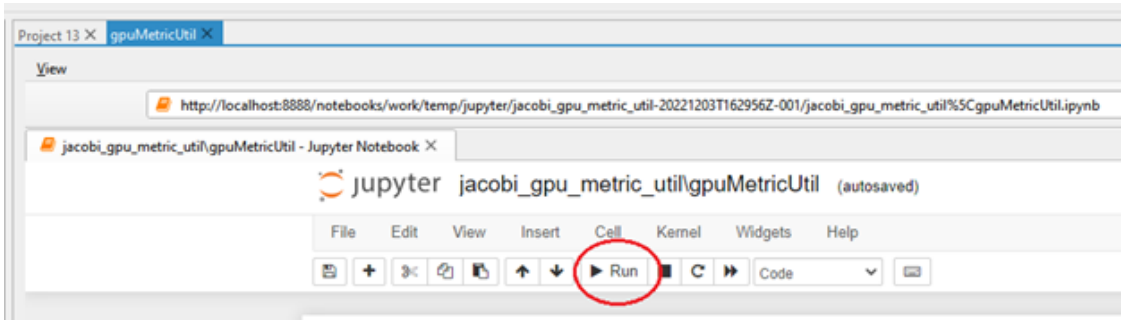
Running the recipe command creates a new analysis file (.nsys-analysis). Open the Nsight Systems GUI and select `File->Open`, and pick your file.



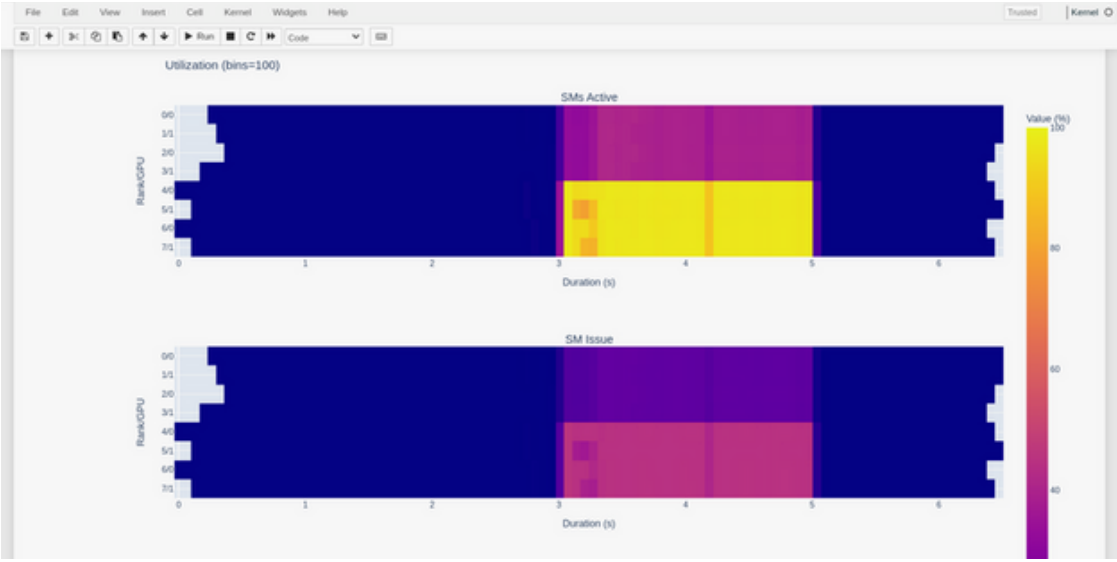
Open the folder icon and click on the notebook icon to open the Jupyter notebook.



Run the Jupyter notebook:



And the output appears on-screen. In this case a heat map of activity running a Jacobi solver.



Configuring Dask#

The multi-report analysis system does not offer options to configure the Dask environment. However, you could achieve this by modifying the recipe script directly or using one of the following from Dask's configuration system:

- YAML files: By default, Dask searches for all YAML files in `~/.config/dask/` or `/etc/dask/`. This search path can be changed using the environment variable `DASK_ROOT_CONFIG` or `DASK_CONFIG`. See the Dask documentation for the complete list of locations and the lookup order. Example:

  ```
  $ cat example.yaml
  'Distributed':
      'scheduler':
        'allowed-failures': 5
  ```

- Environment variables: Dask searches for all environment variables that start with `DASK_`, then transforms keys by converting to lower-case and changing double-underscores to nested structures. See Dask documentation for the complete list of variables. Example:

  ```
  DASK_DISTRIBUTED__SCHEDULER__ALLOWED_FAILURES=5
  ```

Dask Client

With no configuration set, the dask-futures mode option initializes the Dask Client with the default arguments, which results in creating a LocalCluster in the background. The following are the YAML/environment variables that could be set to change the default behavior:

- distributed.comm.timeouts.connect / DASK_DISTRIBUTED__COMM__TIMEOUTS__CONNECT
- client-name / DASK_CLIENT_NAME
- scheduler-address / DASK_SCHEDULER_ADDRESS
- distributed.client.heartbeat / DASK_DISTRIBUTED__CLIENT__HEARTBEAT
- distributed.client.scheduler-info-interval / DASK_DISTRIBUTED__CLIENT__SCHEDULER_INFO_INTERVAL
- distributed.client.preload / DASK_DISTRIBUTED__CLIENT__PRELOAD
- distributed.client.preload-argv / DASK_DISTRIBUTED__CLIENT__PRELOAD_ARGV

Recipe's environment variables

Recipe has its own list of environment variables to make the configuration more complete and flexible. These environment variables are either missing from Dask's configuration system or specific to the recipe system:

- NSYS_DASK_SCHEDULER_FILE: Path to a file with scheduler information. It will be used to initialize the Dask Client.
- NSYS_DIR: Path to the directory of Nsight Systems containing the target and host directories. The nsys executable and the recipe dependencies will be searched in this directory instead of the one deduced from the currently running recipe file path.

gfx_hotspot Recipe#

This recipe's output is different from other recipes and is presented as a web application.

The output can be viewed by passing the `--run-viewer` argument to the recipe - along with the further `--show-`

`viewer` which will automatically open a web browser to the report view.

Alternatively, a previously-executed `gfx_hotspot` recipe's output can be viewed by executing the `run_viewer.py` script from the recipe output folder.

For the best results, run the recipe on a report with resolved symbols.

Threading Analysis

In this tab, an overview of the multi-threading behavior of the target (most active) process is presented.

- Application Statistics:
  This table shows the CPU and thread statistics for the target process.
- CPU Info:
  This table shows information about the CPU hardware.
- Top 5 Processes CPU Utilisation:
  This table shows the most active processes during the sample, to help detect situations where another process is interfering with the target process's execution.
- Threading Health Check:
  This table contains a list of very common CPU-bound application performance indicators. If the target application is GPU bound, the entire table will be shown in green. If it is CPU bound, then each row will be highlighted in green if the value is healthy, in yellow if it requires attention, and in red if it potentially indicates of a threading issue. For unhealthy metrics, the "warning" column will also show steps or investigation angles that may be considered in order to improve the result.
- Thread Utilisation:
  This graph shows the process threads, ordered from most busy to least.
- Thread Concurrency:
  This graph shows the percentage and amount of time an average graphic frame is running each number of threads concurrently. High percentage of low thread counts could indicate excessive serialization in the algorithm, where CPU work could be better parallelized by improving the use of multi-threading.

Hotspot Analysis

In this tab, frames are selected in one of four methods:

- Longest Frame time (Slow Frames)
- Periodic time-based selection (Periodic Frames)
- Frames with highest transfer activity (Bar1 Reads)
- Frames with least GPU activity (GR Idle)

The report view then allows comparing the selected frames to each other and to the median frame in the same metric, helping identify the main differences and possible problem areas in each one.

- Overview:

  These tables show the report overview as well as the frame selection method and other capture-wide statistics and general information. A shorthand list of the “Performance Issues” table for each frame is also shown.

- Frame Times:

  This graph shows a sequence of the graphical frames (CPU time and GPU time derived from GPU Utilisation percentage per CPU frame time) ordered by their index. The selected frames are indicated and labelled. Clicking any of the indicated frames will set it as the left frame for comparison.

- Region / Compare to:

  These controls allow selecting the two frames to be shown for comparison. “Periodic Frames” shows 10 sampled frames (with equally distributed indices), while the other three modes show the 5 frames with the highest value in the chosen metric and the median frame in the same metric. All information from this point onwards is shown per selected frame in each of the two columns, allowing for 1-to-1 comparison. Selecting the same frame for both controls will show just the single frame as the entire width of the view.

- Frame Info:

  This table shows the frame duration and start time, the number of threads that were active during the frame, and the thread IDs of key threads in the frame processing operation which are important for determining likely

performance issues.

- Performance Issues:
  This table shows the key performance limiters and hotspots for the selected frame. Each indicator will have a breakdown of what indicators were present to call out the performance issue during this frame. These indicators are not necessarily the root cause of the problems in the region, but have been flagged for consideration.
- GPU Metrics:
  This table shows the average or total (respectively) values of the GPU metrics collected during the frame time. If GPU Metrics were not collected, this table will not appear.
- System ETW Events (Windows only):
  This chart shows a breakdown of the system process-reported ETW events during the frame. If WDDM trace and Custom ETW trace were not collected, this chart will not appear.
- DxgKrnl Events (Windows only):
  This chart shows a breakdown of the DxgKrnl ETW provider events during the frame. If WDDM trace was not collected, this chart will not appear.
- CPU Thread Utilisation Time:
  This graph shows the time spent inside each thread during the frame. The bars match the two selected frames, and the matching-colored line shows the total frame time. Clicking any of the columns in the graph will select that thread for the following elements in the report.
- Thread:
  This control allows selecting the thread to be shown in the following views.
- Call stacks:
  This control shows the sampled call stacks during the frame. Clicking a call stack frame will filter the view to only show call stacks containing this call stack frame, allowing to drill down into potential problem areas. The title of the control indicates the two modes selected for display, which can be switched with the two toggles in the top right of the control:

- Call stacks - Merged:
  Merges all similar call stacks logically, regardless of when in the frame time the functions appeared. This is useful to see where the cumulative time is spent.
- Call stacks - Over Time:
  Keeps call stacks ordered chronologically, so that repeated calls to the same function appear separately.
- Periodic Sampled Call stacks:
  Only shows call stacks acquired by periodic sampling (matching the orange marks in Nsight Systems's timeline view). This view provides a better statistical overview of where the frame time was spent.
- All Call stacks:
  Shows periodic sampled call stacks as well as call stacks acquired from other sources such as call stacks from ETW events (Windows) and event-based sampling (Linux) (matching both the orange AND the grey marks in Nsight Systems's timeline view).
- Modules in Sampled Call Stacks:
  This graph shows the number of call stacks in the frame that include at least one call stack frame in a function belonging to each module. This helps identify which modules were the most active during the frame.
- ETW Events (Windows only):
  This chart shows a breakdown of the thread-reported ETW events during the frame. If WDDM trace and Custom ETW trace were not collected, this chart will not appear.
- Context Switch Call Stacks:
  This table shows a breakdown of the call stacks that led to context switches for the thread during the frame, indicating where the thread may have stalled. Hovering the mouse cursor over the "Name" column will show the full call stack for each entry.
- DX12 API / Vulkan API:
  These tables show a breakdown of the graphical API functions that appeared in sampled call stacks. If DX12 / Vulkan trace were not collected, these tables will not appear.

- Known Symbols From Sampled Call Stacks:
  This table shows a breakdown of known symbols that often cause performance issues, such as DX12's CreateCommittedResource. If symbols were not resolved for the nsys-rep file, this table will not appear.
- PIX Markers (Windows only):
  This table shows a breakdown of PIX marker ranges that contained sampled call stacks. If WDDM trace and DX11 / DX12 trace were not collected or the target application does not use PIX markers, this table will not appear.

nvtx_cpu_topdown Recipe#

This recipe calculates CPU Topdown methodology metrics for NVTX push/pop ranges based on collected PMU core events for Grace CPU. It can process multiple Nsight Systems reports.

We recommend using this recipe after running the `collect_grace_topdown.sh` script, which simplifies collecting all PMU core event and metric data needed to perform a traditional CPU Topdown analysis of the workload's CPU performance. For more details on this script, refer to the Arm Topdown Analysis section.

If PMU core events other than those required by Topdown are collected, the recipe will calculate available CPU metrics based on them and display those metrics in the output.

Use case

The recipe is expected to be useful when the following is true:

1. The application is run on Grace CPU.
2. The application is instrumented with NVTX push/pop ranges.
3. NVTX range spans a specific CPU algorithm / code section that does not make syscalls or calls to other libraries whose functions take significant time to execute.
4. NVTX ranges with the same name are used to represent the same workload across all threads and all repetitions.
5. Due to the limitation of the maximum sampling rate of ~1 sample/ms, the duration of NVTX ranges should be 5 ms or

more to obtain more accurate results.

Note

For the case of NVTX ranges from multiple threads, only the NVTX ranges from the main thread will be processed.

Usage

[1] mkdir reports && cd reports
[2] <path to target-linux-sbsa-armv8>/cpu/collect_grace_topdown.sh ./myApp
[3] nsys recipe nvtx_cpu_topdown --input .

1. This step creates a new directory to store the reports. We recommend using an empty new directory, because the `collect_grace_topdown.sh` script overwrites the output files and does not currently allow customization of names.
2. This step creates several report files: cpu-td1.nsys-rep, cpu-branch-ipc.nsys-rep, etc.

   Note

   Note that since multiple reports are created, this step can take significant time to complete.
3. This step runs the recipe, uses all reports in the current directory as the input, and produces a `.ipynb` Jupyter notebook, `.parquet` and `.csv` (if `--csv` is used) files as the output.

   Output

   As the main output, the recipe generates the Jupyter notebook `nvtx_cpu_topdown.ipynb` with the following sections:

- NVTX Summary:

  This block displays a summary of NVTX ranges compiled from Nsight Systems reports provided to the recipe.

  For a report selected from the drop-down menu, the block shows the table with NVTX ranges, the number of their instances, and the median duration of the NVTX range instance.

  If NVTX ranges are filtered out, they are grayed out in the table, and a note is displayed in the *Notes* column for

the corresponding range. The following ranges are candidates to be filtered out:

- Ranges that contain fewer than 3 PMU samples in at least one Nsight Systems report provided to the recipe.
- Ranges that are not present in at least one Nsight Systems report provided to the recipe.

If NVTX ranges are not stable across some of the reports, the block will display a warning next to the unstable data and a note in the *Notes* column for the corresponding range.

NVTX Summary

This block displays a summary of NVTX ranges compiled from Nsight Systems reports provided to the recipe.

```
df = pd.read_parquet("nvtx_summary.parquet") •••
```

Report cpu-branch-ipc

	NVTX Range	Count	Instance Duration	Notes
0	parent	1	925.91 ms	
1	└─ loop1	3	154.30 ms ± 0.01 ms	
2	└─ loop2	3	154.27 ms ± 0.02 ms	
3	└─ loop3	3	0.03 ms ± 0.00 ms	This range is filtered out, because it contains fewer than 3 PMU samples in at least one report.

- CPU Topdown Methodology Metrics:

  This block presents the metric results of the CPU Topdown methodology for the selected NVTX range.

  For the range name selected from the drop-down menu, the most appropriate NVTX range instance is identified from the Nsight Systems reports as follows:

  > The NVTX range instance with the median duration from the first report (displayed by default in the *NVTX Summary* section) is considered. The relevant data is then extracted for the same instance index in the subsequent reports.

  The block shows the following tables:

  1. Topdown Level 1 metrics
  2. Frontend Bound metrics

3. Backend Bound metrics

4. Bad Speculation metrics

5. Retiring metrics

6. Miscellaneous metrics

Each table is displayed only when the required data is available.

CPU Topdown Methodology Metrics

This block presents the metric results of the CPU Topdown methodology for the selected NVTX range.

```
df = pd.read_parquet("cpu_metrics.parquet") •••
```

NVTX Range loop1

	Metric	Value	Description	Equation	Report
0	Instructions Per Cycle (IPC)	2.00	Instructions retired per cycle	INST_RETIRED / CPU_CYCLES	cpu-branch-ipc.nsys-rep
1	Frontend Bound	74.93	Percentage of total slots that were stalled because of resource constraints in the processor frontend	(STALL_SLOT_FRONTEND / (CPU_CYCLES * 8) - BR_MIS_PRED / CPU_CYCLES) * 100	cpu-td1.nsys-rep
2	Backend Bound	0.09	Percentage of total slots that were stalled because of resource constraints in the processor backend	(STALL_SLOT_BACKEND / (CPU_CYCLES * 8) - BR_MIS_PRED * 3 / CPU_CYCLES) * 100	cpu-td1.nsys-rep
3	Bad Speculation	0.00	Percentage of total slots that executed operations and didn't retire due to a pipeline flush	100 * ((1-OP_RETIRED/OP_SPEC) * (1-STALL_SLOT/(CPU_CYCLES * 8)) + BR_MIS_PRED * 4 / CPU_CYCLES)	cpu-td1.nsys-rep
4	Retiring	24.98	Percentage of total slots that are retired operations	100 * (OP_RETIRED / OP_SPEC) * (1 - (STALL_SLOT / (CPU_CYCLES * 8)))	cpu-td1.nsys-rep

Frontend Bound

Category		Metric	Value	Description	Equation	Report
Branch Effectiveness	0	Branch MPKI	0.00	The number of branch mispredictions per thousand instructions executed	(BR_MIS_PRED_RETIRED / INST_RETIRED) * 1000	cpu-branch-ipc.nsys-rep
	1	Branch MPKI	0.00	The number of branch mispredictions per thousand instructions executed	(BR_MIS_PRED_RETIRED / INST_RETIRED) * 1000	cpu-dtlb.nsys-rep

- Report Summary:

  This block displays information about the Nsight Systems report files given to the recipe for input, as well as: the PMU core events collected in each specific report, and the CPU core metrics computed for each specific report.

Report Summary

This block displays information about the Nsight Systems report files given to the recipe for input, as well as: the CPU events collected in each specific report, and the CPU metrics computed for each specific report.

```
df = pd.read_parquet("report_summary.parquet") ●●●
```

	Report	CPU events	CPU metrics
0	cpu-branch-ipc.nsys-rep	BR_MIS_PRED_RETIRED BR_RETIRED CPU_CYCLES INST_RETIRED	Instructions Per Cycle (IPC) Branch Misprediction Ratio Branch MPKI
1	cpu-dtlb.nsys-rep	L1D_CACHE L1D_CACHE_REFILL L1D_TLB BR_MIS_PRED_RETIRED INST_RETIRED DTLB_WALK	L1D Cache Miss Ratio L1D Cache MPKI Branch MPKI DTLB MPKI DTLB Walk Ratio

file_access_sum Recipe - Preview Feature#

This recipe provides comprehensive analysis of file access patterns and I/O performance statistics from Nsight Systems reports, supporting single or multiple profiled processes across different machines.

Overview

The file_access_sum recipe generates an interactive Jupyter notebook that analyzes POSIX VFS (Virtual File System) function calls captured during profiling sessions. This analysis helps identify I/O bottlenecks, optimization opportunities, and file access patterns that could impact application performance.

Key Capabilities

The recipe provides insights into:

- **File Access Patterns**: Breakdown of read-only, write-only, and read-write file access patterns.
- **Performance Metrics**: Total bytes transferred, operation counts, and average I/O sizes per operation.
- **Cross-Process Analysis**: File access patterns across multiple hosts, processes, and threads.
- **Temporal Analysis**: Distribution of CPU time by operation type.
- **Hotspot Identification**: Top files by read/write volume and operation frequency.
- **Performance Recommendations**: Automated detection of potentially inefficient I/O patterns with actionable suggestions.

Use Cases

The recipe is particularly valuable for identifying and addressing the following scenarios (but not limited to these):

1. **I/O Patterns**: Understanding application I/O behavior to uncover usage trends and inefficiencies.
2. **Small I/O Operations**: Detection of frequent small read/write operations that could benefit from batching.
3. **Caching Opportunities**: Identification of frequently accessed read-only files that are candidates for local caching.
4. **Metadata Contention**: Identifying cases where frequent metadata operations by one process may cause contention, impacting storage access for other processes.
5. **System File Noise**: Filtering out system files (/dev/, /sys/, etc.) to focus on application-relevant I/O.

Prerequisites

This recipe requires that Nsight Systems reports be collected with specific tracing parameters:

- `--trace=osrt` - Enables OS Runtime API tracing
- `--osrt-file-access=true` - Enables file access tracking
- **Optional:** To enable tracing of MPI rank information, use `--trace=mpi` along with either `--mpi-impl=openmpi` or `--mpi-impl=mpich`.

Usage

[1] Create a reports folder.
[2] Collect nsys-rep reports, using '--trace=osrt' and '--osrt-file-access=true' parameters, and save them to the reports folder.
[3] Run the recipe, using 'nsys recipe file_access_sum --input [reports folder path]'.

Output

As the main output, the recipe generates a Jupyter notebook `file_access_stats.ipynb` with the following sections:

- File Access Summary Table:

File Access Summary Table

This table provides a high-level overview of file access patterns across profiled processes, broken down by access type. It shows aggregated statistics for:

- Total number of files accessed.
- Total bytes read and written.
- Average read and write sizes per operation.

The data is categorized into four types:

1. All files - All files that were accessed during profiling.
2. Read-only files - Files that were only read from (no write operations).
3. Write-only files - Files that were only written to (no read operations).
4. Read-write files - Files that had both read and write operations.

NOTES:

- The statistics are aggregated across all processes and threads.

[6]:
```
display_table(file_access_summary_table, caption="File Access Summary")
```

File Access Summary

	Type	# Of Files Accessed	Total Read Bytes	Avg Read Bytes Per Op	Total Write Bytes	Avg Write Bytes Per Op
0	All files	2122	87.6M	17.9k	18.1k	3.0k
1	Read only files	2106	87.6M	17.9k	0.0	0.0
2	Write only files	4	0.0	0.0	15.7k	3.1k
3	Read and Write files	1	2.4k	1.2k	2.4k	2.4k

- Hottest Read/Write Files Tables:

Hottest Read Files Table

This table identifies the files with the highest total amount of bytes read. For each file, it shows:

- File path and name.
- Number of times the file was opened.
- Total bytes read from the file.
- Average size of read operations.
- Whether the file is read-only (no write operations).

NOTES:

- Files are sorted by total read bytes in descending order.
- This analysis helps identify the files with the highest total amount of bytes read.
- The "Read Only" column indicates if the file was accessed only for reading.
- Only files with read operations (Total Read Bytes > 0) are included.
- Files named "Unknown file #N" represent files that were opened before the Nsight Systems recording session started, so their names could not be resolved.

```
display_table(hottest_read_files, caption="Hottest Read Files")
```

Hottest Read Files

	File	# of times opened	Total Read Bytes	Avg Read Bytes Per Op	Read Only
0	../data/MNIST/raw/train-images-idx3-ubyte	4	188.2M	23.5M	True
1	../data/MNIST/raw/t10k-images-idx3-ubyte	4	31.4M	3.9M	True
2	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/torch/nn/functional.py	33	7.8M	7.6k	True
3	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/pkg_resources/_vendor/pyparsing.py	32	7.4M	7.7k	True
4	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/torch/_meta_registrations.py	29	5.9M	7.5k	True
5	/proc/cpuinfo	39	1.9M	1.7k	True

- All Files Table:

All Files Table

This table provides a detailed breakdown of file access patterns for each individual file in the system. For each file, it shows:

- File path and name.
- Number of times the file was opened.
- Total bytes read and written.
- Average size of read and write operations.
- Process and thread information:
  - Process ID (PID) - the Linux OS id of the process accessing the file, or "multiple" if more than one process accessed the file.
  - Thread ID (TID) - the Linux OS id of the thread accessing the file, or "multiple" if more than one thread accessed the file.
  - Number of processes that accessed the file.
  - Number of threads that accessed the file.

NOTES:

- Files named "Unknown file #N" represent files that were opened before the Nsight Systems recording session started, so their names could not be resolved.

```
all_files_table = all_files_df
display_interactive_table(all_files_table, filename_column='File', default_sortby_column='Total Read Bytes', descending=True, caption="All Files")
```

Sort by: Total Read Bytes

Descending

Search File: Filename (RegEx)

Page Size: 20

Page: 1

All Files

	File	# of times opened	Total Read Bytes	Avg Read Bytes Per Op	Total Write Bytes	Avg Write Bytes Per Op	Process ID	Thread ID	Num Of Processes	Num Of Threads
2	../data/MNIST/raw/train-images-idx3-ubyte	4	188.2M	23.5M	0.0	0.0	4120183	4120183	1	1
0	../data/MNIST/raw/t10k-images-idx3-ubyte	4	31.4M	3.9M	0.0	0.0	4120183	4120183	1	1
1314	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/torch/nn/functional.py	33	7.8M	7.6k	0.0	0.0	4120183	4120183	1	1
191	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/pkg_resources/_vendor/pyparsing.py	32	7.4M	7.7k	0.0	0.0	4120183	4120183	1	1
842	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/torch/_meta_registrations.py	29	5.9M	7.5k	0.0	0.0	4120183	4120183	1	1
1845	/proc/cpuinfo	39	1.9M	1.7k	0.0	0.0	4120183	4120183	1	1

- Read/Write Access Histogram:

Read / Write Access Histogram

The interactive histogram below visualizes the distribution of read and write access sizes for all files accessed during the profiling session.

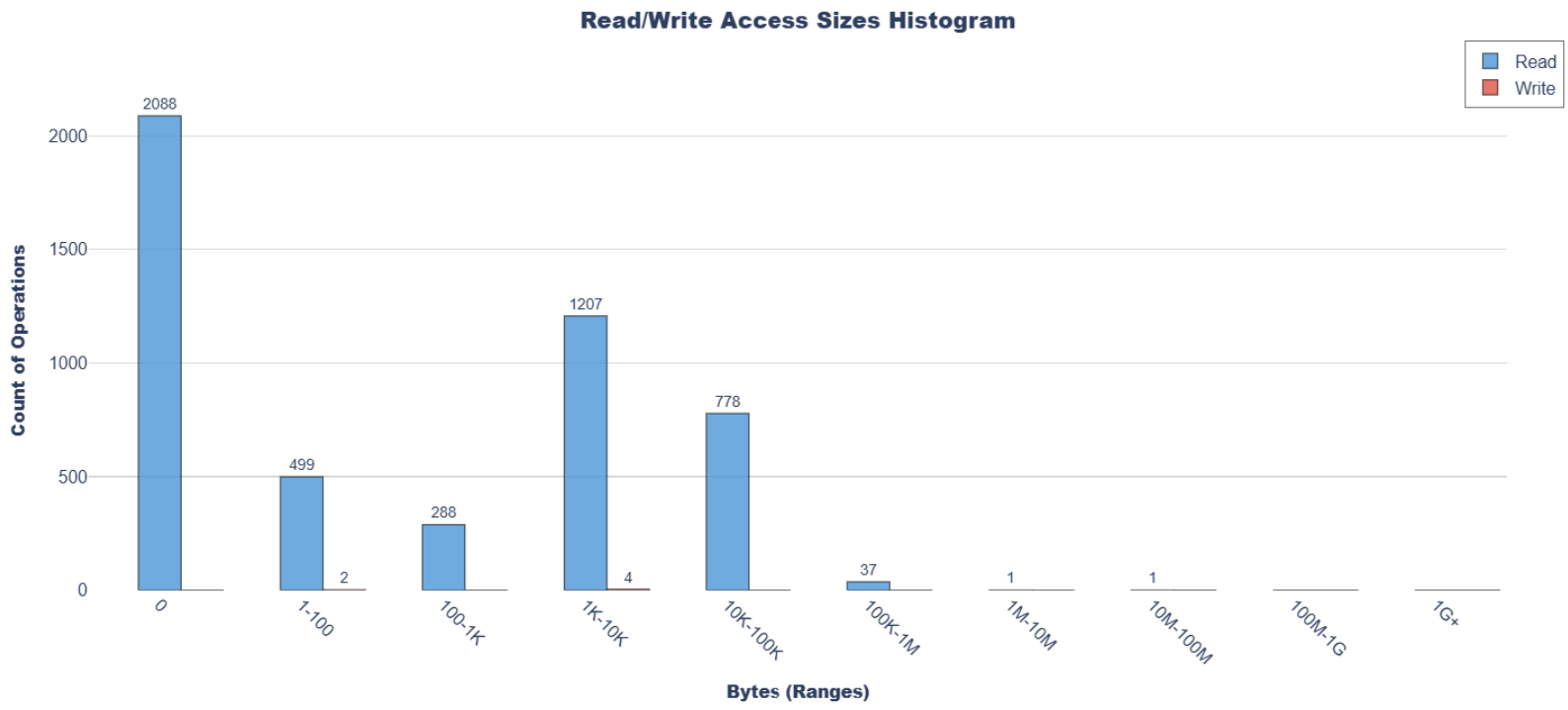
- Use the search box to filter files by name (supports regular expressions).
- The histogram helps identify typical and outlier access sizes, which can be useful for performance analysis and optimization.
- Both read and write operations are included. You can filter them by clicking on the legend.

NOTES:

- Filtering by filename can help focus on specific files of interest.
- The histogram is dynamically updated as you type in the search box.

```
display_interactive_histogram(file_events_filtered_df, plot_access_size_histogram)
```

Display Files: Filename (RegEx)



- CPU Time Graph:

CPU Time Graph

This graph shows the distribution of CPU time spent on different file access operations.

NOTES:

- The chart uses a stacked bar format to show the relative time distribution across different operation types.
- Duration is the total sum of durations of all function calls from that operation type, where each function call's duration is the time elapsed between the call to the function until it returns.

```
# Calculate duration and display chart
file_events_df_modified = file_events_filtered_df.assign(
    duration=lambda x: pd.to_datetime(x['endedAt']) - pd.to_datetime(x['startedAt'])
)

display_interactive_stacked_bar_chart(file_events_df_modified, 'eventType', 'duration', 'CPU Time Chart')
```

Source Report: All

Search Files: Type to search files...

File Name: All

Process ID: All

Thread ID: All

MPI Rank: All

Machine Name: All

- Operations Count Chart:

Operations Count Chart

This chart displays the amount of performed storage operations per category.

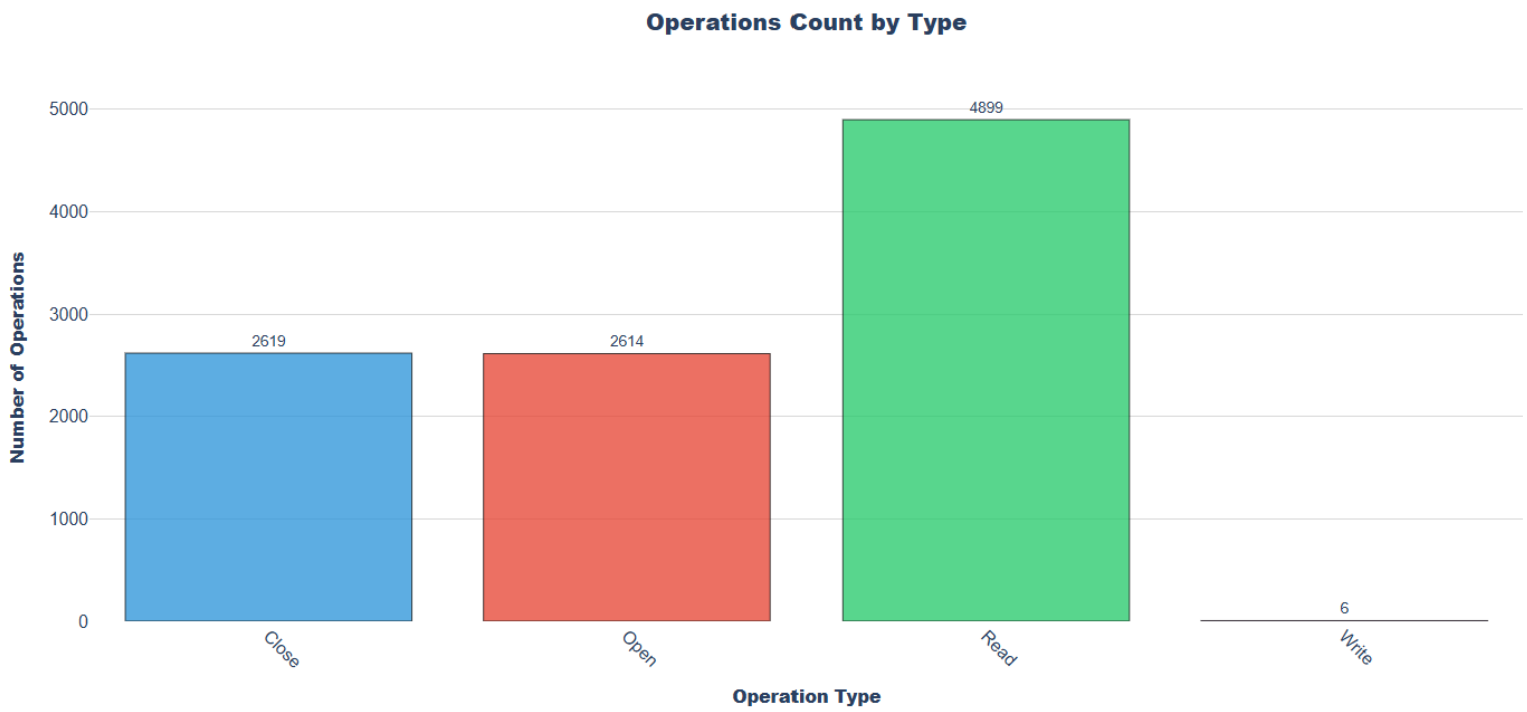
Use the search box to filter files by name (supports regular expressions). The chart helps visualize the distribution of storage operations, making it easy to identify which operations are most frequent and spot usage patterns across different file types.

NOTES:

Filtering by filename can help focus on specific files of interest. The chart is dynamically updated as you type in the search box.

```
display_interactive_histogram(file_events_filtered_df, plot_operations_count_chart)
```

Display Files: Filename (RegEx)



- Performance Analysis:

Performance Analysis Table

This section displays a list of patterns which may cause decrease in I/O performance.

Patterns of potential performance issues include:

- **REPEATED READ ONLY ACCESS**: The file has been accessed more than the set threshold, but has never been written to. This may indicate that the file could benefit from caching to reduce repeated read operations.
- **SMALL AVG READ SIZE**: The file's average read size per operation is less than the set threshold, and there were multiple read operations. Using a larger read chunk size may improve performance.
- **SMALL AVG WRITE SIZE**: The file's average write size per operation is less than the set threshold, and there were multiple write operations. Using a larger write chunk size may improve performance.

How to navigate this table:

- Use the filter options to focus on specific pattern or files of interest.
- Review the "Description" and "Suggested Action" columns for guidance on how to address each issue.

This table helps you quickly identify and prioritize file access inefficiencies that may impact your application's I/O performance.

```
performance_analysis_table = concatenated_performance_analysis_df
display_interactive_table(performance_analysis_table, filename_column='File', columns_to_filter_by=['Pattern'], caption="Performance Analysis")
```

Sort by: Descending

Search File: Filename (RegEx)

Page Size: 10 Page: 1

Filter By Columns:

Pattern: All

Performance Analysis

	Pattern	File	Description	Suggested Action
0	SMALL AVG READ SIZE	/homedir/rmilo/PyTorchExamples/examples/mnist/main.py	Average read size is 1702 bytes, and there were more than 10 read operations.	Consider using larger read operations.
1	SMALL AVG READ SIZE	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/distutils-precedence.pth	Average read size is 76 bytes, and there were more than 10 read operations.	Consider using larger read operations.
2	SMALL AVG READ SIZE	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/torch/cuda/random.py	Average read size is 1762 bytes, and there were more than 10 read operations.	Consider using larger read operations.
3	SMALL AVG READ SIZE	/homedir/rmilo/PyTorchExamples/pytorch_examples_env/lib/python3.10/site-packages/torch/xpu/random.py	Average read size is 1749 bytes, and there were more than 10 read operations.	Consider using larger read operations.
4	SMALL AVG READ SIZE	/proc/cpuinfo	Average read size is 1652 bytes, and there were more than 10 read operations.	Consider using larger read operations.
5	SMALL AVG READ SIZE	/sys/bus/pci/devices/0000:00:00.0/config	Average read size is 48 bytes, and there were more than 10 read operations.	Consider using larger read operations.

Recommended Workflow

- Start by setting file path ignore patterns to exclude system files from analysis.
- Focus on application-specific files during the analysis by using regex filtering.
- View the Nsight Systems report file alongside this analysis to gain a deeper understanding of the application's behavior.

Tutorial: Create a User-Defined Recipe#

The Nsight Systems recipe system is designed to be extensible and we hope that many users will use it to create their own recipes. This short tutorial will highlight the steps needed to create a recipe that is a customized version of

one of the recipes that is included in the Nsight Systems recipe package.

Step 1: Create the recipe directory and script

Make a new directory in the `<install-dir>/target-linux-x64/python/packages/nsys_recipe/recipes` folder based on the name of your new recipe. For this example, we will call our new recipe new_metric_util_map. We will copy the existing gpu_metric_util_map.py script and create a new script called new_metric_util_map.py in the new_metric_util_map directory. We will also copy the heatmap.ipynb and metadata json files into the new_metric_util_map directory. Type these steps in a Linux terminal window:

```
> cd <install-dir>/target-linux-x64/python/packages/nsys_recipe
> mkdir new_metric_util_map
> cp gpu_metric_util_map/metadata.json new_metric_util_map/metadata.json
> cp gpu_metric_util_map/heatmap.ipynb new_metric_util_map/heatmap.ipynb
> cp gpu_metric_util_map/gpu_metric_util_map.py new_metric_util_map/new_metric_util_map.py
```

Replace the module name in `metadata.json` with new_metric_util_map and update the display name and description to your preference. Also, rename the class name `GpuMetricUtilMap` in `new_metric_util_map.py` to `NewMetricUtilMap`. We will discuss the detailed functionality of the new recipe code in the subsequent steps.

Step 2: Modify the mapper function

Many recipes are structured as a map-reduce algorithm. The mapper function is called for every .nsys-rep file in the report directory. The mapper function performs a series of calculations on the events in each Nsight Systems report and produces an intermediate data set. The intermediate results are then combined by the reduce function to produce the final results. The mapper function can be called in parallel, either on multiple cores of a single node (using the concurrent python module), or multiple ranks of a multi-node recipe analysis (using the Dask distributed module).

When we create a new recipe, we need to create a class that derives from the Recipe base class. For our example, that class will be called NewMetricUtilMap (which we had renamed in step 1).

The mapper function is called mapper_func(). It will first convert the .nsys-rep file into a data storage file (SQLite/Parquet/Arrow), if the file does not already exist. It then reads all the necessary tables from the exported file into Pandas Dataframes needed by the recipe. GPU Metric data is stored using a database schema table called `GENERIC_EVENTS`. For extra flexibility, `GENERIC_EVENTS` represents the data as a JSON object, which is stored as a string. The `NewMetricUtilMap` class extracts fields from the JSON object and accumulates them over the histogram bins of the heat map.

The original script retrieved three GPU metrics: SM Active, SM Issue, and Tensor Active. In our new version of the script, we will extract a fourth metric, Unallocated Warps in Active SMs.

1. Find this line (approximately line 44):

   metric_cols = ["SMs Active", "SM Issue", "Tensor Active"]

2. Add the Unallocated Warps in Active SMs metric:

```
metric_cols = [
  "SMs Active",
  "SM Issue",
  "Tensor Active",
  "Unallocated Warps in Active SMs",
]
```

Step 3: Modify the reduce function

Our new mapper function will extract four GPU metrics and return them as a Pandas DataFrame. The reduce function receives a list of DataFrames, one for each .nsys-rep file in the analysis, and combines them into a single DataFrame using the Pandas concat function. Since the reducer function is generic in our case, no modifications are needed. However, if you would like to add any additional post-processing, you can do so in this function.

Step 4: Add a plot to the Jupyter notebook

Our new recipe class will create a Parquet output file with all the data produced by the reducer function, using the `to_parquet()` function. It will also create a Jupyter notebook file using the `create_notebook()` function.

In this step, we will change the `create_notebook()` function to produce a plot for our fourth metric. To do this, we need to change these two lines (located in the second cell of `new_metric_util_map/heatmap.ipynb`):

```
metrics = [
  "SMs Active",
  "SM Issue",
  "Tensor Active",
]
```

To this:

```
metrics = [
   "SMs Active",
   "SM Issue",
   "Tensor Active",
   "Unallocated Warps in Active SMs",
]
```

That completes all the modifications for our NewMetricUtilMap class.

Step 5: Run the new recipe

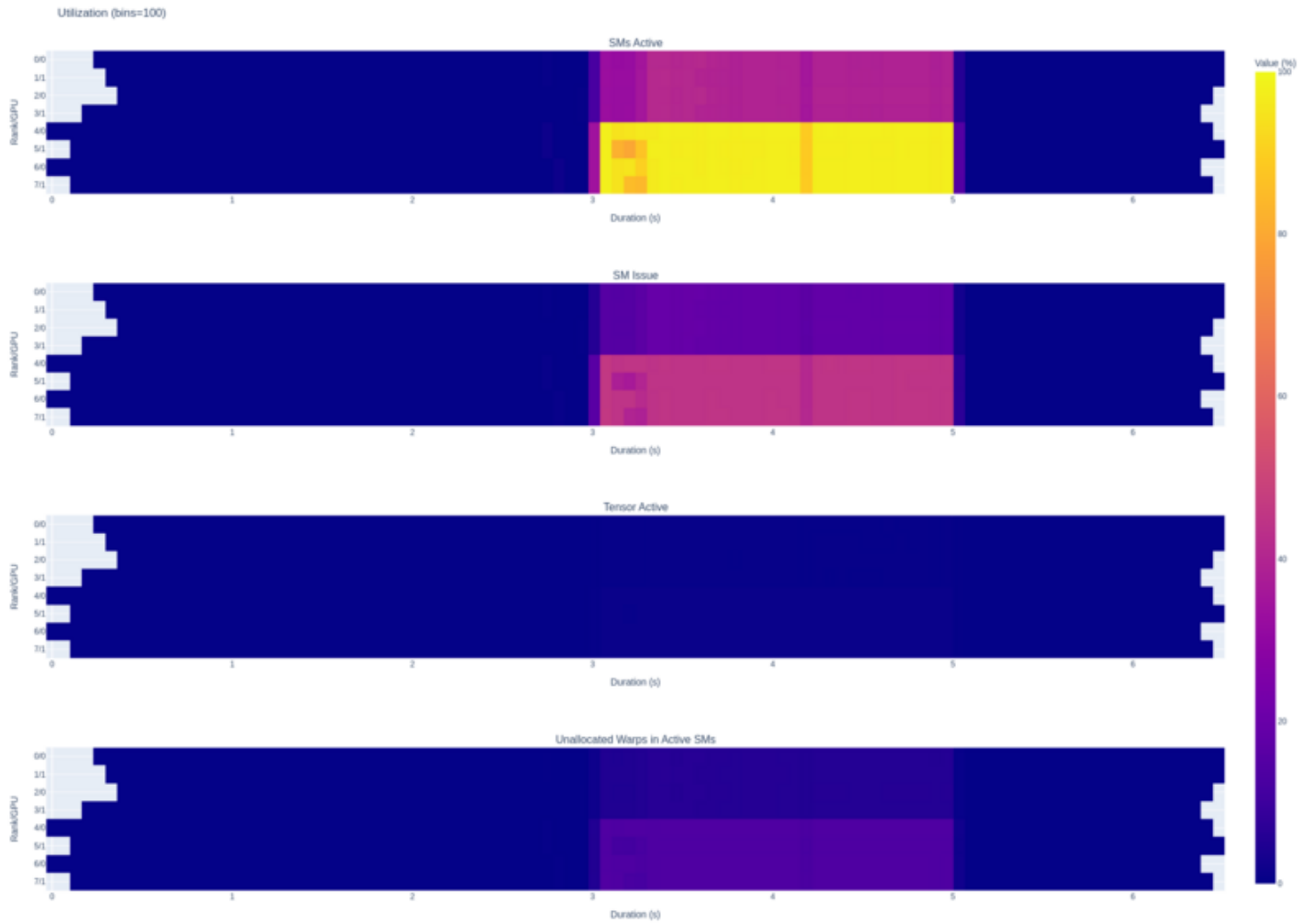
If the new recipe is located in the default recipe directory nsys_recipe/recipes, we can directly run it using the `nsys recipe` command like this:

```
> nsys recipe new_metric_util_map --input <directory of reports>
```

It is also possible to have a recipe located outside of this directory. In this case, you need to set the environment variable `NSYS_RECIPE_PATH` to the directory containing the recipe when running the `nsys recipe` command.

When successful, the recipe should produce a new recipe result directory called `new_metric_util_map-1`.

If we open the Jupyter notebook in that recipe and execute the code, we should see our new heatmap along with the three plots produced by the original version of the recipe. Here is an example:



Available Export Formats#

You may want to create your own analysis system interrogating the data produced by Nsight Systems. However, the .nsys-rep output format of Nsight Systems is not designed to be accessed by users, and we cannot guarantee its stability. Therefore you will want to export into a good format for your further processing.

SQLite Schema Reference#

Nsight Systems has the ability to export SQLite database files from the .nsys-rep results file. From the CLI, use `nsys export`. From the GUI, call `File->Export...`.

Note

The .nsys-rep report format is the only data format for Nsight Systems that should be considered forward-compatible. The SQLite schema can and will change in the future.

The schema for a concrete database can be obtained with the sqlite3 tool built-in command `.schema`. The sqlite3 tool can be located in the Target or Host directory of your Nsight Systems installation.

Note

Currently, tables are created lazily, and therefore not every table described in the documentation will be present in a particular database. This will change in a future version of the product. If you want a full schema of all possible tables, use `nsys export --lazy=false` during the export phase.

Currently, a table is created for each data type in the exported database. Since usage patterns for exported data may vary greatly and no default use cases have been established, no indexes or extra constraints are created. Instead, refer to the SQLite Examples section for a list of common recipes. This may change in a future version of the product.

To check the version of your exported SQLite file, check the value of `EXPORT_SCHEMA_VERSION` in the `META_DATA_EXPORT` table. The schema version is a common three-value major/minor/micro version number. The first value, or major value, indicates the overall format of the database, and is only changed if there is a major re-write or re-factor of the entire database format. It is assumed that if the major version changes, all scripts or queries will break. The middle, or minor, version is changed anytime there is a more localized, but potentially breaking change, such as renaming an existing column, or changing the type of an existing column. The last, or micro version is changed any time there are additions, such as a new table or column, that should not introduce any breaking change when used with well-written, best-practices queries.

The changes between schema versions are documented in `<install_dir>/host*/exporter/`

`export_schema_version_notes.txt`.

This is the schema as of the 2025.5 release, schema version 3.23.2.

```
CREATE TABLE StringIds (
  -- Consolidation of repetitive string values.

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- ID reference value.
  value             TEXT    NOT NULL              -- String value.
);
CREATE TABLE ANALYSIS_FILE (
  -- Analysis file content

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- ID reference value.
  filename          TEXT,                       -- File path
  contentId         INTEGER,                    -- REFERENCES StringIds(id) -- File content
  globalPid         INTEGER  NOT NULL           -- Serialized GlobalId.
);
CREATE TABLE ThreadNames (
  nameId            INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Thread name
  priority          INTEGER,                    -- Priority of the thread.
  globalTid         INTEGER                     -- Serialized GlobalId.
);
CREATE TABLE ProcessStreams (
  globalPid         INTEGER  NOT NULL,          -- Serialized GlobalId.
  filenameId        INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- File name
  contentId         INTEGER  NOT NULL           -- REFERENCES StringIds(id) -- Stream content
);
CREATE TABLE TARGET_INFO_SYSTEM_ENV (
  globalVid         INTEGER,                    -- Serialized GlobalId.
```

```
  devStateName        TEXT     NOT NULL,          -- Device state name.
  name                TEXT     NOT NULL,          -- Property name.
  nameEnum            INTEGER  NOT NULL,          -- Property enum value.
  value               TEXT     NOT NULL           -- Property value.
);
CREATE TABLE TARGET_INFO_NIC_INFO (
  GUID                INTEGER  NOT NULL,          -- Network interface GUID
  stateName           TEXT     NOT NULL,          -- Device state name
  nicId               INTEGER  NOT NULL,          -- Network interface Id
  name                TEXT     NOT NULL,          -- Network interface name
  deviceId            INTEGER  NOT NULL,          -- REFERENCES ENUM_NET_DEVICE_ID(id)
  vendorId            INTEGER  NOT NULL,          -- REFERENCES ENUM_NET_VENDOR_ID(id)
  linkLayer           INTEGER  NOT NULL           -- REFERENCES ENUM_NET_LINK_TYPE(id)
);
CREATE TABLE NIC_ID_MAP (
  -- Map between NIC info nicId and NIC metric globalId

  nicId               INTEGER  NOT NULL,          -- REFERENCES TARGET_INFO_NIC_INFO(nicId)
  globalId            INTEGER  NOT NULL           -- REFERENCES NET_NIC_METRIC(globalId)
);
CREATE TABLE TARGET_INFO_SESSION_START_TIME (
  utcEpochNs          INTEGER,                    -- UTC Epoch timestamp at start of the capture (ns).
  utcTime             TEXT,                       -- Start of the capture in UTC.
  localTime           TEXT                        -- Start of the capture in local time of target.
);
CREATE TABLE ANALYSIS_DETAILS (
  -- Details about the analysis session.
```

```
  globalVid          INTEGER  NOT NULL,          -- Serialized GlobalId.
  duration           INTEGER  NOT NULL,          -- The total time span of the entire trace (ns).
  startTime          INTEGER  NOT NULL,          -- Trace start timestamp in nanoseconds.
  stopTime           INTEGER  NOT NULL           -- Trace stop timestamp in nanoseconds.
);
CREATE TABLE PMU_EVENT_REQUESTS (
  -- PMU event requests

  id                 INTEGER  NOT NULL,          -- PMU event request.
  eventid            INTEGER,                    -- PMU counter event id.
  source             INTEGER  NOT NULL,          -- REFERENCES ENUM_PMU_EVENT_SOURCE(id)
  unit_type          INTEGER  NOT NULL,          -- REFERENCES ENUM_PMU_UNIT_TYPE(id)
  event_name         TEXT,                       -- PMU counter unique name

  PRIMARY KEY (id)
);
CREATE TABLE TARGET_INFO_GPU (
  vmId               INTEGER  NOT NULL,          -- Serialized GlobalId.
  id                 INTEGER  NOT NULL,          -- Device ID.
  name               TEXT,                       -- Device name.
  busLocation        TEXT,                       -- PCI bus location.
  isDiscrete         INTEGER,                    -- True if discrete, false if integrated.
  l2CacheSize        INTEGER,                    -- Size of L2 cache (B).
  totalMemory        INTEGER,                    -- Total amount of memory on the device (B).
  memoryBandwidth    INTEGER,                    -- Amount of memory transferred (B).
  clockRate          INTEGER,                    -- Clock frequency (Hz).
  smCount            INTEGER,                    -- Number of multiprocessors on the device.
  pwGpuId            INTEGER,                    -- PerfWorks GPU ID.
```

```
uuid                TEXT,                    -- Device UUID.
luid                INTEGER,                 -- Device LUID.
chipName            TEXT,                    -- Chip name.
cuDevice            INTEGER,                 -- CUDA device ID.
ctxswDevPath        TEXT,                    -- GPU context switch device node path.
ctrlDevPath         TEXT,                    -- GPU control device node path.
revision            INTEGER,                 -- Revision number.
nodeMask            INTEGER,                 -- Device node mask.
constantMemory      INTEGER,                 -- Memory available on device for __constant__ variables (B).
maxIPC              INTEGER,                 -- Maximum instructions per count.
maxRegistersPerBlock    INTEGER,             -- Maximum number of 32-bit registers available per block.
maxShmemPerBlock        INTEGER,             -- Maximum optin shared memory per block.
maxShmemPerBlockOptin   INTEGER,             -- Maximum optin shared memory per block.
maxShmemPerSm           INTEGER,             -- Maximum shared memory available per multiprocessor
(B).
maxRegistersPerSm       INTEGER,             -- Maximum number of 32-bit registers available per
multiprocessor.
threadsPerWarp          INTEGER,             -- Warp size in threads.
asyncEngines            INTEGER,             -- Number of asynchronous engines.
maxWarpsPerSm           INTEGER,             -- Maximum number of warps per multiprocessor.
maxBlocksPerSm          INTEGER,             -- Maximum number of blocks per multiprocessor.
maxThreadsPerBlock      INTEGER,             -- Maximum number of threads per block.
maxBlockDimX            INTEGER,             -- Maximum X-dimension of a block.
maxBlockDimY            INTEGER,             -- Maximum Y-dimension of a block.
maxBlockDimZ            INTEGER,             -- Maximum Z-dimension of a block.
maxGridDimX             INTEGER,             -- Maximum X-dimension of a grid.
maxGridDimY             INTEGER,             -- Maximum Y-dimension of a grid.
maxGridDimZ             INTEGER,             -- Maximum Z-dimension of a grid.
```

```
  computeMajor        INTEGER,                -- Major compute capability version number.
  computeMinor        INTEGER,                -- Minor compute capability version number.
  smMajor             INTEGER,                -- Major multiprocessor version number.
  smMinor             INTEGER                 -- Minor multiprocessor version number.
);
CREATE TABLE TARGET_INFO_XMC_SPEC (
  vmId                INTEGER  NOT NULL,      -- Serialized GlobalId.
  clientId            INTEGER  NOT NULL,      -- Client ID.
  type                TEXT     NOT NULL,      -- Client type.
  name                TEXT     NOT NULL,      -- Client name.
  groupId             TEXT     NOT NULL       -- Client group ID.
);
CREATE TABLE TARGET_INFO_CUDA_DEVICE (
  gpuId               INTEGER,                -- GPU ID.
  cudaId              INTEGER  NOT NULL,      -- CUDA device ID.
  pid                 INTEGER  NOT NULL       -- Process ID.
);
CREATE TABLE TARGET_INFO_PROCESS (
  processId           INTEGER  NOT NULL,      -- Process ID.
  openGlVersion       TEXT     NOT NULL,      -- OpenGL version.
  correlationId       INTEGER  NOT NULL,      -- Correlation ID of the kernel.
  nameId              INTEGER  NOT NULL       -- REFERENCES StringIds(id) -- Function name
);
CREATE TABLE TARGET_INFO_NVTX_CUDA_DEVICE (
  name                TEXT     NOT NULL,      -- CUDA device name assigned using NVTX.
  hwId                INTEGER  NOT NULL,      -- Hardware ID.
  vmId                INTEGER  NOT NULL,      -- VM ID.
  deviceId            INTEGER  NOT NULL       -- Device ID.
```

```
);
CREATE TABLE TARGET_INFO_NVTX_CUDA_CONTEXT (
  name            TEXT    NOT NULL,        -- CUDA context name assigned using NVTX.
  hwId            INTEGER NOT NULL,        -- Hardware ID.
  vmId            INTEGER NOT NULL,        -- VM ID.
  processId       INTEGER NOT NULL,        -- Process ID.
  deviceId        INTEGER NOT NULL,        -- Device ID.
  contextId       INTEGER NOT NULL         -- Context ID.
);
CREATE TABLE TARGET_INFO_NVTX_CUDA_STREAM (
  name            TEXT    NOT NULL,        -- CUDA stream name assigned using NVTX.
  hwId            INTEGER NOT NULL,        -- Hardware ID.
  vmId            INTEGER NOT NULL,        -- VM ID.
  processId       INTEGER NOT NULL,        -- Process ID.
  deviceId        INTEGER NOT NULL,        -- Device ID.
  contextId       INTEGER NOT NULL,        -- Context ID.
  streamId        INTEGER NOT NULL         -- Stream ID.
);
CREATE TABLE TARGET_INFO_CUDA_CONTEXT_INFO (
  nullStreamId    INTEGER NOT NULL,        -- Stream ID.
  hwId            INTEGER NOT NULL,        -- Hardware ID.
  vmId            INTEGER NOT NULL,        -- VM ID.
  processId       INTEGER NOT NULL,        -- Process ID.
  deviceId        INTEGER NOT NULL,        -- Device ID.
  contextId       INTEGER NOT NULL,        -- Context ID.
  parentContextId INTEGER,                 -- For green context, this is the parent context id.
  isGreenContext  INTEGER                  -- Is this a Green Context?
);
```

```
CREATE TABLE TARGET_INFO_CUDA_STREAM (
  streamId          INTEGER  NOT NULL,          -- Stream ID.
  hwId              INTEGER  NOT NULL,          -- Hardware ID.
  vmId              INTEGER  NOT NULL,          -- VM ID.
  processId         INTEGER  NOT NULL,          -- Process ID.
  contextId         INTEGER  NOT NULL,          -- Context ID.
  priority          INTEGER  NOT NULL,          -- Priority of the stream.
  flag              INTEGER  NOT NULL           -- REFERENCES ENUM_CUPTI_STREAM_TYPE(id)
);
CREATE TABLE TARGET_INFO_WDDM_CONTEXTS (
  context           INTEGER  NOT NULL,
  engineType        INTEGER  NOT NULL,
  nodeOrdinal       INTEGER  NOT NULL,
  friendlyName      TEXT     NOT NULL
);
CREATE TABLE TARGET_INFO_PERF_METRIC (
  id                INTEGER  NOT NULL,          -- Event or Metric ID value
  name              TEXT     NOT NULL,          -- Event or Metric name
  description       TEXT     NOT NULL,          -- Event or Metric description
  unit              TEXT     NOT NULL,          -- Event or Metric measurement unit
  nameSuffix        TEXT                        -- Event or Metric name suffix
);
CREATE TABLE TARGET_INFO_NETWORK_METRICS (
  metricsListId     INTEGER  NOT NULL,          -- Metric list ID
  metricsIdx        INTEGER  NOT NULL,          -- List index of metric
  name              TEXT     NOT NULL,          -- Name of metric
  description       TEXT     NOT NULL,          -- Description of metric
  unit              TEXT     NOT NULL           -- Measurement unit of metric
```

```
);
CREATE TABLE TARGET_INFO_COMPONENT (
  componentId          INTEGER  NOT NULL,          -- Component ID
  name                 TEXT     NOT NULL,          -- Component name
  instance             INTEGER,                    -- Component instance
  parentId             INTEGER                     -- Parent Component ID
);
CREATE TABLE NET_IB_DEVICE_INFO (
  networkId            INTEGER  NOT NULL,          -- The Device's Network ID
  guid                 INTEGER,                    -- Device Guid
  name                 TEXT,                       -- Device Name
  des                  TEXT,                       -- Device description
  lid                  INTEGER                     -- Device Lid
);
CREATE TABLE NET_IB_DEVICE_PORT_INFO (
  guid                 INTEGER,                    -- REFERENCES NET_IB_DEVICE_INFO(guid) -- Device Global
Identifier
  portNumber           INTEGER  NOT NULL,          -- Internal Port Number
  portLabel            TEXT     NOT NULL,          -- Port Label
  portLid              INTEGER  NOT NULL           -- Port Lid
);
CREATE TABLE NET_IB_DEVICE_TYPE_MAP (
  guid                 INTEGER,                    -- REFERENCES NET_IB_DEVICE_INFO(guid) -- Device Global
Identifier
  deviceType           INTEGER  NOT NULL           -- REFERENCES ENUM_NET_IB_DEVICE_TYPE(id)
);
CREATE TABLE META_DATA_CAPTURE (
  -- information about nsys capture parameters
```

```
  name                TEXT    NOT NULL,             -- Name of meta-data record
  value               TEXT                          -- Value of meta-data record
);
CREATE TABLE META_DATA_EXPORT (
  -- information about nsys export process

  name                TEXT    NOT NULL,             -- Name of meta-data record
  value               TEXT                          -- Value of meta-data record
);
CREATE TABLE ENUM_NSYS_EVENT_TYPE (
  -- Nsys event type labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name                TEXT,                           -- Enum symbol name.
  label               TEXT                            -- Enum human name.
);
CREATE TABLE ENUM_NSYS_EVENT_CLASS (
  -- Nsys event class labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name                TEXT,                           -- Enum symbol name.
  label               TEXT                            -- Enum human name.
);
CREATE TABLE ENUM_NSYS_GENERIC_EVENT_SOURCE (
  -- Nsys generic event source labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NSYS_GENERIC_EVENT_GROUP (
  -- Nsys generic event group labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NSYS_GENERIC_EVENT_FIELD_TYPE (
  -- Nsys generic event field type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NSYS_GENERIC_EVENT_FIELD_ETW_PROPERTY (
  -- Nsys generic event field ETW property flag labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NSYS_GENERIC_EVENT_FIELD_ETW_TYPE (
  -- Nsys generic event field ETW type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NSYS_GENERIC_EVENT_FIELD_ETW_FLAGS (
  -- Nsys generic event field ETW map info flag labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_GPU_CTX_SWITCH (
  -- GPU context switch labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_MEMCPY_OPER (
  -- CUDA memcpy operation labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_MEM_KIND (
  -- CUDA memory kind labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_MEMPOOL_TYPE (
  -- CUDA mempool type labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_MEMPOOL_OPER (
  -- CUDA mempool operation labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_DEV_MEM_EVENT_OPER (
  -- CUDA device mem event operation labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_KERNEL_LAUNCH_TYPE (
  -- CUDA kernel launch type labels

  id                INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_SHARED_MEM_LIMIT_CONFIG (
  -- CUDA shared memory limit config labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_UNIF_MEM_MIGRATION (
  -- CUDA unified memory migration cause labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_UNIF_MEM_ACCESS_TYPE (
  -- CUDA unified memory access type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUDA_FUNC_CACHE_CONFIG (
  -- CUDA function cache config labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUPTI_STREAM_TYPE (
  -- CUPTI stream type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_CUPTI_SYNC_TYPE (
  -- CUPTI synchronization type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_STACK_UNWIND_METHOD (
  -- Stack unwind method labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_SAMPLING_THREAD_STATE (
  -- Sampling thread state labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_SCHEDULING_THREAD_BLOCK (
  -- Scheduling thread block labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_OPENGL_DEBUG_SOURCE (
  -- OpenGL debug source labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_OPENGL_DEBUG_TYPE (
  -- OpenGL debug type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_OPENGL_DEBUG_SEVERITY (
  -- OpenGL debug severity labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_OSRT_FILE_ACCESS_EVENT_TYPE (
  -- OSRT File Access event type

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_VULKAN_PIPELINE_CREATION_FLAGS (
  -- Vulkan pipeline creation feedback flag labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_D3D12_HEAP_TYPE (
  -- D3D12 heap type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_D3D12_PAGE_PROPERTY (
  -- D3D12 CPU page property labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_D3D12_HEAP_FLAGS (
  -- D3D12 heap flag labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_D3D12_CMD_LIST_TYPE (
  -- D3D12 command list type labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENACC_DEVICE (
  -- OpenACC device type labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENACC_EVENT_KIND (
  -- OpenACC event type labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
```

```
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_EVENT_KIND (
  -- OpenMP event kind labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_THREAD (
  -- OpenMP thread labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_DISPATCH (
  -- OpenMP dispatch labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_SYNC_REGION (
  -- OpenMP sync region labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
```

```
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_WORK (
  -- OpenMP work labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_MUTEX (
  -- OpenMP mutex labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_TASK_FLAG (
  -- OpenMP task flags labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_OPENMP_TASK_STATUS (
  -- OpenMP task status labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_DXGI_FORMAT (
  -- DXGI image format labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NVDRIVER_EVENT_ID (
  -- NV-Driver event it labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_WDDM_PAGING_QUEUE_TYPE (
  -- WDDM paging queue type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_WDDM_PACKET_TYPE (
  -- WDDM packet type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_WDDM_ENGINE_TYPE (
  -- WDDM engine type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_WDDM_INTERRUPT_TYPE (
  -- WDDM DMA interrupt type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_WDDM_VIDMM_OP_TYPE (
  -- WDDM VidMm operation type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NET_LINK_TYPE (
  -- NIC link layer labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_NET_DEVICE_ID (
  -- NIC PCIe device id labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_NET_VENDOR_ID (
  -- NIC PCIe vendor id labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_ETW_MEMORY_TRANSFER_TYPE (
  -- memory transfer type labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_PMU_EVENT_SOURCE (
  -- PMU event source labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
```

```
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_PMU_UNIT_TYPE (
  -- PMU unit type labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_VIDEO_ENGINE_TYPE (
  -- Video engine type id labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_VIDEO_ENGINE_CODEC (
  -- Video engine codec labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
  name          TEXT,              -- Enum symbol name.
  label         TEXT               -- Enum human name.
);
CREATE TABLE ENUM_DIAGNOSTIC_SEVERITY_LEVEL (
  -- Diagnostic message severity level labels

  id            INTEGER  NOT NULL  PRIMARY KEY,   -- Enum numerical value.
```

```
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_DIAGNOSTIC_SOURCE_TYPE (
  -- Diagnostic message source type labels

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_DIAGNOSTIC_TIMESTAMP_SOURCE (
  -- Diagnostic message timestamp source lables

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NET_IB_DEVICE_TYPE (
  -- network device types

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
  name              TEXT,                   -- Enum symbol name.
  label             TEXT                    -- Enum human name.
);
CREATE TABLE ENUM_NET_IB_CONGESTION_EVENT_TYPE (
  -- IB Switch congestion event types

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- Enum numerical value.
```

```
  name                TEXT,                    -- Enum symbol name.
  label               TEXT                     -- Enum human name.
);
CREATE TABLE GENERIC_EVENT_SOURCES (
  -- Generic event source modules

  sourceId            INTEGER  NOT NULL  PRIMARY KEY,    -- Serialized GlobalId.
  nameId              INTEGER  NOT NULL,                 -- REFERENCES StringIds(id) -- Event source name
  timeSourceId        INTEGER  NOT NULL,                 -- REFERENCES
ENUM_NSYS_GENERIC_EVENT_SOURCE(id)
  sourceGroupId       INTEGER  NOT NULL,                 -- REFERENCES
ENUM_NSYS_GENERIC_EVENT_GROUP(id)
  hyperType           TEXT,                    -- Hypervisor Type
  hyperVersion        TEXT,                    -- Hypervisor Version
  hyperStructPrefix   TEXT,                    -- Hypervisor Struct Prefix
  hyperMacroPrefix    TEXT,                    -- Hypervisor Macro Prefix
  hyperFilterFlags    INTEGER,                 -- Hypervisor Custom Filter Flags
  hyperDomain         TEXT,                    -- Hypervisor Domain
  data                TEXT                     -- JSON encoded generic event source description.
);
CREATE TABLE GENERIC_EVENT_TYPES (
  -- Generic event type/schema descriptions.

  typeId              INTEGER  NOT NULL  PRIMARY KEY,    -- Serialized GlobalId.
  sourceId            INTEGER  NOT NULL,                 -- REFERENCES GENERIC_EVENT_SOURCES(sourceId)
  nameId              INTEGER  NOT NULL,                 -- REFERENCES StringIds(id) -- Event type name
  hyperComment        TEXT,                    -- Event Type Hypervisor Comment
  ftraceFormat        TEXT,                    -- Event Type FTrace Format
```

```
  etwProviderId          INTEGER,                    -- Event Type ETW Provider Id
  etwProviderNameId        INTEGER,                    -- Event Type ETW Provider Name Id
  etwTaskId              INTEGER,                    -- Event Type ETW Task Id
  etwTaskNameId          INTEGER,                    -- Event Type ETW Task Name Id
  etwEventId             INTEGER,                    -- Event Type ETW Event Id
  etwVersion             INTEGER,                    -- Event Type ETW Version
  etwGuidHigh            INTEGER,                    -- Event Type ETW GUID high
  etwGuidLow             INTEGER,                    -- Event Type ETW GUID low
  etwGuid                TEXT,                       -- ETW Provider GUID.
  data                   TEXT                        -- JSON encoded generic event type description.
);
CREATE TABLE GENERIC_EVENT_TYPE_FIELDS (
  -- Generic event type/schema individual data field descriptions.

  typeId                 INTEGER  NOT NULL,          -- Serialized GlobalId.
  fieldIdx               INTEGER  NOT NULL,          -- Index of type field
  fieldNameId            INTEGER  NOT NULL,          -- Name of field.
  offset                 INTEGER  NOT NULL,          -- Field alignment offset size, in bytes.
  size                   INTEGER  NOT NULL,          -- Field size, in bytes.
  type                   INTEGER  NOT NULL,          -- REFERENCES
ENUM_NSYS_GENERIC_EVENT_FIELD_TYPE(id)
  hyperTypeName          TEXT,                       -- Event Field Hypervisor Type Name
  hyperFormat            TEXT,                       -- Event Field Hypervisor Format
  hyperComment           TEXT,                       -- Event Field Hypervisor Comment
  ftracePrefix           TEXT,                       -- Event Field FTrace Prefix
  ftraceSuffix           TEXT,                       -- Event Field FTrace Suffix
  etwFlags               INTEGER,                    -- REFERENCES
ENUM_NSYS_GENERIC_EVENT_FIELD_ETW_PROPERTY(id)
```

```
  etwCountFieldIndex       INTEGER,                    -- Event Field ETW Count Field Index
  etwLengthFieldIndex      INTEGER,                    -- Event Field ETW Length Field Index
  etwType                  INTEGER,                    -- REFERENCES
ENUM_NSYS_GENERIC_EVENT_FIELD_ETW_TYPE(id)
  etwMapInfoFlags          INTEGER,                    -- REFERENCES
ENUM_NSYS_GENERIC_EVENT_FIELD_ETW_FLAGS(id)
  etwOrderedFieldIndex     INTEGER                     -- Event Field ETW Ordered Field Index
);
CREATE TABLE GENERIC_EVENT_TYPE_FIELD_MAP (
  -- Generic event ENUM data.  Mostly used by ETW.

  typeId                   INTEGER  NOT NULL,          -- Serialized GlobalId.
  fieldIdx                 INTEGER  NOT NULL,          -- Index of type field
  enum                     INTEGER  NOT NULL,          -- Event Field ETW Map Info enum.
  name                     TEXT     NOT NULL,          -- Event Field ETW Map Info Name.
  nameId                   INTEGER  NOT NULL           -- Event Field ETW Map Info Name Id.
);
CREATE TABLE GENERIC_EVENTS (
  -- Dynamic or unstructured event data.

  genericEventId           INTEGER  NOT NULL  PRIMARY KEY,  -- Id of particular generic event
  rawTimestamp             INTEGER  NOT NULL,          -- Raw event timestamp recorded during profiling.
  timestamp                INTEGER  NOT NULL,          -- Event timestamp converted to the profiling session
timeline.
  typeId                   INTEGER  NOT NULL,          -- REFERENCES GENERIC_EVENT_TYPES(typeId)
  globalTid                INTEGER,                    -- Serialized GlobalId.
  data                     TEXT                        -- JSON encoded event data.
);
```

```
CREATE TABLE GENERIC_EVENT_DATA (
  -- GENERIC_EVENTS data values.

  genericEventId        INTEGER  NOT NULL,          -- REFERENCES GENERIC_EVENTS(genericEventId)
  fieldIdx              INTEGER  NOT NULL,          -- Index of type field
  intVal                INTEGER,                    -- Integer value, signed
  uintVal               INTEGER,                    -- Integer value, unsigned
  floatVal              REAL,                       -- Floating point value, 32-bit
  doubleVal             REAL                        -- Floating point value, 64-bit
);
CREATE TABLE ETW_PROVIDERS (
  -- Names and identifiers of ETW providers captured in the report.

  providerId            INTEGER  NOT NULL  PRIMARY KEY,   -- Provider ID.
  providerNameId        INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Provider name
  guid                  TEXT     NOT NULL           -- ETW Provider GUID.
);
CREATE TABLE ETW_TASKS (
  -- Names and identifiers of ETW tasks captured in the report.

  taskNameId            INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Task name
  taskId                INTEGER  NOT NULL,          -- The event task ID.
  providerId            INTEGER  NOT NULL           -- Provider ID.
);
CREATE TABLE ETW_EVENTS (
  -- Raw ETW events captured in the report.

  rawTimestamp          INTEGER  NOT NULL,          -- Raw event timestamp recorded during profiling.
```

```
  timestamp          INTEGER  NOT NULL,          -- Event start timestamp (ns).
  typeId             INTEGER  NOT NULL,          -- REFERENCES GENERIC_EVENT_TYPES(typeId)
  globalTid          INTEGER,                    -- Serialized GlobalId.
  opcode             INTEGER,                    -- The event opcode.
  data               TEXT     NOT NULL           -- JSON encoded event data.
);
CREATE TABLE TARGET_INFO_GPU_METRICS (
  -- GPU Metrics, metric names and ids.

  typeId             INTEGER  NOT NULL,          -- REFERENCES GENERIC_EVENT_TYPES(typeId)
  sourceId           INTEGER  NOT NULL,          -- REFERENCES GENERIC_EVENT_SOURCES(sourceId)
  typeName           TEXT     NOT NULL,          -- Name of event type.
  metricId           INTEGER  NOT NULL,          -- Id of metric in event; not assumed to be stable.
  metricName         TEXT     NOT NULL           -- Definitive name of metric.
);
CREATE TABLE GPU_METRICS (
  -- GPU Metrics, events and values.

  rawTimestamp       INTEGER  NOT NULL,          -- Raw event timestamp recorded during profiling.
  timestamp          INTEGER  NOT NULL,          -- Event timestamp (ns).
  typeId             INTEGER  NOT NULL,          -- REFERENCES TARGET_INFO_GPU_METRICS(typeId)
and GENERIC_EVENT_TYPES(typeId)
  metricId           INTEGER  NOT NULL,          -- REFERENCES TARGET_INFO_GPU_METRICS(metricId)
  value              INTEGER  NOT NULL           -- Counter data value
);
CREATE TABLE TARGET_INFO_SOC_METRICS (
  -- SoC Metrics, metric names and ids.
```

```
  typeId          INTEGER  NOT NULL,        -- REFERENCES GENERIC_EVENT_TYPES(typeId)
  sourceId        INTEGER  NOT NULL,        -- REFERENCES GENERIC_EVENT_SOURCES(sourceId)
  typeName        TEXT     NOT NULL,        -- Name of event type.
  metricId        INTEGER  NOT NULL,        -- Id of metric in event; not assumed to be stable.
  metricName      TEXT     NOT NULL         -- Definitive name of metric.
);
CREATE TABLE SOC_METRICS (
  -- SoC Metrics, events and values.

  rawTimestamp    INTEGER  NOT NULL,        -- Raw event timestamp recorded during profiling.
  timestamp       INTEGER  NOT NULL,        -- Event timestamp (ns).
  typeId          INTEGER  NOT NULL,        -- REFERENCES TARGET_INFO_SOC_METRICS(typeId)
and GENERIC_EVENT_TYPES(typeId)
  metricId        INTEGER  NOT NULL,        -- REFERENCES TARGET_INFO_GPU_METRICS(metricId)
  value           INTEGER  NOT NULL         -- Counter data value
);
CREATE TABLE MPI_COMMUNICATORS (
  -- Identification of MPI communication groups.

  rank            INTEGER,                  -- Active MPI rank
  timestamp       INTEGER,                  -- Time of MPI communicator creation.
  commHandle      INTEGER,                  -- MPI communicator handle.
  parentHandle    INTEGER,                  -- MPI communicator handle.
  localRank       INTEGER,                  -- Local MPI rank in a communicator.
  size            INTEGER,                  -- MPI communicator size.
  groupRoot       INTEGER,                  -- Root rank (global) in MPI communicator.
  groupRootUid    INTEGER,                  -- Group root's communicator ID.
  members         TEXT                      -- MPI communicator members (index is global rank).
```

```
);
CREATE TABLE NVTX_PAYLOAD_SCHEMAS (
  -- NVTX payload schema attributes.

  domainId            INTEGER,                -- User-controlled ID that can be used to group events.
  schemaId            INTEGER,                -- Identifier of the payload schema.
  name                TEXT,                   -- Schema name.
  type                INTEGER,                -- Schema type.
  flags               INTEGER,                -- Schema flags.
  numEntries          INTEGER,                -- Number of payload schema entries.
  payloadSize         INTEGER,                -- Size of the static payload.
  alignTo             INTEGER                 -- Field alignment in bytes.
);
CREATE TABLE NVTX_PAYLOAD_SCHEMA_ENTRIES (
  -- NVTX payload schema entries.

  domainId            INTEGER  NOT NULL,      -- User-controlled ID that can be used to group events.
  schemaId            INTEGER  NOT NULL,      -- Identifier of the payload schema.
  idx                 INTEGER  NOT NULL,      -- Index of the entry in the payload schema.
  flags               INTEGER,                -- Payload entry flags.
  type                INTEGER,                -- Payload entry type.
  name                TEXT,                   -- Label of the payload entry.
  description         TEXT,                   -- Description of the payload entry.
  arrayOrUnionDetail  INTEGER,                -- Array length (index) or selected union member.
  offset              INTEGER                 -- Entry offset in the binary data in bytes.
);
CREATE TABLE NVTX_PAYLOAD_ENUMS (
  -- NVTX payload enum attributes.
```

```
  domainId          INTEGER,              -- User-controlled ID that can be used to group events.
  schemaId          INTEGER,              -- Identifier of the payload schema.
  name              TEXT,                 -- Schema name.
  numEntries        INTEGER,              -- Number of entries in the enum.
  size              INTEGER               -- Size of enumeration type in bytes.
);
CREATE TABLE NVTX_PAYLOAD_ENUM_ENTRIES (
  -- NVTX payload enum entries.

  domainId          INTEGER  NOT NULL,    -- User-controlled ID that can be used to group events.
  schemaId          INTEGER  NOT NULL,    -- Identifier of the payload schema.
  idx               INTEGER  NOT NULL,    -- Index of the entry in the payload schema.
  name              TEXT,                 -- Name of the enum value.
  value             INTEGER,              -- Value of the enum entry.
  isFlag            INTEGER               -- Indicates that the entry sets a specific set of bits, which can be used to
define bitsets.
);
CREATE TABLE NVTX_SCOPES (
  -- NVTX scopes.

  domainId          INTEGER,              -- User-controlled ID that can be used to group events.
  scopeId           INTEGER,              -- Scope ID.
  parentScopeId     INTEGER,              -- Parent scope ID.
  path              TEXT                  -- Scope path.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_MEMCPY (
  start             INTEGER  NOT NULL,    -- Event start timestamp (ns).
```

```
  end                 INTEGER   NOT NULL,          -- Event end timestamp (ns).
  deviceId            INTEGER   NOT NULL,          -- Device ID.
  contextId           INTEGER   NOT NULL,          -- Context ID.
  greenContextId      INTEGER,                     -- Green context ID.
  streamId            INTEGER   NOT NULL,          -- Stream ID.
  correlationId       INTEGER,                     -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  globalPid           INTEGER,                     -- Serialized GlobalId.
  bytes               INTEGER   NOT NULL,          -- Number of bytes transferred (B).
  copyKind            INTEGER   NOT NULL,          -- REFERENCES ENUM_CUDA_MEMCPY_OPER(id)
  deprecatedSrcId     INTEGER,                     -- Deprecated, use srcDeviceId instead.
  srcKind             INTEGER,                     -- REFERENCES ENUM_CUDA_MEM_KIND(id)
  dstKind             INTEGER,                     -- REFERENCES ENUM_CUDA_MEM_KIND(id)
  srcDeviceId         INTEGER,                     -- Source device ID.
  srcContextId        INTEGER,                     -- Source context ID.
  dstDeviceId         INTEGER,                     -- Destination device ID.
  dstContextId        INTEGER,                     -- Destination context ID.
  migrationCause      INTEGER,                     -- REFERENCES ENUM_CUDA_UNIF_MEM_MIGRATION(id)
  graphNodeId         INTEGER,                     -- REFERENCES
CUDA_GRAPH_NODE_EVENTS(graphNodeId)
  virtualAddress      INTEGER,                     -- Virtual base address of the page/s being transferred.
  copyCount           INTEGER                      -- The total number of memcopy operations traced in this record. In
CUDA MemcpyBatchAsync APIs, multiple memcpy operations may be batched together for optimization purposes
based on certain heuristics.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_MEMSET (
  start               INTEGER   NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER   NOT NULL,          -- Event end timestamp (ns).
```

```
  deviceId            INTEGER  NOT NULL,          -- Device ID.
  contextId           INTEGER  NOT NULL,          -- Context ID.
  greenContextId        INTEGER,                  -- Green context ID.
  streamId            INTEGER  NOT NULL,          -- Stream ID.
  correlationId         INTEGER,                  -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  globalPid           INTEGER,                    -- Serialized GlobalId.
  value               INTEGER  NOT NULL,          -- Value assigned to memory.
  bytes               INTEGER  NOT NULL,          -- Number of bytes set (B).
  graphNodeId           INTEGER,                  -- REFERENCES
CUDA_GRAPH_NODE_EVENTS(graphNodeId)
  memKind               INTEGER                   -- REFERENCES ENUM_CUDA_MEM_KIND(id)
);
CREATE TABLE CUPTI_ACTIVITY_KIND_KERNEL (
  start               INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,          -- Event end timestamp (ns).
  deviceId            INTEGER  NOT NULL,          -- Device ID.
  contextId           INTEGER  NOT NULL,          -- Context ID.
  greenContextId        INTEGER,                  -- Green context ID.
  streamId            INTEGER  NOT NULL,          -- Stream ID.
  correlationId         INTEGER,                  -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  globalPid           INTEGER,                    -- Serialized GlobalId.
  demangledName         INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Kernel function name
w/ templates
  shortName           INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Base kernel function
name
  mangledName           INTEGER,                  -- REFERENCES StringIds(id) -- Raw C++ mangled kernel
```

```
function name
  launchType            INTEGER,                    -- REFERENCES ENUM_CUDA_KERNEL_LAUNCH_TYPE(id)
  cacheConfig           INTEGER,                    -- REFERENCES ENUM_CUDA_FUNC_CACHE_CONFIG(id)
  registersPerThread    INTEGER  NOT NULL,          -- Number of registers required for each thread
executing the kernel.
  gridX                 INTEGER  NOT NULL,          -- X-dimension grid size.
  gridY                 INTEGER  NOT NULL,          -- Y-dimension grid size.
  gridZ                 INTEGER  NOT NULL,          -- Z-dimension grid size.
  blockX                INTEGER  NOT NULL,          -- X-dimension block size.
  blockY                INTEGER  NOT NULL,          -- Y-dimension block size.
  blockZ                INTEGER  NOT NULL,          -- Z-dimension block size.
  staticSharedMemory    INTEGER  NOT NULL,          -- Static shared memory allocated for the kernel (B).
  dynamicSharedMemory   INTEGER  NOT NULL,          -- Dynamic shared memory reserved for the kernel
(B).
  localMemoryPerThread  INTEGER  NOT NULL,          -- Amount of local memory reserved for each thread
(B).
  localMemoryTotal      INTEGER  NOT NULL,          -- Total amount of local memory reserved for the kernel
(B).
  gridId                INTEGER  NOT NULL,          -- Unique grid ID of the kernel assigned at runtime.
  sharedMemoryExecuted  INTEGER,                    -- Shared memory size set by the driver.
  graphNodeId           INTEGER,                    -- REFERENCES
CUDA_GRAPH_NODE_EVENTS(graphNodeId)
  sharedMemoryLimitConfig  INTEGER                  -- REFERENCES
ENUM_CUDA_SHARED_MEM_LIMIT_CONFIG(id)
);
CREATE TABLE CUPTI_ACTIVITY_KIND_SYNCHRONIZATION (
  start                 INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,          -- Event end timestamp (ns).
```

```
  deviceId              INTEGER   NOT NULL,              -- Device ID.
  contextId             INTEGER   NOT NULL,              -- Context ID.
  greenContextId          INTEGER,                     -- Green context ID.
  streamId              INTEGER   NOT NULL,              -- Stream ID.
  correlationId           INTEGER,                     -- Correlation ID of the synchronization API to which this result is
associated.
  globalPid             INTEGER,                     -- Serialized GlobalId.
  deprecatedSyncType        INTEGER,                     -- Deprecated, use syncType instead. For older report,
REFERENCES ENUM_CUPTI_SYNC_TYPE(id)
  syncType              INTEGER   NOT NULL,              -- REFERENCES ENUM_CUPTI_SYNC_TYPE(id)
  eventId               INTEGER   NOT NULL,              -- Event ID for which the synchronization API is called.
  eventSyncId             INTEGER                      -- CUDA Event Sync ID to link the synchronization API to
associated event record API.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_CUDA_EVENT (
  timestamp             INTEGER   NOT NULL,              -- Event timestamp (ns).
  deviceId              INTEGER   NOT NULL,              -- Device ID.
  contextId             INTEGER   NOT NULL,              -- Context ID.
  greenContextId          INTEGER,                     -- Green context ID.
  streamId              INTEGER   NOT NULL,              -- Stream ID.
  correlationId           INTEGER,                     -- Correlation ID of the event record API to which this result is
associated.
  globalPid             INTEGER,                     -- Serialized GlobalId.
  eventId               INTEGER   NOT NULL,              -- Event ID for which the event record API is called.
  eventSyncId             INTEGER                      -- CUDA Event Sync ID to link event record API to related
synchronization APIs.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_GRAPH_TRACE (
```

```
  start              INTEGER   NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER   NOT NULL,          -- Event end timestamp (ns).
  deviceId           INTEGER   NOT NULL,          -- Device ID.
  contextId          INTEGER   NOT NULL,          -- Context ID.
  greenContextId     INTEGER,                     -- Green context ID.
  streamId           INTEGER   NOT NULL,          -- Stream ID.
  correlationId      INTEGER,                     -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  globalPid          INTEGER,                     -- Serialized GlobalId.
  graphId            INTEGER   NOT NULL,          -- REFERENCES CUDA_GRAPH_EVENTS(graphId)
  graphExecId        INTEGER   NOT NULL           -- REFERENCES
CUDA_GRAPH_EVENTS(graphExecId)
);
CREATE TABLE CUPTI_ACTIVITY_KIND_RUNTIME (
  start              INTEGER   NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER   NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER   NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                     -- Serialized GlobalId.
  correlationId      INTEGER,                     -- ID used to identify events that this function call has triggered.
  nameId             INTEGER   NOT NULL,          -- REFERENCES StringIds(id) -- Function name
  returnValue        INTEGER   NOT NULL,          -- Return value of the function call.
  callchainId        INTEGER                      -- REFERENCES CUDA_CALLCHAINS(id)
);
CREATE TABLE CUDNN_EVENTS (
  start              INTEGER   NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER   NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER   NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                     -- Serialized GlobalId.
```

```
  nameId              INTEGER   NOT NULL              -- REFERENCES StringIds(id) -- Function name
);
CREATE TABLE CUBLAS_EVENTS (
  start               INTEGER   NOT NULL,             -- Event start timestamp (ns).
  end                 INTEGER   NOT NULL,             -- Event end timestamp (ns).
  eventClass          INTEGER   NOT NULL,             -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                        -- Serialized GlobalId.
  nameId              INTEGER   NOT NULL              -- REFERENCES StringIds(id) -- Function name
);
CREATE TABLE CUDA_GRAPH_NODE_EVENTS (
  start               INTEGER   NOT NULL,             -- Event start timestamp (ns).
  end                 INTEGER   NOT NULL,             -- Event end timestamp (ns).
  eventClass          INTEGER   NOT NULL,             -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                        -- Serialized GlobalId.
  nameId              INTEGER   NOT NULL,             -- REFERENCES StringIds(id) -- Function name
  graphNodeId         INTEGER   NOT NULL,             -- Graph node ID.
  originalGraphNodeId INTEGER                         -- Reference to the original graph node ID, if cloned node.
);
CREATE TABLE CUDA_GRAPH_EVENTS (
  start               INTEGER   NOT NULL,             -- Event start timestamp (ns).
  end                 INTEGER   NOT NULL,             -- Event end timestamp (ns).
  eventClass          INTEGER   NOT NULL,             -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                        -- Serialized GlobalId.
  nameId              INTEGER   NOT NULL,             -- REFERENCES StringIds(id) -- Function name
  graphId             INTEGER,                        -- Graph ID.
  originalGraphId     INTEGER,                        -- Reference to the original graph ID, if cloned.
  graphExecId         INTEGER                         -- Executable graph ID.
);
```

```
CREATE TABLE CUPTI_ACTIVITY_KIND_BLOCK_TRACE (
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  deviceId           INTEGER,                    -- Device ID.
  correlationId      INTEGER,                    -- Correlation ID of the event record API to which this result is
associated.
  nodeId             INTEGER,                    -- Node ID of the event record API to which this result is associated.
  SMId               INTEGER,                    -- SM ID of the event on which the particular event was running.
  globalPid          INTEGER,                    -- Serialized GlobalId.
  BlockID            INTEGER  NOT NULL,          -- Block ID.
  UGPUId             INTEGER,                    -- uGPU ID of the event on which the particular event was running.
  CGAId              INTEGER                     -- CGA ID of the event on which the particular event was running.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_BLOCK_PHASE_TRACE (
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  deviceId           INTEGER,                    -- Device ID.
  correlationId      INTEGER,                    -- Correlation ID of the event record API to which this result is
associated.
  nodeId             INTEGER,                    -- Node ID of the event record API to which this result is associated.
  SMId               INTEGER,                    -- SM ID of the event on which the particular event was running.
  globalPid          INTEGER,                    -- Serialized GlobalId.
  BlockID            INTEGER  NOT NULL,          -- Block ID.
  phase1Timestamp    INTEGER  NOT NULL,          -- Phase start timestamp.
  phase2Timestamp    INTEGER  NOT NULL,          -- Phase stop timestamp.
  UGPUId             INTEGER,                    -- uGPU ID of the event on which the particular event was running.
  CGAId              INTEGER                     -- CGA ID of the event on which the particular event was running.
);
```

```
CREATE TABLE CUPTI_ACTIVITY_KIND_WARP_TRACE (
   start                INTEGER   NOT NULL,              -- Event start timestamp (ns).
   end                  INTEGER   NOT NULL,              -- Event end timestamp (ns).
   deviceId             INTEGER,                         -- Device ID.
   correlationId        INTEGER,                         -- Correlation ID of the event record API to which this result is
associated.
   nodeId               INTEGER,                         -- Node ID of the event record API to which this result is associated.
   SMId                 INTEGER,                         -- SM ID of the event on which the particular event was running.
   globalPid            INTEGER,                         -- Serialized GlobalId.
   BlockID              INTEGER   NOT NULL,              -- Block ID.
   WarpID               INTEGER   NOT NULL,              -- Warp ID.
   UGPUId               INTEGER,                         -- uGPU ID of the event on which the particular event was running.
   CGAId                INTEGER                          -- CGA ID of the event on which the particular event was running.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_WARP_PHASE_TRACE (
   start                INTEGER   NOT NULL,              -- Event start timestamp (ns).
   end                  INTEGER   NOT NULL,              -- Event end timestamp (ns).
   deviceId             INTEGER,                         -- Device ID.
   correlationId        INTEGER,                         -- Correlation ID of the event record API to which this result is
associated.
   nodeId               INTEGER,                         -- Node ID of the event record API to which this result is associated.
   SMId                 INTEGER,                         -- SM ID of the event on which the particular event was running.
   globalPid            INTEGER,                         -- Serialized GlobalId.
   BlockID              INTEGER   NOT NULL,              -- Block ID.
   WarpID               INTEGER   NOT NULL,              -- Warp ID.
   WarpEventIds         TEXT,                            -- Warp event IDs.
   WarpEventTimestampOffsets   TEXT,                     -- Warp event timestamp offsets.
   UGPUId               INTEGER,                         -- uGPU ID of the event on which the particular event was running.
```

```
  CGAId                INTEGER                    -- CGA ID of the event on which the particular event was running.
);
CREATE TABLE CUDA_UM_CPU_PAGE_FAULT_EVENTS (
  start                INTEGER  NOT NULL,         -- Event start timestamp (ns).
  globalPid            INTEGER  NOT NULL,         -- Serialized GlobalId.
  address              INTEGER  NOT NULL,         -- Virtual address of the page that faulted.
  originalFaultPc      INTEGER,                   -- Program counter of the CPU instruction that caused the page
fault.
  CpuInstruction       INTEGER  NOT NULL,         -- REFERENCES StringIds(id) -- Function name
  module               INTEGER  NOT NULL,         -- REFERENCES StringIds(id) -- Module name
  unresolvedFaultPc    INTEGER,                   -- True if the program counter was not resolved.
  sourceFile           INTEGER,                   -- Source file where the page fault occurred.
  sourceLine           INTEGER                    -- Source line number that caused the page fault in the source file.
);
CREATE TABLE CUDA_UM_GPU_PAGE_FAULT_EVENTS (
  start                INTEGER  NOT NULL,         -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,         -- Event end timestamp (ns).
  globalPid            INTEGER  NOT NULL,         -- Serialized GlobalId.
  deviceId             INTEGER  NOT NULL,         -- Device ID.
  address              INTEGER  NOT NULL,         -- Virtual address of the page that faulted.
  numberOfPageFaults   INTEGER  NOT NULL,         -- Number of page faults for the same page.
  faultAccessType      INTEGER  NOT NULL          -- REFERENCES
ENUM_CUDA_UNIF_MEM_ACCESS_TYPE(id)
);
CREATE TABLE CUDA_GPU_MEMORY_USAGE_EVENTS (
  start                INTEGER  NOT NULL,         -- Event start timestamp (ns).
  globalPid            INTEGER  NOT NULL,         -- Serialized GlobalId.
  deviceId             INTEGER  NOT NULL,         -- Device ID.
```

```
  contextId              INTEGER  NOT NULL,              -- Context ID.
  address                INTEGER  NOT NULL,              -- Virtual address of the allocation/deallocation.
  pc                     INTEGER  NOT NULL,              -- Program counter of the allocation/deallocation.
  bytes                  INTEGER  NOT NULL,              -- Number of bytes allocated/deallocated (B).
  memKind                INTEGER  NOT NULL,              -- REFERENCES ENUM_CUDA_MEM_KIND(id)
  memoryOperationType    INTEGER  NOT NULL,              -- REFERENCES
ENUM_CUDA_DEV_MEM_EVENT_OPER(id)
  name                   TEXT,                           -- Variable name, if available.
  correlationId          INTEGER,                        -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  streamId               INTEGER,                        -- Stream ID.
  localMemoryPoolAddress     INTEGER,                    -- Base address of the local memory pool used
  localMemoryPoolReleaseThreshold  INTEGER,              -- Release threshold of the local memory pool used
  localMemoryPoolSize        INTEGER,                    -- Size of the local memory pool used
  localMemoryPoolUtilizedSize  INTEGER,                  -- Utilized size of the local memory pool used
  importedMemoryPoolAddress  INTEGER,                    -- Base address of the imported memory pool used
  importedMemoryPoolProcessId  INTEGER                   -- Process ID of the imported memory pool used
);
CREATE TABLE CUDA_GPU_MEMORY_POOL_EVENTS (
  start                  INTEGER  NOT NULL,              -- Event start timestamp (ns).
  globalPid              INTEGER  NOT NULL,              -- Serialized GlobalId.
  deviceId               INTEGER  NOT NULL,              -- Device ID.
  address                INTEGER  NOT NULL,              -- The base virtual address of the memory pool.
  operationType          INTEGER  NOT NULL,              -- REFERENCES ENUM_CUDA_MEMPOOL_OPER(id)
  poolType               INTEGER  NOT NULL,              -- REFERENCES ENUM_CUDA_MEMPOOL_TYPE(id)
  correlationId          INTEGER,                        -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  minBytesToKeep         INTEGER,                        -- Minimum number of bytes to keep of the memory pool.
```

```
  localMemoryPoolReleaseThreshold   INTEGER,                -- Release threshold of the local memory pool used
  localMemoryPoolSize       INTEGER,                    -- Size of the local memory pool used
  localMemoryPoolUtilizedSize   INTEGER                   -- Utilized size of the local memory pool used
);
CREATE TABLE CUDA_CALLCHAINS (
  id                  INTEGER   NOT NULL,             -- Part of PRIMARY KEY (id, stackDepth).
  symbol              INTEGER   NOT NULL,             -- REFERENCES StringIds(id) -- Function name
  module              INTEGER   NOT NULL,             -- REFERENCES StringIds(id) -- Module name
  unresolved          INTEGER,                        -- True if the symbol was not resolved.
  originalIP          INTEGER,                        -- Instruction pointer value.
  stackDepth          INTEGER   NOT NULL,             -- Zero-base index of the given function in call stack.

  PRIMARY KEY (id, stackDepth)
);
CREATE TABLE MPI_RANKS (
  -- Mapping of global thread IDs (gtid) to MPI ranks

  globalTid           INTEGER   NOT NULL,             -- Serialized GlobalId.
  rank                INTEGER   NOT NULL              -- MPI rank
);
CREATE TABLE MPI_P2P_EVENTS (
  start               INTEGER   NOT NULL,             -- Event start timestamp (ns).
  end                 INTEGER,                        -- Event end timestamp (ns).
  globalTid           INTEGER,                        -- Serialized GlobalId.
  textId              INTEGER,                        -- REFERENCES StringIds(id) -- Registered NVTX domain/string
  commHandle          INTEGER,                        -- MPI communicator handle.
  tag                 INTEGER,                        -- MPI message tag
  remoteRank          INTEGER,                        -- MPI remote rank (destination or source)
```

```
  size            INTEGER,            -- MPI message size in bytes
  requestHandle       INTEGER             -- MPI request handle.
);
CREATE TABLE MPI_COLLECTIVES_EVENTS (
  start           INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end             INTEGER,            -- Event end timestamp (ns).
  globalTid           INTEGER,            -- Serialized GlobalId.
  textId          INTEGER,            -- REFERENCES StringIds(id) -- Registered NVTX domain/string
  commHandle          INTEGER,            -- MPI communicator handle.
  rootRank            INTEGER,            -- root rank in the collective
  size            INTEGER,            -- MPI message size in bytes (send size for bidirectional ops)
  recvSize            INTEGER,            -- MPI receive size in bytes
  requestHandle       INTEGER             -- MPI request handle.
);
CREATE TABLE MPI_START_WAIT_EVENTS (
  -- MPI_Start*, MPI_Test* and MPI_Wait*

  start           INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end             INTEGER,            -- Event end timestamp (ns).
  globalTid           INTEGER,            -- Serialized GlobalId.
  textId          INTEGER,            -- REFERENCES StringIds(id) -- Registered NVTX domain/string
  requestHandle       INTEGER             -- MPI request handle.
);
CREATE TABLE MPI_OTHER_EVENTS (
  -- MPI events without additional parameters

  start           INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end             INTEGER,            -- Event end timestamp (ns).
```

```
  globalTid          INTEGER,                -- Serialized GlobalId.
  textId             INTEGER                 -- REFERENCES StringIds(id) -- Registered NVTX domain/string
);
CREATE TABLE UCP_WORKERS (
  globalTid          INTEGER  NOT NULL,      -- Serialized GlobalId.
  workerUid          INTEGER  NOT NULL       -- UCP worker UID
);
CREATE TABLE UCP_SUBMIT_EVENTS (
  start              INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                INTEGER,                -- Event end timestamp (ns).
  globalTid          INTEGER,                -- Serialized GlobalId.
  textId             INTEGER,                -- REFERENCES StringIds(id) -- Registered NVTX domain/string
  bufferAddr         INTEGER,                -- Address of the message buffer
  packedSize         INTEGER,                -- Message size (packed) in bytes
  peerWorkerUid      INTEGER,                -- Peer's UCP worker UID
  tag                INTEGER                 -- UCP message tag
);
CREATE TABLE UCP_PROGRESS_EVENTS (
  start              INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                INTEGER,                -- Event end timestamp (ns).
  globalTid          INTEGER,                -- Serialized GlobalId.
  textId             INTEGER,                -- REFERENCES StringIds(id) -- Registered NVTX domain/string
  bufferAddr         INTEGER,                -- Address of the message buffer
  packedSize         INTEGER,                -- Message size (packed) in bytes
  peerWorkerUid      INTEGER,                -- Peer's UCP worker UID
  tag                INTEGER                 -- UCP message tag
);
CREATE TABLE UCP_EVENTS (
```

```
  -- UCP events without additional parameters

  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER,                  -- Event end timestamp (ns).
  globalTid      INTEGER,                  -- Serialized GlobalId.
  textId         INTEGER                   -- REFERENCES StringIds(id) -- Registered NVTX domain/string
);
CREATE TABLE NVTX_EVENTS (
  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER,                  -- Event end timestamp (ns).
  eventType      INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_TYPE(id)
  rangeId        INTEGER,                  -- Correlation ID returned from a nvtxRangeStart call.
  category       INTEGER,                  -- User-controlled ID that can be used to group events.
  color          INTEGER,                  -- Encoded ARGB color value.
  text           TEXT,                     -- Explicit name/text (non-registered string)
  globalTid      INTEGER,                  -- Serialized GlobalId.
  endGlobalTid   INTEGER,                  -- Serialized GlobalId.
  textId         INTEGER,                  -- REFERENCES StringIds(id) -- Registered NVTX domain/string
  domainId       INTEGER,                  -- User-controlled ID that can be used to group events.
  uint64Value    INTEGER,                  -- One of possible payload value union members.
  int64Value     INTEGER,                  -- One of possible payload value union members.
  doubleValue    REAL,                     -- One of possible payload value union members.
  uint32Value    INTEGER,                  -- One of possible payload value union members.
  int32Value     INTEGER,                  -- One of possible payload value union members.
  floatValue     REAL,                     -- One of possible payload value union members.
  jsonTextId     INTEGER,                  -- One of possible payload value union members.
  jsonText       TEXT,                     -- One of possible payload value union members.
  binaryData     TEXT                      -- Binary payload. See docs for format.
```

```
);
CREATE TABLE OPENGL_API (
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_TYPE(id)
  globalTid          INTEGER,                    -- Serialized GlobalId.
  endGlobalTid       INTEGER,                    -- Serialized GlobalId.
  correlationId      INTEGER,                    -- First ID matching an API call to GPU workloads.
  endCorrelationId   INTEGER,                    -- Last ID matching an API call to GPU workloads.
  nameId             INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- First function name
  endNameId          INTEGER,                    -- REFERENCES StringIds(id) -- Last function name
  returnValue        INTEGER  NOT NULL,          -- Return value of the function call.
  frameId            INTEGER,                    -- Index of the graphics frame starting from 1.
  contextId          INTEGER,                    -- Context ID.
  gpu                INTEGER,                    -- GPU index.
  display            INTEGER                     -- Display ID.
);
CREATE TABLE OPENGL_WORKLOAD (
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_TYPE(id)
  globalTid          INTEGER,                    -- Serialized GlobalId.
  endGlobalTid       INTEGER,                    -- Serialized GlobalId.
  correlationId      INTEGER,                    -- First ID matching an API call to GPU workloads.
  endCorrelationId   INTEGER,                    -- Last ID matching an API call to GPU workloads.
  nameId             INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- First function name
  endNameId          INTEGER,                    -- REFERENCES StringIds(id) -- Last function name
  returnValue        INTEGER  NOT NULL,          -- Return value of the function call.
```

```
  frameId          INTEGER,              -- Index of the graphics frame starting from 1.
  contextId        INTEGER,              -- Context ID.
  gpu              INTEGER,              -- GPU index.
  display          INTEGER               -- Display ID.
);
CREATE TABLE KHR_DEBUG_EVENTS (
  eventClass       INTEGER  NOT NULL,    -- REFERENCES ENUM_NSYS_EVENT_TYPE(id)
  start            INTEGER  NOT NULL,    -- Event start timestamp (ns).
  end              INTEGER,              -- Event end timestamp (ns).
  textId           INTEGER,              -- REFERENCES StringIds(id) -- Debug marker/group text
  globalTid        INTEGER,              -- Serialized GlobalId.
  source           INTEGER,              -- REFERENCES ENUM_OPENGL_DEBUG_SOURCE(id)
  khrdType         INTEGER,              -- REFERENCES ENUM_OPENGL_DEBUG_TYPE(id)
  id               INTEGER,              -- KHR event ID.
  severity         INTEGER,              -- REFERENCES ENUM_OPENGL_DEBUG_SEVERITY(id)
  correlationId    INTEGER,              -- ID used to correlate KHR CPU trace to GPU trace.
  context          INTEGER               -- Context ID.
);
CREATE TABLE OSRT_API (
  -- OS runtime libraries traced to gather information about low-level userspace APIs.

  start            INTEGER  NOT NULL,    -- Event start timestamp (ns).
  end              INTEGER  NOT NULL,    -- Event end timestamp (ns).
  eventClass       INTEGER  NOT NULL,    -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid        INTEGER,              -- Serialized GlobalId.
  nameId           INTEGER  NOT NULL,    -- REFERENCES StringIds(id) -- Function name
  returnValue      INTEGER  NOT NULL,    -- Return value of the function call.
  nestingLevel     INTEGER,              -- Zero-base index of the nesting level.
```

```
  callchainId          INTEGER  NOT NULL              -- REFERENCES OSRT_CALLCHAINS(id)
);
CREATE TABLE OSRT_CALLCHAINS (
  -- Callchains attached to OSRT events, depending on selected profiling settings.

  id                 INTEGER  NOT NULL,            -- Part of PRIMARY KEY (id, stackDepth).
  symbol             INTEGER  NOT NULL,            -- REFERENCES StringIds(id) -- Function name
  module             INTEGER  NOT NULL,            -- REFERENCES StringIds(id) -- Module name
  kernelMode         INTEGER,                      -- True if kernel mode.
  thumbCode          INTEGER,                      -- True if thumb code.
  unresolved         INTEGER,                      -- True if the symbol was not resolved.
  specialEntry       INTEGER,                      -- True if artifical entry added during processing callchain.
  originalIP         INTEGER,                      -- Instruction pointer value.
  unwindMethod       INTEGER,                      -- REFERENCES ENUM_STACK_UNWIND_METHOD(id)
  stackDepth         INTEGER  NOT NULL,            -- Zero-base index of the given function in call stack.

  PRIMARY KEY (id, stackDepth)
);
CREATE TABLE OSRT_FILE_ACCESS_DESCRIPTORS (
  -- Metadata of all file accesses that were made by the OS during the recording.

  fileAccessId       INTEGER  NOT NULL,            -- File Access Id.
  processId          INTEGER  NOT NULL,            -- Process ID.
  openedAt           INTEGER  NOT NULL,            -- The time when the file was opened (ns).
  closedAt           INTEGER  NOT NULL,            -- The time when the file was closed (ns).
  filePath           TEXT     NOT NULL,            -- The opened file path.
);
CREATE TABLE OSRT_FILE_ACCESS_EVENTS (
```

```
  -- OS Runtime events related to file accesses (opening, closing, reading, and writing).

  fileAccessId          INTEGER  NOT NULL,            -- REFERENCES
OSRT_FILE_ACCESS_DESCRIPTORS(fileAccessId)
  threadId              INTEGER  NOT NULL,            -- Thread ID.
  startedAt             INTEGER  NOT NULL,            -- Event start timestamp (ns).
  endedAt               INTEGER  NOT NULL,            -- Event end timestamp (ns).
  eventType             INTEGER  NOT NULL,            -- REFERENCES
ENUM_OSRT_FILE_ACCESS_EVENT_TYPE(id)
  apiCallId             INTEGER  NOT NULL,            -- REFERENCES StringIds(id) -- Function name
  bytesProcessed        INTEGER  NOT NULL             -- Actual bytes read/written.
);
CREATE TABLE PROFILER_OVERHEAD (
  start                 INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid             INTEGER,                      -- Serialized GlobalId.
  nameId                INTEGER  NOT NULL,            -- REFERENCES StringIds(id) -- Function name
  returnValue           INTEGER  NOT NULL             -- Return value of the function call.
);
CREATE TABLE SCHED_EVENTS (
  -- Thread scheduling events.

  start                 INTEGER  NOT NULL,            -- Event start timestamp (ns).
  cpu                   INTEGER  NOT NULL,            -- ID of CPU this thread was scheduled in or out.
  isSchedIn             INTEGER  NOT NULL,            -- 0 if thread was scheduled out, non-zero otherwise.
  globalTid             INTEGER,                      -- Serialized GlobalId.
  threadState           INTEGER,                      -- REFERENCES ENUM_SAMPLING_THREAD_STATE(id)
  threadBlock           INTEGER                       -- REFERENCES ENUM_SCHEDULING_THREAD_BLOCK(id)
```

```
);
CREATE TABLE COMPOSITE_EVENTS (
  -- Thread sampling events.

  id              INTEGER  NOT NULL  PRIMARY KEY,    -- ID of the composite event.
  start           INTEGER  NOT NULL,                 -- Event start timestamp (ns).
  cpu             INTEGER,                           -- ID of CPU this thread was running on.
  threadState     INTEGER,                           -- REFERENCES ENUM_SAMPLING_THREAD_STATE(id)
  globalTid       INTEGER,                           -- Serialized GlobalId.
  cpuCycles       INTEGER  NOT NULL                  -- Value of Performance Monitoring Unit (PMU) counter.
);
CREATE TABLE SAMPLING_CALLCHAINS (
  -- Callchain entries obtained from composite events, used to construct function table views.

  id              INTEGER  NOT NULL,                 -- REFERENCES COMPOSITE_EVENTS(id)
  symbol          INTEGER  NOT NULL,                 -- REFERENCES StringIds(id) -- Function name
  module          INTEGER  NOT NULL,                 -- REFERENCES StringIds(id) -- Module name
  kernelMode      INTEGER,                           -- True if kernel mode.
  thumbCode       INTEGER,                           -- True if thumb code.
  unresolved      INTEGER,                           -- True if the symbol was not resolved.
  specialEntry    INTEGER,                           -- True if artifical entry added during processing callchain.
  originalIP      INTEGER,                           -- Instruction pointer value.
  unwindMethod    INTEGER,                           -- REFERENCES ENUM_STACK_UNWIND_METHOD(id)
  stackDepth      INTEGER  NOT NULL,                 -- Zero-base index of the given function in call stack.

  PRIMARY KEY (id, stackDepth)
);
CREATE TABLE PERF_EVENT_SOC_OR_CPU_RAW_EVENT (
```

```
  -- SoC and CPU raw event values from Sampled Performance Counters.

  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  vmId               INTEGER,                    -- VM ID.
  componentId        INTEGER,                    -- REFERENCES TARGET_INFO_COMPONENT(componentId)
  eventId            INTEGER,                    -- REFERENCES TARGET_INFO_PERF_METRIC(id)
  count              INTEGER                     -- Counter data value
);
CREATE TABLE PERF_EVENT_SOC_OR_CPU_METRIC_EVENT (
  -- SoC and CPU metric values from Sampled Performance Counters.

  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  vmId               INTEGER,                    -- VM ID.
  componentId        INTEGER,                    -- REFERENCES TARGET_INFO_COMPONENT(componentId)
  metricId           INTEGER,                    -- REFERENCES TARGET_INFO_PERF_METRIC(id)
  value              REAL                        -- Metric data value
);
CREATE TABLE DX12_API (
  id                 INTEGER  NOT NULL  PRIMARY KEY,
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                    -- Serialized GlobalId.
  correlationId      INTEGER,                    -- First ID matching an API call to GPU workloads.
  endCorrelationId   INTEGER,                    -- Last ID matching an API call to GPU workloads.
  nameId             INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Function name
```

```
  shortContextId        INTEGER,                -- Short form of the COM interface object address.
  frameId               INTEGER,                -- Index of the graphics frame starting from 1.
  color                 INTEGER,                -- Encoded ARGB color value.
  textId                INTEGER,                -- REFERENCES StringIds(id) -- PIX marker text
  commandListType       INTEGER,                -- REFERENCES ENUM_D3D12_CMD_LIST_TYPE(id)
  objectNameId          INTEGER,                -- REFERENCES StringIds(id) -- D3D12 object name
  longContextId         INTEGER                 -- Long form of the COM interface object address.
);
CREATE TABLE DX12_WORKLOAD (
  start                 INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass            INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid             INTEGER,                -- Serialized GlobalId.
  correlationId         INTEGER,                -- First ID matching an API call to GPU workloads.
  endCorrelationId      INTEGER,                -- Last ID matching an API call to GPU workloads.
  nameId                INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Function name
  shortContextId        INTEGER,                -- Short form of the COM interface object address.
  frameId               INTEGER,                -- Index of the graphics frame starting from 1.
  gpu                   INTEGER,                -- GPU index.
  color                 INTEGER,                -- Encoded ARGB color value.
  textId                INTEGER,                -- REFERENCES StringIds(id) -- PIX marker text
  commandListType       INTEGER,                -- REFERENCES ENUM_D3D12_CMD_LIST_TYPE(id)
  objectNameId          INTEGER,                -- REFERENCES StringIds(id) -- D3D12 object name
  longContextId         INTEGER                 -- Long form of the COM interface object address.
);
CREATE TABLE DX12_MEMORY_OPERATION (
  gpu                   INTEGER,                -- GPU index.
  rangeStart            INTEGER,                -- Offset denoting the beginning of a memory range (B).
```

```
  rangeEnd            INTEGER,                -- Offset denoting the end of a memory range (B).
  subresourceId       INTEGER,                -- Subresource index.
  heapType            INTEGER,                -- REFERENCES ENUM_D3D12_HEAP_TYPE(id)
  heapFlags           INTEGER,                -- REFERENCES ENUM_D3D12_HEAP_FLAGS(id)
  cpuPageProperty     INTEGER,                -- REFERENCES ENUM_D3D12_PAGE_PROPERTY(id)
  nvApiFlags          INTEGER,                -- NV specific flags. See docs for specifics.
  traceEventId        INTEGER  NOT NULL       -- REFERENCES DX12_API(id)
);
CREATE TABLE VULKAN_API (
  id                  INTEGER  NOT NULL  PRIMARY KEY,
  start               INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass          INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                -- Serialized GlobalId.
  correlationId       INTEGER,                -- First ID matching an API call to GPU workloads.
  endCorrelationId    INTEGER,                -- Last ID matching an API call to GPU workloads.
  nameId              INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Function name
  contextId           INTEGER                 -- Short form of the interface object address.
);
CREATE TABLE VULKAN_WORKLOAD (
  start               INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass          INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                -- Serialized GlobalId.
  correlationId       INTEGER,                -- First ID matching an API call to GPU workloads.
  endCorrelationId    INTEGER,                -- Last ID matching an API call to GPU workloads.
  nameId              INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Function name
  gpu                 INTEGER,                -- GPU index.
```

```
  contextId          INTEGER,                -- Short form of the interface object address.
  color              INTEGER,                -- Encoded ARGB color value.
  textId             INTEGER                 -- REFERENCES StringIds(id) -- Vulkan CPU debug marker string
);
CREATE TABLE VULKAN_DEBUG_API (
  start              INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                -- Serialized GlobalId.
  correlationId      INTEGER,                -- First ID matching an API call to GPU workloads.
  endCorrelationId   INTEGER,                -- Last ID matching an API call to GPU workloads.
  nameId             INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Function name
  contextId          INTEGER,                -- Short form of the interface object address.
  color              INTEGER,                -- Encoded ARGB color value.
  textId             INTEGER                 -- REFERENCES StringIds(id) -- Vulkan CPU debug marker string
);
CREATE TABLE VULKAN_PIPELINE_CREATION_EVENTS (
  id                 INTEGER  NOT NULL  PRIMARY KEY,  -- ID of the pipeline creation event.
  duration           INTEGER,                -- Event duration (ns).
  flags              INTEGER,                -- REFERENCES
ENUM_VULKAN_PIPELINE_CREATION_FLAGS(id)
  traceEventId       INTEGER  NOT NULL       -- REFERENCES VULKAN_API(id) -- ID of the attached
vulkan API.
);
CREATE TABLE VULKAN_PIPELINE_STAGE_EVENTS (
  id                 INTEGER  NOT NULL  PRIMARY KEY,  -- ID of the pipeline stage event.
  duration           INTEGER,                -- Event duration (ns).
  flags              INTEGER,                -- REFERENCES
```

```
ENUM_VULKAN_PIPELINE_CREATION_FLAGS(id)
  creationEventId        INTEGER  NOT NULL              -- REFERENCES
VULKAN_PIPELINE_CREATION_EVENTS(id) -- ID of the attached pipeline creation event.
);
CREATE TABLE GPU_CONTEXT_SWITCH_EVENTS (
  tag                INTEGER  NOT NULL,             -- REFERENCES ENUM_GPU_CTX_SWITCH(id)
  vmId               INTEGER  NOT NULL,             -- VM ID.
  seqNo              INTEGER  NOT NULL,             -- Sequential event number.
  contextId          INTEGER  NOT NULL,             -- Context ID.
  timestamp          INTEGER  NOT NULL,             -- Event start timestamp (ns).
  globalPid          INTEGER,                       -- Serialized GlobalId.
  gpuId              INTEGER                        -- GPU index.
);
CREATE TABLE OPENMP_EVENT_KIND_THREAD (
  start              INTEGER  NOT NULL,             -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,             -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,             -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                       -- Serialized GlobalId.
  correlationId      INTEGER,                       -- Currently unused.
  nameId             INTEGER  NOT NULL,             -- REFERENCES StringIds(id) -- Event name
  eventKind          INTEGER,                       -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  threadId           INTEGER,                       -- Internal thread sequence starting from 1.
  threadType         INTEGER                        -- REFERENCES ENUM_OPENMP_THREAD(id)
);
CREATE TABLE OPENMP_EVENT_KIND_PARALLEL (
  start              INTEGER  NOT NULL,             -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,             -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,             -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
```

```
  globalTid          INTEGER,                    -- Serialized GlobalId.
  correlationId      INTEGER,                    -- Currently unused.
  nameId             INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  eventKind          INTEGER,                    -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parallelId         INTEGER,                    -- Internal parallel region sequence starting from 1.
  parentTaskId       INTEGER                     -- ID for task that creates this parallel region.
);
CREATE TABLE OPENMP_EVENT_KIND_SYNC_REGION_WAIT (
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                    -- Serialized GlobalId.
  correlationId      INTEGER,                    -- Currently unused.
  nameId             INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  eventKind          INTEGER,                    -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parallelId         INTEGER,                    -- ID of the parallel region that this event belongs to.
  taskId             INTEGER,                    -- ID of the task that this event belongs to.
  kind               INTEGER                     -- REFERENCES ENUM_OPENMP_SYNC_REGION(id)
);
CREATE TABLE OPENMP_EVENT_KIND_SYNC_REGION (
  start              INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                    -- Serialized GlobalId.
  correlationId      INTEGER,                    -- Currently unused.
  nameId             INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  eventKind          INTEGER,                    -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parallelId         INTEGER,                    -- ID of the parallel region that this event belongs to.
```

```
  taskId          INTEGER,                -- ID of the task that this event belongs to.
  kind            INTEGER                 -- REFERENCES ENUM_OPENMP_SYNC_REGION(id)
);
CREATE TABLE OPENMP_EVENT_KIND_TASK (
  start           INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end             INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass      INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid       INTEGER,                -- Serialized GlobalId.
  correlationId   INTEGER,                -- Currently unused.
  nameId          INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Event name
  eventKind       INTEGER,                -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parallelId      INTEGER,                -- ID of the parallel region that this event belongs to.
  taskId          INTEGER,                -- ID of the task that this event belongs to.
  kind            INTEGER                 -- REFERENCES ENUM_OPENMP_TASK_FLAG(id)
);
CREATE TABLE OPENMP_EVENT_KIND_MASTER (
  start           INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end             INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass      INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid       INTEGER,                -- Serialized GlobalId.
  correlationId   INTEGER,                -- Currently unused.
  nameId          INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Event name
  eventKind       INTEGER,                -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parallelId      INTEGER,                -- ID of the parallel region that this event belongs to.
  taskId          INTEGER                 -- ID of the task that this event belongs to.
);
CREATE TABLE OPENMP_EVENT_KIND_REDUCTION (
  start           INTEGER  NOT NULL,      -- Event start timestamp (ns).
```

```
  end             INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass        INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid         INTEGER,                    -- Serialized GlobalId.
  correlationId     INTEGER,                    -- Currently unused.
  nameId            INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  eventKind         INTEGER,                    -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parallelId        INTEGER,                    -- ID of the parallel region that this event belongs to.
  taskId            INTEGER                     -- ID of the task that this event belongs to.
);
CREATE TABLE OPENMP_EVENT_KIND_TASK_CREATE (
  start           INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end             INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass        INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid         INTEGER,                    -- Serialized GlobalId.
  correlationId     INTEGER,                    -- Currently unused.
  nameId            INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  eventKind         INTEGER,                    -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  parentTaskId      INTEGER,                    -- ID of the parent task that is creating a new task.
  newTaskId         INTEGER                     -- ID of the new task that is being created.
);
CREATE TABLE OPENMP_EVENT_KIND_TASK_SCHEDULE (
  start           INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end             INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass        INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid         INTEGER,                    -- Serialized GlobalId.
  correlationId     INTEGER,                    -- Currently unused.
  nameId            INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  eventKind         INTEGER,                    -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
```

```
  parallelId          INTEGER,                -- ID of the parallel region that this event belongs to.
  priorTaskId         INTEGER,                -- ID of the task that is being switched out.
  priorTaskStatus     INTEGER,                -- REFERENCES ENUM_OPENMP_TASK_STATUS(id)
  nextTaskId          INTEGER                 -- ID of the task that is being switched in.
);
CREATE TABLE OPENMP_EVENT_KIND_CANCEL (
  start               INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass          INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                -- Serialized GlobalId.
  correlationId       INTEGER,                -- Currently unused.
  nameId              INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Event name
  eventKind           INTEGER,                -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  taskId              INTEGER                 -- ID of the task that is being cancelled.
);
CREATE TABLE OPENMP_EVENT_KIND_MUTEX_WAIT (
  start               INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,      -- Event end timestamp (ns).
  eventClass          INTEGER  NOT NULL,      -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                -- Serialized GlobalId.
  correlationId       INTEGER,                -- Currently unused.
  nameId              INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Event name
  eventKind           INTEGER,                -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind                INTEGER,                -- REFERENCES ENUM_OPENMP_MUTEX(id)
  waitId              INTEGER,                -- ID indicating the object being waited.
  taskId              INTEGER                 -- ID of the task that this event belongs to.
);
CREATE TABLE OPENMP_EVENT_KIND_CRITICAL_SECTION (
```

```
  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass     INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid      INTEGER,                  -- Serialized GlobalId.
  correlationId  INTEGER,                  -- Currently unused.
  nameId         INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind      INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind           INTEGER,                  -- REFERENCES ENUM_OPENMP_MUTEX(id)
  waitId         INTEGER                   -- ID indicating the object being held.
);
CREATE TABLE OPENMP_EVENT_KIND_MUTEX_RELEASED (
  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass     INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid      INTEGER,                  -- Serialized GlobalId.
  correlationId  INTEGER,                  -- Currently unused.
  nameId         INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind      INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind           INTEGER,                  -- REFERENCES ENUM_OPENMP_MUTEX(id)
  waitId         INTEGER,                  -- ID indicating the object being released.
  taskId         INTEGER                   -- ID of the task that this event belongs to.
);
CREATE TABLE OPENMP_EVENT_KIND_LOCK_INIT (
  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass     INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid      INTEGER,                  -- Serialized GlobalId.
  correlationId  INTEGER,                  -- Currently unused.
```

```
  nameId          INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind       INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind            INTEGER,                  -- REFERENCES ENUM_OPENMP_MUTEX(id)
  waitId          INTEGER                   -- ID indicating object being created/destroyed.
);
CREATE TABLE OPENMP_EVENT_KIND_LOCK_DESTROY (
  start           INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end             INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass      INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid       INTEGER,                  -- Serialized GlobalId.
  correlationId   INTEGER,                  -- Currently unused.
  nameId          INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind       INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind            INTEGER,                  -- REFERENCES ENUM_OPENMP_MUTEX(id)
  waitId          INTEGER                   -- ID indicating object being created/destroyed.
);
CREATE TABLE OPENMP_EVENT_KIND_WORKSHARE (
  start           INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end             INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass      INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid       INTEGER,                  -- Serialized GlobalId.
  correlationId   INTEGER,                  -- Currently unused.
  nameId          INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind       INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind            INTEGER,                  -- REFERENCES ENUM_OPENMP_WORK(id)
  parallelId      INTEGER,                  -- ID of the parallel region that this event belongs to.
  taskId          INTEGER,                  -- ID of the task that this event belongs to.
  count           INTEGER                   -- Measure of the quantity of work involved in the region.
```

```
);
CREATE TABLE OPENMP_EVENT_KIND_DISPATCH (
  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass     INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid      INTEGER,                  -- Serialized GlobalId.
  correlationId  INTEGER,                  -- Currently unused.
  nameId         INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind      INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  kind           INTEGER,                  -- REFERENCES ENUM_OPENMP_DISPATCH(id)
  parallelId     INTEGER,                  -- ID of the parallel region that this event belongs to.
  taskId         INTEGER                   -- ID of the task that this event belongs to.
);
CREATE TABLE OPENMP_EVENT_KIND_FLUSH (
  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER  NOT NULL,        -- Event end timestamp (ns).
  eventClass     INTEGER  NOT NULL,        -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid      INTEGER,                  -- Serialized GlobalId.
  correlationId  INTEGER,                  -- Currently unused.
  nameId         INTEGER  NOT NULL,        -- REFERENCES StringIds(id) -- Event name
  eventKind      INTEGER,                  -- REFERENCES ENUM_OPENMP_EVENT_KIND(id)
  threadId       INTEGER                   -- ID of the thread that this event belongs to.
);
CREATE TABLE D3D11_PIX_DEBUG_API (
  -- D3D11 debug marker events.

  start          INTEGER  NOT NULL,        -- Event start timestamp (ns).
  end            INTEGER  NOT NULL,        -- Event end timestamp (ns).
```

```
  globalTid            INTEGER,                    -- Serialized GlobalId.
  correlationId        INTEGER,                    -- First ID matching an API call to GPU workloads.
  endCorrelationId     INTEGER,                    -- Last ID matching an API call to GPU workloads.
  nameId               INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Function name
  shortContextId       INTEGER,                    -- Short form of the COM interface object address.
  frameId              INTEGER,                    -- Index of the graphics frame starting from 1.
  color                INTEGER,                    -- Encoded ARGB color value.
  textId               INTEGER                     -- REFERENCES StringIds(id) -- PIX marker text
);
CREATE TABLE D3D12_PIX_DEBUG_API (
  -- D3D12 debug marker events.

  start                INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,          -- Event end timestamp (ns).
  globalTid            INTEGER,                    -- Serialized GlobalId.
  correlationId        INTEGER,                    -- First ID matching an API call to GPU workloads.
  endCorrelationId     INTEGER,                    -- Last ID matching an API call to GPU workloads.
  nameId               INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Function name
  shortContextId       INTEGER,                    -- Short form of the COM interface object address.
  frameId              INTEGER,                    -- Index of the graphics frame starting from 1.
  color                INTEGER,                    -- Encoded ARGB color value.
  textId               INTEGER,                    -- REFERENCES StringIds(id) -- PIX marker text
  commandListType      INTEGER,                    -- REFERENCES ENUM_D3D12_CMD_LIST_TYPE(id)
  objectNameId         INTEGER,                    -- REFERENCES StringIds(id) -- D3D12 object name
  longContextId        INTEGER                     -- Long form of the COM interface object address.
);
CREATE TABLE WDDM_EVICT_ALLOCATION_EVENTS (
  -- Raw ETW EvictAllocation events.
```

```
  start                INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid            INTEGER,                      -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,            -- GPU index.
  allocationHandle     INTEGER  NOT NULL             -- Global allocation handle.
);
CREATE TABLE WDDM_PAGING_QUEUE_PACKET_START_EVENTS (
  -- Raw ETW PagingQueuePacketStart events.

  start                INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid            INTEGER,                      -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,            -- GPU index.
  dxgDevice            INTEGER,                      -- Address of an IDXGIDevice.
  dxgAdapter           INTEGER,                      -- Address of an IDXGIAdapter.
  pagingQueue          INTEGER  NOT NULL,            -- Address of the paging queue.
  pagingQueuePacket    INTEGER  NOT NULL,            -- Address of the paging queue packet.
  sequenceId           INTEGER  NOT NULL,            -- Internal sequence starting from 0.
  alloc                INTEGER,                      -- Allocation handle.
  vidMmOpType          INTEGER  NOT NULL,            -- REFERENCES
ENUM_WDDM_VIDMM_OP_TYPE(id)
  pagingQueueType      INTEGER  NOT NULL             -- REFERENCES
ENUM_WDDM_PAGING_QUEUE_TYPE(id)
);
CREATE TABLE WDDM_PAGING_QUEUE_PACKET_STOP_EVENTS (
  -- Raw ETW PagingQueuePacketStop events.
```

```
  start                INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid            INTEGER,                      -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,            -- GPU index.
  pagingQueue          INTEGER  NOT NULL,            -- Address of the paging queue.
  pagingQueuePacket    INTEGER  NOT NULL,            -- Address of the paging queue packet.
  sequenceId           INTEGER  NOT NULL             -- Internal sequence starting from 0.
);
CREATE TABLE WDDM_PAGING_QUEUE_PACKET_INFO_EVENTS (
  -- Raw ETW PagingQueuePacketInfo events.

  start                INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid            INTEGER,                      -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,            -- GPU index.
  pagingQueue          INTEGER  NOT NULL,            -- Address of the paging queue.
  pagingQueuePacket    INTEGER  NOT NULL,            -- Address of the paging queue packet.
  sequenceId           INTEGER  NOT NULL             -- Internal sequence starting from 0.
);
CREATE TABLE WDDM_QUEUE_PACKET_START_EVENTS (
  -- Raw ETW QueuePacketStart events.

  start                INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid            INTEGER,                      -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,            -- GPU index.
  context              INTEGER  NOT NULL,            -- The context ID of WDDM queue.
  dmaBufferSize        INTEGER  NOT NULL,            -- The dma buffer size.
```

```
  dmaBuffer              INTEGER   NOT NULL,              -- The reported address of dma buffer.
  queuePacket            INTEGER   NOT NULL,              -- The address of queue packet.
  progressFenceValue     INTEGER   NOT NULL,              -- The fence value.
  packetType             INTEGER   NOT NULL,              -- REFERENCES ENUM_WDDM_PACKET_TYPE(id)
  submitSequence         INTEGER   NOT NULL,              -- Internal sequence starting from 1.
  allocationListSize     INTEGER   NOT NULL,              -- The number of allocations referenced.
  patchLocationListSize  INTEGER   NOT NULL,              -- The number of patch locations.
  present                INTEGER   NOT NULL,              -- True or False if the packet is a present packet.
  engineType             INTEGER   NOT NULL,              -- REFERENCES ENUM_WDDM_ENGINE_TYPE(id)
  syncObject             INTEGER                          -- The address of fence object.
);
CREATE TABLE WDDM_QUEUE_PACKET_STOP_EVENTS (
  -- Raw ETW QueuePacketStop events.

  start                  INTEGER   NOT NULL,              -- Event start timestamp (ns).
  end                    INTEGER   NOT NULL,              -- Event end timestamp (ns).
  globalTid              INTEGER,                         -- Serialized GlobalId.
  gpu                    INTEGER   NOT NULL,              -- GPU index.
  context                INTEGER   NOT NULL,              -- The context ID of WDDM queue.
  queuePacket            INTEGER   NOT NULL,              -- The address of queue packet.
  packetType             INTEGER   NOT NULL,              -- REFERENCES ENUM_WDDM_PACKET_TYPE(id)
  submitSequence         INTEGER   NOT NULL,              -- Internal sequence starting from 1.
  preempted              INTEGER   NOT NULL,              -- True or False if the packet is preempted.
  timeouted              INTEGER   NOT NULL,              -- True or False if the packet is timeouted.
  engineType             INTEGER   NOT NULL               -- REFERENCES ENUM_WDDM_ENGINE_TYPE(id)
);
CREATE TABLE WDDM_QUEUE_PACKET_INFO_EVENTS (
  -- Raw ETW QueuePacketInfo events.
```

```
  start                INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,          -- Event end timestamp (ns).
  globalTid            INTEGER,                    -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,          -- GPU index.
  context              INTEGER  NOT NULL,          -- The context ID of WDDM queue.
  packetType           INTEGER  NOT NULL,          -- REFERENCES ENUM_WDDM_PACKET_TYPE(id)
  submitSequence       INTEGER  NOT NULL,          -- Internal sequence starting from 1.
  engineType           INTEGER  NOT NULL           -- REFERENCES ENUM_WDDM_ENGINE_TYPE(id)
);
CREATE TABLE WDDM_DMA_PACKET_START_EVENTS (
  -- Raw ETW DmaPacketStart events.

  start                INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,          -- Event end timestamp (ns).
  globalTid            INTEGER,                    -- Serialized GlobalId.
  gpu                  INTEGER  NOT NULL,          -- GPU index.
  context              INTEGER  NOT NULL,          -- The context ID of WDDM queue.
  queuePacketContext   INTEGER  NOT NULL,          -- The queue packet context.
  uliSubmissionId      INTEGER  NOT NULL,          -- The queue packet submission ID.
  dmaBuffer            INTEGER  NOT NULL,          -- The reported address of dma buffer.
  packetType           INTEGER  NOT NULL,          -- REFERENCES ENUM_WDDM_PACKET_TYPE(id)
  ulQueueSubmitSequence INTEGER NOT NULL,          -- Internal sequence starting from 1.
  quantumStatus        INTEGER  NOT NULL,          -- The quantum Status.
  engineType           INTEGER  NOT NULL           -- REFERENCES ENUM_WDDM_ENGINE_TYPE(id)
);
CREATE TABLE WDDM_DMA_PACKET_STOP_EVENTS (
  -- Raw ETW DmaPacketStop events.
```

```
  start                 INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid             INTEGER,                      -- Serialized GlobalId.
  gpu                   INTEGER  NOT NULL,            -- GPU index.
  context               INTEGER  NOT NULL,            -- The context ID of WDDM queue.
  uliCompletionId       INTEGER  NOT NULL,            -- The queue packet completion ID.
  packetType            INTEGER  NOT NULL,            -- REFERENCES ENUM_WDDM_PACKET_TYPE(id)
  ulQueueSubmitSequence INTEGER  NOT NULL,            -- Internal sequence starting from 1.
  preempted             INTEGER  NOT NULL,            -- True or False if the packet is preempted.
  engineType            INTEGER  NOT NULL             -- REFERENCES ENUM_WDDM_ENGINE_TYPE(id)
);
CREATE TABLE WDDM_DMA_PACKET_INFO_EVENTS (
  -- Raw ETW DmaPacketInfo events.

  start                 INTEGER  NOT NULL,            -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,            -- Event end timestamp (ns).
  globalTid             INTEGER,                      -- Serialized GlobalId.
  gpu                   INTEGER  NOT NULL,            -- GPU index.
  context               INTEGER  NOT NULL,            -- The context ID of WDDM queue.
  uliCompletionId       INTEGER  NOT NULL,            -- The queue packet completion ID.
  faultedVirtualAddress INTEGER  NOT NULL,            -- The virtual address of faulted process.
  faultedProcessHandle  INTEGER  NOT NULL,            -- The address of faulted process.
  packetType            INTEGER  NOT NULL,            -- REFERENCES ENUM_WDDM_PACKET_TYPE(id)
  ulQueueSubmitSequence INTEGER  NOT NULL,            -- Internal sequence starting from 1.
  interruptType         INTEGER  NOT NULL,            -- REFERENCES ENUM_WDDM_INTERRUPT_TYPE(id)
  quantumStatus         INTEGER  NOT NULL,            -- The quantum Status.
  pageFaultFlags        INTEGER  NOT NULL,            -- The page fault flag ID.
```

```
  engineType          INTEGER  NOT NULL          -- REFERENCES ENUM_WDDM_ENGINE_TYPE(id)
);
CREATE TABLE WDDM_HW_QUEUE_EVENTS (
  -- Raw ETW HwQueueStart events.

  start               INTEGER  NOT NULL,         -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,         -- Event end timestamp (ns).
  globalTid           INTEGER,                   -- Serialized GlobalId.
  gpu                 INTEGER  NOT NULL,         -- GPU index.
  context             INTEGER  NOT NULL,         -- The context ID of WDDM queue.
  hwQueue             INTEGER  NOT NULL,         -- The address of HW queue.
  parentDxgHwQueue    INTEGER  NOT NULL          -- The address of parent Dxg HW queue.
);
CREATE TABLE NVVIDEO_ENCODER_API (
  -- NV Video Encoder API traced to gather information about NVIDIA Video Codek SDK Encoder APIs.

  start               INTEGER  NOT NULL,         -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,         -- Event end timestamp (ns).
  eventClass          INTEGER  NOT NULL,         -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                   -- Serialized GlobalId.
  nameId              INTEGER  NOT NULL,         -- REFERENCES StringIds(id) -- Function name
  apiId               INTEGER                    -- REFERENCES GPU_VIDEO_ENGINE_WORKLOAD(apiId)
);
CREATE TABLE NVVIDEO_DECODER_API (
  -- NV Video Encoder API traced to gather information about NVIDIA Video Codek SDK Decoder APIs.

  start               INTEGER  NOT NULL,         -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,         -- Event end timestamp (ns).
```

```
  eventClass          INTEGER   NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                     -- Serialized GlobalId.
  nameId              INTEGER   NOT NULL,          -- REFERENCES StringIds(id) -- Function name
  apiId               INTEGER                      -- REFERENCES GPU_VIDEO_ENGINE_WORKLOAD(apiId)
);
CREATE TABLE NVVIDEO_JPEG_API (
  -- NV Video Encoder API traced to gather information about NVIDIA Video Codek SDK JPEG APIs.

  start               INTEGER   NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER   NOT NULL,          -- Event end timestamp (ns).
  eventClass          INTEGER   NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                     -- Serialized GlobalId.
  nameId              INTEGER   NOT NULL           -- REFERENCES StringIds(id) -- Function name
);
CREATE TABLE GPU_VIDEO_ENGINE_WORKLOAD (
  -- Video engine workload events

  start               INTEGER   NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER   NOT NULL,          -- Event end timestamp (ns).
  globalEngineId      INTEGER   NOT NULL,          -- Serialized GlobalId.
  engineType          INTEGER   NOT NULL,          -- REFERENCES ENUM_VIDEO_ENGINE_TYPE(id)
  engineId            INTEGER   NOT NULL,
  vmId                INTEGER   NOT NULL,          -- Driver provided ID.
  contextId           INTEGER,                     -- Context ID.
  globalPid           INTEGER,                     -- Serialized GlobalId.
  apiId               INTEGER   NOT NULL,          -- ID used to correlate API and workload trace.
  codecId             INTEGER                      -- REFERENCES ENUM_VIDEO_ENGINE_CODEC(id)
);
```

```
CREATE TABLE GPU_VIDEO_ENGINE_MISSING (
  -- Video engine missing ranges

  start                INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,          -- Event end timestamp (ns).
  globalEngineId       INTEGER  NOT NULL,          -- Serialized GlobalId.
  rangeCount           INTEGER  NOT NULL           -- Number of missing ranges.
);
CREATE TABLE MEMORY_TRANSFER_EVENTS (
  -- Raw ETW Memory Transfer events.

  start                INTEGER  NOT NULL,          -- Event start timestamp (ns).
  globalTid            INTEGER,                    -- Serialized GlobalId.
  gpu                  INTEGER,                    -- GPU index.
  taskId               INTEGER  NOT NULL,          -- The event task ID.
  eventId              INTEGER  NOT NULL,          -- Event ID.
  allocationGlobalHandle  INTEGER  NOT NULL,       -- Address of the global allocation handle.
  dmaBuffer            INTEGER  NOT NULL,          -- The reported address of dma buffer.
  size                 INTEGER  NOT NULL,          -- The size of the dma buffer in bytes.
  offset               INTEGER  NOT NULL,          -- The offset from the start of the reported dma buffer in bytes.
  memoryTransferType   INTEGER  NOT NULL           -- REFERENCES
ENUM_ETW_MEMORY_TRANSFER_TYPE(id)
);
CREATE TABLE NV_LOAD_BALANCE_MASTER_EVENTS (
  -- Raw ETW NV-wgf2um LoadBalanceMaster events.

  start                INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                  INTEGER  NOT NULL,          -- Event end timestamp (ns).
```

```
  globalTid              INTEGER   NOT NULL,              -- Serialized GlobalId.
  gpu                    INTEGER   NOT NULL,              -- GPU index.
  eventId                INTEGER   NOT NULL,              -- Event ID.
  task                   TEXT      NOT NULL,              -- The task name.
  frameCount             INTEGER   NOT NULL,              -- The frame ID.
  frameTime              REAL      NOT NULL,              -- Frame duration.
  averageFrameTime       REAL      NOT NULL,              -- Average of frame duration.
  averageLatency         REAL      NOT NULL,              -- Average of latency.
  minLatency             REAL      NOT NULL,              -- The minimum latency.
  averageQueuedFrames    REAL      NOT NULL,              -- Average number of queued frames.
  totalActiveMs          REAL      NOT NULL,              -- Total active time in milliseconds.
  totalIdleMs            REAL      NOT NULL,              -- Total idle time in milliseconds.
  idlePercent            REAL      NOT NULL,              -- The percentage of idle time.
  isGPUAlmostOneFrameAhead   INTEGER   NOT NULL            -- True or False if GPU is almost one frame
ahead.
);
CREATE TABLE NV_LOAD_BALANCE_EVENTS (
  -- Raw ETW NV-wgf2um LoadBalance events.

  start                  INTEGER   NOT NULL,              -- Event start timestamp (ns).
  end                    INTEGER   NOT NULL,              -- Event end timestamp (ns).
  globalTid              INTEGER   NOT NULL,              -- Serialized GlobalId.
  gpu                    INTEGER   NOT NULL,              -- GPU index.
  eventId                INTEGER   NOT NULL,              -- Event ID.
  task                   TEXT      NOT NULL,              -- The task name.
  averageFPS             REAL      NOT NULL,              -- Average frame per second.
  queuedFrames           REAL      NOT NULL,              -- The amount of queued frames.
  averageQueuedFrames    REAL      NOT NULL,              -- Average number of queued frames.
```

```
  currentCPUTime          REAL     NOT NULL,              -- The current CPU time.
  averageCPUTime          REAL     NOT NULL,              -- Average CPU time.
  averageStallTime        REAL     NOT NULL,              -- Average of stall time.
  averageCPUIdleTime      REAL     NOT NULL,              -- Average CPU idle time.
  isGPUAlmostOneFrameAhead   INTEGER   NOT NULL              -- True or False if GPU is almost one frame
ahead.
);
CREATE TABLE PROCESSES (
  -- Names and identifiers of processes captured in the report.

  globalPid             INTEGER,                    -- Serialized GlobalId.
  pid                   INTEGER,                    -- The process ID.
  name                  TEXT                        -- The process name.
);
CREATE TABLE CUPTI_ACTIVITY_KIND_OPENACC_DATA (
  -- OpenACC data events collected using CUPTI.

  start                 INTEGER   NOT NULL,             -- Event start timestamp (ns).
  end                   INTEGER   NOT NULL,             -- Event end timestamp (ns).
  nameId                INTEGER   NOT NULL,             -- REFERENCES StringIds(id) -- Event name
  globalTid             INTEGER,                    -- Serialized GlobalId.
  eventKind             INTEGER   NOT NULL,             -- REFERENCES ENUM_OPENACC_EVENT_KIND(id)
  DeviceType            INTEGER   NOT NULL,             -- REFERENCES ENUM_OPENACC_DEVICE(id)
  lineNo                INTEGER   NOT NULL,             -- Line number of the directive or program construct.
  cuDeviceId            INTEGER   NOT NULL,             -- CUDA device ID. Valid only if deviceType is
acc_device_nvidia.
  cuContextId           INTEGER   NOT NULL,             -- CUDA context ID. Valid only if deviceType is
acc_device_nvidia.
```

```
  cuStreamId          INTEGER  NOT NULL,          -- CUDA stream ID. Valid only if deviceType is
acc_device_nvidia.
  srcFile             INTEGER,                    -- REFERENCES StringIds(id) -- Source file name or path
  funcName            INTEGER,                    -- REFERENCES StringIds(id) -- Function in which event occurred
  correlationId       INTEGER,                    -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  bytes               INTEGER,                    -- Number of bytes.
  varName             INTEGER                     -- REFERENCES StringIds(id) -- Variable name
);
CREATE TABLE CUPTI_ACTIVITY_KIND_OPENACC_LAUNCH (
  -- OpenACC launch events collected using CUPTI.

  start               INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,          -- Event end timestamp (ns).
  nameId              INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Event name
  globalTid           INTEGER,                    -- Serialized GlobalId.
  eventKind           INTEGER  NOT NULL,          -- REFERENCES ENUM_OPENACC_EVENT_KIND(id)
  DeviceType          INTEGER  NOT NULL,          -- REFERENCES ENUM_OPENACC_DEVICE(id)
  lineNo              INTEGER  NOT NULL,          -- Line number of the directive or program construct.
  cuDeviceId          INTEGER  NOT NULL,          -- CUDA device ID. Valid only if deviceType is
acc_device_nvidia.
  cuContextId         INTEGER  NOT NULL,          -- CUDA context ID. Valid only if deviceType is
acc_device_nvidia.
  cuStreamId          INTEGER  NOT NULL,          -- CUDA stream ID. Valid only if deviceType is
acc_device_nvidia.
  srcFile             INTEGER,                    -- REFERENCES StringIds(id) -- Source file name or path
  funcName            INTEGER,                    -- REFERENCES StringIds(id) -- Function in which event occurred
  correlationId       INTEGER,                    -- REFERENCES
```

```
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
  numGangs          INTEGER,                -- Number of gangs created for this kernel launch.
  numWorkers        INTEGER,                -- Number of workers created for this kernel launch.
  vectorLength      INTEGER,                -- Number of vector lanes created for this kernel launch.
  kernelName        INTEGER                 -- REFERENCES StringIds(id) -- Kernel name
);
CREATE TABLE CUPTI_ACTIVITY_KIND_OPENACC_OTHER (
  -- OpenACC other events collected using CUPTI.

  start             INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end               INTEGER  NOT NULL,      -- Event end timestamp (ns).
  nameId            INTEGER  NOT NULL,      -- REFERENCES StringIds(id) -- Event name
  globalTid         INTEGER,                -- Serialized GlobalId.
  eventKind         INTEGER  NOT NULL,      -- REFERENCES ENUM_OPENACC_EVENT_KIND(id)
  DeviceType        INTEGER  NOT NULL,      -- REFERENCES ENUM_OPENACC_DEVICE(id)
  lineNo            INTEGER  NOT NULL,      -- Line number of the directive or program construct.
  cuDeviceId        INTEGER  NOT NULL,      -- CUDA device ID. Valid only if deviceType is
acc_device_nvidia.
  cuContextId       INTEGER  NOT NULL,      -- CUDA context ID. Valid only if deviceType is
acc_device_nvidia.
  cuStreamId        INTEGER  NOT NULL,      -- CUDA stream ID. Valid only if deviceType is
acc_device_nvidia.
  srcFile           INTEGER,                -- REFERENCES StringIds(id) -- Source file name or path
  funcName          INTEGER,                -- REFERENCES StringIds(id) -- Function in which event occurred
  correlationId     INTEGER                 -- REFERENCES
CUPTI_ACTIVITY_KIND_RUNTIME(correlationId)
);
CREATE TABLE NET_NIC_METRIC (
```

```
  start                INTEGER   NOT NULL,              -- Event start timestamp (ns).
  end                  INTEGER   NOT NULL,              -- Event end timestamp (ns).
  globalId             INTEGER   NOT NULL,              -- Serialized GlobalId.
  portId               INTEGER   NOT NULL,              -- REFERENCES
NET_IB_DEVICE_PORT_INFO(portNumber) -- Port ID
  metricsListId        INTEGER   NOT NULL,              -- REFERENCES
TARGET_INFO_NETWORK_METRICS(metricsListId)
  metricsIdx           INTEGER   NOT NULL,              -- REFERENCES
TARGET_INFO_NETWORK_METRICS(metricsIdx)
  value                INTEGER   NOT NULL               -- Counter data value
);
CREATE TABLE NET_IB_SWITCH_METRIC (
  start                INTEGER   NOT NULL,              -- Event start timestamp (ns).
  end                  INTEGER   NOT NULL,              -- Event end timestamp (ns).
  globalId             INTEGER   NOT NULL,              -- Serialized GlobalId.
  portId               INTEGER   NOT NULL,              -- REFERENCES
NET_IB_DEVICE_PORT_INFO(portNumber) -- Port ID
  metricsListId        INTEGER   NOT NULL,              -- REFERENCES
TARGET_INFO_NETWORK_METRICS(metricsListId)
  metricsIdx           INTEGER   NOT NULL,              -- REFERENCES
TARGET_INFO_NETWORK_METRICS(metricsIdx)
  value                INTEGER   NOT NULL               -- Counter data value
);
CREATE TABLE NET_IB_SWITCH_CONGESTION_EVENT (
  start                INTEGER   NOT NULL,              -- Event start timestamp (ns).
  globalId             INTEGER   NOT NULL,              -- Serialized GlobalId (view in hex).
  congestionType       INTEGER,                         -- REFERENCES
ENUM_NET_IB_CONGESTION_EVENT_TYPE(id)
```

```
  packetSLID            INTEGER,                -- Packet Source LID
  packetDLID            INTEGER,                -- Packet Destination LID
  packetSL              INTEGER,                -- Packet Service Level
  packetOpCode          INTEGER,                -- Packet Operation Code
  packetSourceQP        INTEGER,                -- Packet Source Queue Pair
  packetDestinationQP   INTEGER,                -- Packet Destination Queue Pair
  switchIngressPort     INTEGER,                -- Packet's Ingress Switch Port
  switchEgressPort      INTEGER                 -- Packet's Egress Switch Port
);
CREATE TABLE PMU_EVENTS (
  -- CPU Core events.

  start                 INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,      -- Event end timestamp (ns).
  globalVm              INTEGER  NOT NULL,      -- Serialized GlobalId.
  cpu                   INTEGER  NOT NULL,      -- CPU ID
  counter_id            INTEGER                 -- REFERENCES PMU_EVENT_COUNTERS(id)
);
CREATE TABLE PMU_EVENT_COUNTERS (
  -- CPU Core events counters.

  id                    INTEGER  NOT NULL,
  idx                   INTEGER  NOT NULL,      -- REFERENCES PMU_EVENT_REQUESTS(id).
  value                 INTEGER  NOT NULL       -- Counter data value
);
CREATE TABLE TRACE_PROCESS_EVENT_NVMEDIA (
  start                 INTEGER  NOT NULL,      -- Event start timestamp (ns).
  end                   INTEGER  NOT NULL,      -- Event end timestamp (ns).
```

```
  eventClass          INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                    -- Serialized GlobalId.
  nameId              INTEGER  NOT NULL,          -- REFERENCES StringIds(id) -- Function name
  correlationId       INTEGER                     -- First ID matching an API call to GPU workloads.
);
CREATE TABLE TEGRA_INTERNAL_API_CALLS (
  start               INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,          -- Event end timestamp (ns).
  eventClass          INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid           INTEGER,                    -- Serialized GlobalId.
  nameId              INTEGER  NOT NULL           -- REFERENCES StringIds(id) -- Function name
);
CREATE TABLE UNCORE_PMU_EVENTS (
  -- PMU Uncore events.

  start               INTEGER  NOT NULL,          -- Event start timestamp (ns).
  end                 INTEGER  NOT NULL,          -- Event end timestamp (ns).
  globalVm            INTEGER  NOT NULL,          -- Serialized GlobalId.
  clusterId           INTEGER,                    -- Cluster ID.
  counterId           INTEGER                     -- REFERENCES UNCORE_PMU_EVENT_VALUES(id).
);
CREATE TABLE UNCORE_PMU_EVENT_VALUES (
  -- Uncore events values.

  id                  INTEGER  NOT NULL,
  type                INTEGER  NOT NULL,          -- REFERENCES ENUM_NSYS_EVENT_TYPE(id)
  value               INTEGER  NOT NULL,          -- Event value.
  rawId               INTEGER  NOT NULL,          -- Event value raw ID.
```

```
  clusterId          INTEGER                    -- Cluster ID.
);
CREATE TABLE DIAGNOSTIC_EVENT (
  timestamp          INTEGER  NOT NULL,         -- Event timestamp (ns).
  timestampType      INTEGER  NOT NULL,         -- REFERENCES
ENUM_DIAGNOSTIC_TIMESTAMP_SOURCE(id)
  source             INTEGER  NOT NULL,         -- REFERENCES
ENUM_DIAGNOSTIC_SOURCE_TYPE(id)
  severity           INTEGER  NOT NULL,         -- REFERENCES
ENUM_DIAGNOSTIC_SEVERITY_LEVEL(id)
  text               TEXT     NOT NULL,         -- Diagnostic message text
  globalPid          INTEGER                    -- Serialized GlobalId.
);
CREATE TABLE SYSCALL (
  start              INTEGER  NOT NULL,         -- Event start timestamp (ns).
  end                INTEGER  NOT NULL,         -- Event end timestamp (ns).
  eventClass         INTEGER  NOT NULL,         -- REFERENCES ENUM_NSYS_EVENT_CLASS(id)
  globalTid          INTEGER,                   -- Serialized GlobalId.
  nameId             INTEGER  NOT NULL,         -- REFERENCES StringIds(id) -- Function name
  callchainId        INTEGER  NOT NULL          -- REFERENCES SYSCALL_CALLCHAINS(id)
);
CREATE TABLE SYSCALL_CALLCHAINS (
  -- Callchains attached to syscall events, depending on selected profiling settings.

  id                 INTEGER  NOT NULL,         -- Part of PRIMARY KEY (id, stackDepth).
  stackDepth         INTEGER  NOT NULL,         -- Zero-base index of the given function in call stack.
  symbol             INTEGER  NOT NULL,         -- REFERENCES StringIds(id) -- Function name
  module             INTEGER  NOT NULL,         -- REFERENCES StringIds(id) -- Module name
```

```
  unresolved          INTEGER,                    -- True if the symbol was not resolved.

  PRIMARY KEY (id, stackDepth)
);
```

Note

GENERIC_EVENTS.typeId is a composite bit field that combines HW ID, VM ID, source ID, and type ID with the following structure:

<Hardware ID:8><VM ID:8><Source ID:16><Type ID:32>

The type ID is yet another composite bit field that combines the GPU metrics event tag and the GPU ID. To extract the latter, you need to get the lower 8 bits:

```
SELECT typeId & 0xFF AS gpuId FROM GENERIC_EVENTS
```

Some event types have been deprecated and are no longer supported by Nsight Systems. While tables for these event will no longer appear in exported SQL databases, databases exported by older versions of Nsight Systems may still contain them.

```
CREATE TABLE ETW_EVENTS_DEPRECATED_TABLE (
  [...]
);
CREATE TABLE GPU_MEMORY_BUDGET_EVENTS (
  -- Raw ETW VidMmProcessBudgetChange events (deprecated).

  [...]
);
CREATE TABLE GPU_MEMORY_USAGE_EVENTS (
  -- Raw ETW VidMmProcessUsageChange events (deprecated).

  [...]
```

```
);
CREATE TABLE DEMOTED_BYTES_EVENTS (
  -- Raw ETW VidMmProcessDemotedCommitmentChange events (deprecated).

  [...]
);
CREATE TABLE TOTAL_BYTES_RESIDENT_IN_SEGMENT_EVENTS (
  -- Raw ETW TotalBytesResidentInSegment events (deprecated).

  [...]
);
```

SQLite Schema Event Values#

Here are the set values stored in enums in the Nsight Systems SQLite schema

CUDA Memcopy Kind

0 - CUDA_MEMCPY_KIND_UNKNOWN
1 - CUDA_MEMCPY_KIND_HTOD
2 - CUDA_MEMCPY_KIND_DTOH
3 - CUDA_MEMCPY_KIND_HTOA
4 - CUDA_MEMCPY_KIND_ATOH
5 - CUDA_MEMCPY_KIND_ATOA
6 - CUDA_MEMCPY_KIND_ATOD
7 - CUDA_MEMCPY_KIND_DTOA
8 - CUDA_MEMCPY_KIND_DTOD
9 - CUDA_MEMCPY_KIND_HTOH
10 - CUDA_MEMCPY_KIND_PTOP
11 - CUDA_MEMCPY_KIND_UVM_HTOD

12 - CUDA_MEMCPY_KIND_UVM_DTOH
13 - CUDA_MEMCPY_KIND_UVM_DTOD

CUDA Memory Operations Memory Kind

0 - CUDA_MEMOPR_MEMORY_KIND_PAGEABLE
1 - CUDA_MEMOPR_MEMORY_KIND_PINNED
2 - CUDA_MEMOPR_MEMORY_KIND_DEVICE
3 - CUDA_MEMOPR_MEMORY_KIND_ARRAY
4 - CUDA_MEMOPR_MEMORY_KIND_MANAGED
5 - CUDA_MEMOPR_MEMORY_KIND_DEVICE_STATIC
6 - CUDA_MEMOPR_MEMORY_KIND_MANAGED_STATIC
7 - CUDA_MEMOPR_MEMORY_KIND_UNKNOWN

CUDA Event Class Values

0 - TRACE_PROCESS_EVENT_CUDA_RUNTIME
1 - TRACE_PROCESS_EVENT_CUDA_DRIVER
13 - TRACE_PROCESS_EVENT_CUDA_EGL_DRIVER
28 - TRACE_PROCESS_EVENT_CUDNN
29 - TRACE_PROCESS_EVENT_CUBLAS
33 - TRACE_PROCESS_EVENT_CUDNN_START
34 - TRACE_PROCESS_EVENT_CUDNN_FINISH
35 - TRACE_PROCESS_EVENT_CUBLAS_START
36 - TRACE_PROCESS_EVENT_CUBLAS_FINISH
67 - TRACE_PROCESS_EVENT_CUDABACKTRACE
77 - TRACE_PROCESS_EVENT_CUDA_GRAPH_NODE_CREATION

See CUPTI documentation for detailed information on collected event and data types.

NVTX Event Type Values

33 - NvtxCategory

34 - NvtxMark
39 - NvtxThread
59 - NvtxPushPopRange
60 - NvtxStartEndRange
75 - NvtxDomainCreate
76 - NvtxDomainDestroy

The difference between text and textId columns is that if an NVTX event message was passed via call to nvtxDomainRegisterString function, then the message will be available through textId field, otherwise the text field will contain the message if it was provided.

OpenGL Events

KHR event class values

62 - KhrDebugPushPopRange
63 - KhrDebugGpuPushPopRange

KHR source kind values

0x8249 - GL_DEBUG_SOURCE_THIRD_PARTY
0x824A - GL_DEBUG_SOURCE_APPLICATION

KHR type values

0x824C - GL_DEBUG_TYPE_ERROR
0x824D - GL_DEBUG_TYPE_DEPRECATED_BEHAVIOR
0x824E - GL_DEBUG_TYPE_UNDEFINED_BEHAVIOR
0x824F - GL_DEBUG_TYPE_PORTABILITY
0x8250 - GL_DEBUG_TYPE_PERFORMANCE
0x8251 - GL_DEBUG_TYPE_OTHER
0x8268 - GL_DEBUG_TYPE_MARKER
0x8269 - GL_DEBUG_TYPE_PUSH_GROUP

0x826A - GL_DEBUG_TYPE_POP_GROUP

KHR severity values

0x826B - GL_DEBUG_SEVERITY_NOTIFICATION
0x9146 - GL_DEBUG_SEVERITY_HIGH
0x9147 - GL_DEBUG_SEVERITY_MEDIUM
0x9148 - GL_DEBUG_SEVERITY_LOW

OSRT Event Class Values

OS runtime libraries can be traced to gather information about low-level userspace APIs. This traces the system call wrappers and thread synchronization interfaces exposed by the C runtime and POSIX Threads (pthread) libraries. This does not perform a complete runtime library API trace, but instead focuses on the functions that can take a long time to execute, or could potentially cause your thread be unscheduled from the CPU while waiting for an event to complete.

OSRT events may have callchains attached to them, depending on selected profiling settings. In such cases, one can use callchainId column to select relevant callchains from OSRT_CALLCHAINS table

OSRT event class values

27 - TRACE_PROCESS_EVENT_OS_RUNTIME
31 - TRACE_PROCESS_EVENT_OS_RUNTIME_START
32 - TRACE_PROCESS_EVENT_OS_RUNTIME_FINISH

DX12 Event Class Values

41 - TRACE_PROCESS_EVENT_DX12_API
42 - TRACE_PROCESS_EVENT_DX12_WORKLOAD
43 - TRACE_PROCESS_EVENT_DX12_START
44 - TRACE_PROCESS_EVENT_DX12_FINISH
52 - TRACE_PROCESS_EVENT_DX12_DISPLAY
59 - TRACE_PROCESS_EVENT_DX12_CREATE_OBJECT

PIX Event Class Values

65 - TRACE_PROCESS_EVENT_DX12_DEBUG_API
75 - TRACE_PROCESS_EVENT_DX11_DEBUG_API

Vulkan Event Class Values

53 - TRACE_PROCESS_EVENT_VULKAN_API
54 - TRACE_PROCESS_EVENT_VULKAN_WORKLOAD
55 - TRACE_PROCESS_EVENT_VULKAN_START
56 - TRACE_PROCESS_EVENT_VULKAN_FINISH
60 - TRACE_PROCESS_EVENT_VULKAN_CREATE_OBJECT
66 - TRACE_PROCESS_EVENT_VULKAN_DEBUG_API

Vulkan Flags

VALID_BIT = 0x00000001
CACHE_HIT_BIT = 0x00000002
BASE_PIPELINE_ACCELERATION_BIT = 0x00000004

WDDM Event Values

VIDMM operation type values

0 - None
101 - RestoreSegments
102 - PurgeSegments
103 - CleanupPrimary
104 - AllocatePagingBufferResources
105 - FreePagingBufferResources
106 - ReportVidMmState
107 - RunApertureCoherencyTest
108 - RunUnmapToDummyPageTest

109 - DeferredCommand
110 - SuspendMemorySegmentAccess
111 - ResumeMemorySegmentAccess
112 - EvictAndFlush
113 - CommitVirtualAddressRange
114 - UncommitVirtualAddressRange
115 - DestroyVirtualAddressAllocator
116 - PageInDevice
117 - MapContextAllocation
118 - InitPagingProcessVaSpace
200 - CloseAllocation
202 - ComplexLock
203 - PinAllocation
204 - FlushPendingGpuAccess
205 - UnpinAllocation
206 - MakeResident
207 - Evict
208 - LockInAperture
209 - InitContextAllocation
210 - ReclaimAllocation
211 - DiscardAllocation
212 - SetAllocationPriority
1000 - EvictSystemMemoryOfferList

Paging queue type values

0 - VIDMM_PAGING_QUEUE_TYPE_UMD
1 - VIDMM_PAGING_QUEUE_TYPE_Default
2 - VIDMM_PAGING_QUEUE_TYPE_Evict
3 - VIDMM_PAGING_QUEUE_TYPE_Reclaim

Packet type values

0 - DXGKETW_RENDER_COMMAND_BUFFER
1 - DXGKETW_DEFERRED_COMMAND_BUFFER
2 - DXGKETW_SYSTEM_COMMAND_BUFFER
3 - DXGKETW_MMIOFLIP_COMMAND_BUFFER
4 - DXGKETW_WAIT_COMMAND_BUFFER
5 - DXGKETW_SIGNAL_COMMAND_BUFFER
6 - DXGKETW_DEVICE_COMMAND_BUFFER
7 - DXGKETW_SOFTWARE_COMMAND_BUFFER

Engine type values

0 - DXGK_ENGINE_TYPE_OTHER
1 - DXGK_ENGINE_TYPE_3D
2 - DXGK_ENGINE_TYPE_VIDEO_DECODE
3 - DXGK_ENGINE_TYPE_VIDEO_ENCODE
4 - DXGK_ENGINE_TYPE_VIDEO_PROCESSING
5 - DXGK_ENGINE_TYPE_SCENE_ASSEMBLY
6 - DXGK_ENGINE_TYPE_COPY
7 - DXGK_ENGINE_TYPE_OVERLAY
8 - DXGK_ENGINE_TYPE_CRYPTO

DMA interrupt type values

1 = DXGK_INTERRUPT_DMA_COMPLETED
2 = DXGK_INTERRUPT_DMA_PREEMPTED
4 = DXGK_INTERRUPT_DMA_FAULTED
9 = DXGK_INTERRUPT_DMA_PAGE_FAULTED

Queue type values

0 = Queue_Packet

1 = Dma_Packet
2 = Paging_Queue_Packet

Driver Events

Load balance event type values

1 - LoadBalanceEvent_GPU
8 - LoadBalanceEvent_CPU
21 - LoadBalanceMasterEvent_GPU
22 - LoadBalanceMasterEvent_CPU

OpenMP Events

OpenMP event class values

78 - TRACE_PROCESS_EVENT_OPENMP
79 - TRACE_PROCESS_EVENT_OPENMP_START
80 - TRACE_PROCESS_EVENT_OPENMP_FINISH

OpenMP event kind values

15 - OPENMP_EVENT_KIND_TASK_CREATE
16 - OPENMP_EVENT_KIND_TASK_SCHEDULE
17 - OPENMP_EVENT_KIND_CANCEL
20 - OPENMP_EVENT_KIND_MUTEX_RELEASED
21 - OPENMP_EVENT_KIND_LOCK_INIT
22 - OPENMP_EVENT_KIND_LOCK_DESTROY
25 - OPENMP_EVENT_KIND_DISPATCH
26 - OPENMP_EVENT_KIND_FLUSH
27 - OPENMP_EVENT_KIND_THREAD
28 - OPENMP_EVENT_KIND_PARALLEL
29 - OPENMP_EVENT_KIND_SYNC_REGION_WAIT

30 - OPENMP_EVENT_KIND_SYNC_REGION
31 - OPENMP_EVENT_KIND_TASK
32 - OPENMP_EVENT_KIND_MASTER
33 - OPENMP_EVENT_KIND_REDUCTION
34 - OPENMP_EVENT_KIND_MUTEX_WAIT
35 - OPENMP_EVENT_KIND_CRITICAL_SECTION
36 - OPENMP_EVENT_KIND_WORKSHARE

OpenMP thread type values

1 - OpenMP Initial Thread
2 - OpenMP Worker Thread
3 - OpenMP Internal Thread
4 - Unknown

OpenMP sync region kind values

1 - Barrier
2 - Implicit barrier
3 - Explicit barrier
4 - Implementation-dependent barrier
5 - Taskwait
6 - Taskgroup

OpenMP task kind values

1 - Initial task
2 - Implicit task
3 - Explicit task

OpenMP prior task status values

1 - Task completed

2 - Task yielded to another task
3 - Task was cancelled
7 - Task was switched out for other reasons

OpenMP mutex kind values

1 - Waiting for lock
2 - Testing lock
3 - Waiting for nested lock
4 - Tesing nested lock
5 - Waitng for entering critical section region
6 - Waiting for entering atomic region
7 - Waiting for entering ordered region

OpenMP critical section kind values

5 - Critical section region
6 - Atomic region
7 - Ordered region

OpenMP workshare kind values

1 - Loop region
2 - Sections region
3 - Single region (executor)
4 - Single region (waiting)
5 - Workshare region
6 - Distrubute region
7 - Taskloop region

OpenMP dispatch kind values

1 - Iteration

Common SQLite Examples#

Common Helper Commands

When utilizing the sqlite3 command line tool, it's helpful to have data printed as named columns, this can be done with:

```
.mode column
.headers on
```

The default column width is determined by the data in the first row of results. If this doesn't work out well, you can specify widths manually.

```
.width 10 20 50
```

Obtaining Sample Report

The CLI interface of Nsight Systems was used to profile the radixSortThrust CUDA sample, then the resulting .nsys-rep file was exported using the nsys export.

```
nsys profile --trace=cuda,osrt radixSortThrust
nsys export --type sqlite report1.nsys-rep
```

Serialized Process and Thread Identifiers

Note

The globalTid field is a 64-bit identifier that encodes multiple components into a single value: <Hardware ID:8><VM ID:8><Process ID:24><Thread ID:24> It follows the structure: - Thread ID: bits 0–23 - Process ID: bits 24–47 - VM ID: bits 48–55 - Hardware ID: bits 56–63

Goal: Extract readable process ID (PID) and thread ID (TID) values from Nsight Systems's serialized identifier format.

Nsight Systems stores identifiers where events originated in serialized form to efficiently pack multiple values into a single field. This example shows how to decode them back to standard PID/TID format.

What the code does: Uses bit shifting and modulo operations to extract the embedded PID and TID values from the globalTid field.

For events that have globalTid or globalPid fields exported, use the following code to extract numeric TID and PID.

```
SELECT globalTid / 0x1000000 % 0x1000000 AS PID, globalTid % 0x1000000 AS TID FROM TABLE_NAME;
```

```
# Python equivalent:
def extract_pid_tid(global_tid):
    PID = (global_tid // 0x1000000) % 0x1000000
    TID = global_tid % 0x1000000
    return PID, TID
```

Note

globalTid field includes both TID and PID values, while globalPid only contains the PID value.

Understanding Event Types and Tags

Many tables in the nsys SQLite export use numeric codes for event types, tags, and classes. These codes represent different categories of events:

- **Event Types**: Identify the kind of operation or marker (e.g., NVTX marks vs ranges)
- **Tags**: Specify event phases like BEGIN/END for state transitions
- **Event Classes**: Categorize broad event categories (e.g., different types of GPU operations)

When you see numeric values in queries, refer to the documentation sections above or query the relevant string tables to understand their meanings.

Correlate CUDA Kernel Launches With CUDA API Kernel Launches

Goal: Link CUDA runtime API calls to the actual GPU kernels they launch, enabling analysis of which API calls

resulted in the longest-running kernels.

What the code does: 1. Adds human-readable columns to the runtime table 2. Joins runtime API calls with GPU kernel executions using correlation IDs 3. Populates kernel names and API function names from the string table 4. Finds the 10 longest API calls that resulted in kernel execution

```
ALTER TABLE CUPTI_ACTIVITY_KIND_RUNTIME ADD COLUMN name TEXT;
ALTER TABLE CUPTI_ACTIVITY_KIND_RUNTIME ADD COLUMN kernelName TEXT;

UPDATE CUPTI_ACTIVITY_KIND_RUNTIME SET kernelName =
  (SELECT value FROM StringIds
  JOIN CUPTI_ACTIVITY_KIND_KERNEL AS cuda_gpu
    ON cuda_gpu.shortName = StringIds.id
    AND CUPTI_ACTIVITY_KIND_RUNTIME.correlationId = cuda_gpu.correlationId);

UPDATE CUPTI_ACTIVITY_KIND_RUNTIME SET name =
  (SELECT value FROM StringIds WHERE nameId = StringIds.id);
```

Select the 10 longest CUDA API ranges that resulted in kernel execution.

```
SELECT name, kernelName, start, end FROM CUPTI_ACTIVITY_KIND_RUNTIME
  WHERE kernelName IS NOT NULL ORDER BY end - start LIMIT 10;
```

```
# Python equivalent workflow:
# 1. Load runtime and kernel data
# 2. Join on correlation_id to match API calls with kernels
# 3. Add readable names from string table
# 4. Sort by duration and get top 10

runtime_with_kernels = runtime.merge(kernels, on='correlationId')
runtime_with_kernels['duration'] = runtime_with_kernels['end'] - runtime_with_kernels['start']
```

```
top_10_longest = runtime_with_kernels.nlargest(10, 'duration')
```

Results:

```
name                 kernelName           start     end
--------------------- ---------------------- ---------- ----------
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 658863435  658868490
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 609755015  609760075
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 632683286  632688349
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 606495356  606500439
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 603114486  603119586
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 802729785  802734906
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 593381170  593386294
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 658759955  658765090
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 681549917  681555059
cudaLaunchKernel_v7000  RadixSortScanBinsKernel 717812527  717817671
```

Remove Ranges Overlapping With Overhead

Goal: Identify and remove CUDA API calls that overlap with profiler overhead to get cleaner performance measurements.

What the code does: Uses spatial overlap detection to find CUDA runtime ranges that intersect with profiler overhead periods. The query checks for three types of overlap: range starts within overhead, range ends within overhead, or range completely encompasses overhead.

Use the this query to count CUDA API ranges overlapping with the overhead ones.

Replace “SELECT COUNT(*)” with “DELETE” to remove such ranges.

```
SELECT COUNT(*) FROM CUPTI_ACTIVITY_KIND_RUNTIME WHERE rowid IN
(
  SELECT cuda.rowid
```

```
  FROM PROFILER_OVERHEAD as overhead
  INNER JOIN CUPTI_ACTIVITY_KIND_RUNTIME as cuda ON
  (cuda.start BETWEEN overhead.start and overhead.end)
  OR (cuda.end BETWEEN overhead.start and overhead.end)
  OR (cuda.start < overhead.start AND cuda.end > overhead.end)
);

# Python equivalent for finding overlaps:
def ranges_overlap(range1_start, range1_end, range2_start, range2_end):
  return (range1_start <= range2_end and range1_end >= range2_start)

overlapping_ranges = []
for cuda_range in cuda_ranges:
  for overhead_range in overhead_ranges:
    if ranges_overlap(cuda_range.start, cuda_range.end,
                      overhead_range.start, overhead_range.end):
      overlapping_ranges.append(cuda_range)
```

Results:

```
COUNT(*)
----------
1095
```

Find CUDA API Calls that Resulted in the Original Graph Node Creation

Goal: Identify which CUDA API calls were responsible for creating the original nodes in CUDA graphs (as opposed to cloned or instantiated nodes).

What the code does: 1. Filters graph nodes to find only original creations (those without originalGraphNodeId) 2. Groups by graphNodeId to get the first occurrence 3. Correlates with CUDA runtime API calls that were active when the graph node was created 4. Joins with string table to get readable API function names

```
SELECT graph.graphNodeId, api.start, graph.start as graphStart, api.end,
  api.globalTid, api.correlationId, api.globalTid,
  (SELECT value FROM StringIds where api.nameId == id) as name
FROM CUPTI_ACTIVITY_KIND_RUNTIME as api
JOIN
  (
    SELECT start, graphNodeId, globalTid from CUDA_GRAPH_NODE_EVENTS
    GROUP BY graphNodeId
    HAVING COUNT(originalGraphNodeId) = 0
  ) as graph
ON api.globalTid == graph.globalTid AND api.start < graph.start AND api.end > graph.start
ORDER BY graphNodeId;
```

Results:

```
graphNodeId  start      graphStart  end        globalTid       correlationId  globalTid       name
-----------  ----------  ----------  ----------  ---------------  -------------  ---------------  ------------------------------
1            584366518   584378040   584379102   281560221750233  109            281560221750233
cudaGraphAddMemcpyNode_v10000
2            584379402   584382428   584383139   281560221750233  110            281560221750233
cudaGraphAddMemsetNode_v10000
3            584390663   584395352   584396053   281560221750233  111            281560221750233
cudaGraphAddKernelNode_v10000
4            584396314   584397857   584398438   281560221750233  112            281560221750233
cudaGraphAddMemsetNode_v10000
5            584398759   584400311   584400812   281560221750233  113            281560221750233
cudaGraphAddKernelNode_v10000
6            584401083   584403047   584403527   281560221750233  114            281560221750233
cudaGraphAddMemcpyNode_v10000
7            584403928   584404920   584405491   281560221750233  115            281560221750233
```

cudaGraphAddHostNode_v10000
29 632107852 632117921 632121407 281560221750233 144 281560221750233
cudaMemcpyAsync_v3020
30 632122168 632125545 632127989 281560221750233 145 281560221750233
cudaMemsetAsync_v3020
31 632131546 632133339 632135584 281560221750233 147 281560221750233
cudaMemsetAsync_v3020
34 632162514 632167393 632169297 281560221750233 151 281560221750233
cudaMemcpyAsync_v3020
35 632170068 632173334 632175388 281560221750233 152 281560221750233
cudaLaunchHostFunc_v10000

Backtraces for OSRT Ranges

Goal: Analyze operating system runtime (OSRT) function calls with their full call stacks to understand performance bottlenecks and call patterns.

What the code does: 1. Adds human-readable columns for function names, symbol names, and module names 2. Populates these from the string table for better readability 3. Shows how to query the longest OSRT call with its complete backtrace

Adding text columns makes results of the query below more human-readable.

```
ALTER TABLE OSRT_API ADD COLUMN name TEXT;
UPDATE OSRT_API SET name = (SELECT value FROM StringIds WHERE OSRT_API.nameId = StringIds.id);

ALTER TABLE OSRT_CALLCHAINS ADD COLUMN symbolName TEXT;
UPDATE OSRT_CALLCHAINS SET symbolName = (SELECT value FROM StringIds WHERE symbol =
StringIds.id);

ALTER TABLE OSRT_CALLCHAINS ADD COLUMN moduleName TEXT;
```

```
UPDATE OSRT_CALLCHAINS SET moduleName = (SELECT value FROM StringIds WHERE module =
StringIds.id);
```

Print backtrace of the longest OSRT range.

```
SELECT globalTid / 0x1000000 % 0x1000000 AS PID, globalTid % 0x1000000 AS TID,
   start, end, name, callchainId, stackDepth, symbolName, moduleName
FROM OSRT_API LEFT JOIN OSRT_CALLCHAINS ON callchainId == OSRT_CALLCHAINS.id
WHERE OSRT_API.rowid IN (SELECT rowid FROM OSRT_API ORDER BY end - start DESC LIMIT 1)
ORDER BY stackDepth LIMIT 10;
```

```
# Python equivalent for finding longest call with backtrace:
longest_call = osrt_api.loc[osrt_api['duration'].idxmax()]
backtrace = osrt_callchains[osrt_callchains['id'] == longest_call['callchainId']]
backtrace_ordered = backtrace.sort_values('stackDepth')
```

Results:

```
PID        TID        start      end        name                 callchainId  stackDepth  symbolName                  moduleName
---------- ---------- ---------- ---------- -------------------- ----------- ---------- ------------------------------
---------------------------------------
19163      19176      360897690  860966851  pthread_cond_timedwait 88        0
pthread_cond_timedwait@GLIBC_2  /lib/x86_64-linux-gnu/libpthread-2.27.so
19163      19176      360897690  860966851  pthread_cond_timedwait 88        1        0x7fbc983b7227              /
usr/lib/x86_64-linux-gnu/libcuda.so.418
19163      19176      360897690  860966851  pthread_cond_timedwait 88        2        0x7fbc9835d5c7              /
usr/lib/x86_64-linux-gnu/libcuda.so.418
19163      19176      360897690  860966851  pthread_cond_timedwait 88        3        0x7fbc983b64a8              /
usr/lib/x86_64-linux-gnu/libcuda.so.418
19163      19176      360897690  860966851  pthread_cond_timedwait 88        4        start_thread                /lib/
x86_64-linux-gnu/libpthread-2.27.so
```

19163 19176 360897690 860966851 pthread_cond_timedwait 88 5 __clone /lib/x86_64-linux-gnu/libc-2.27.so

Profiled processes output streams.

Goal: Access stdout and stderr output from profiled processes to correlate application output with performance data.

What the code does: Resolves file paths and content from string IDs to show the captured stdout/stderr streams from profiled applications.

```
ALTER TABLE ProcessStreams ADD COLUMN filename TEXT;
UPDATE ProcessStreams SET filename = (SELECT value FROM StringIds WHERE ProcessStreams.filenameId = StringIds.id);

ALTER TABLE ProcessStreams ADD COLUMN content TEXT;
UPDATE ProcessStreams SET content = (SELECT value FROM StringIds WHERE ProcessStreams.contentId = StringIds.id);
```

Select all collected stdout and stderr streams.

```
select globalPid / 0x1000000 % 0x1000000 AS PID, filename, content from ProcessStreams;
```

Results:

```
PID       filename                                  content
----------  ----------------------------------------------------
-----------------------------------------------------------------------------------------------------------------
19163     /tmp/nvidia/nsight_systems/streams/pid_19163_stdout.log  /home/user_name/NVIDIA_CUDA-10.1_Samples/6_Advanced/radixSortThrust/radixSortThrust Starting...

GPU Device 0: "Quadro P2000" with compute capability 6.1
```

Sorting 1048576 32-bit unsigned int keys and values

radixSortThrust, Throughput = 401.0872 MElements/s, Time = 0.00261 s, Size = 1048576 elements
Test passed

19163 /tmp/nvidia/nsight_systems/streams/pid_19163_stderr.log

Thread Summary

Goal: Calculate CPU utilization statistics per thread to identify which threads are consuming the most CPU resources.

Note that Nsight Systems applies additional logic during sampling events processing to work around lost events. This means that the results of the below query might differ slightly from the ones shown in “Analysis Summary” tab.

Approach 1: Using CPU Cycles (when available)

What this code does: Calculates thread CPU utilization using hardware performance counter data (CPU cycles) which provides the most accurate measurement of actual CPU usage per thread.

```
SELECT
  globalTid / 0x1000000 % 0x1000000 AS PID,
  globalTid % 0x1000000 AS TID,
  ROUND(100.0 * SUM(cpuCycles) /
    (
      SELECT SUM(cpuCycles) FROM COMPOSITE_EVENTS
      GROUP BY globalTid / 0x1000000000000 % 0x100
    ),
    2
  ) as CPU_utilization,
  (SELECT value FROM StringIds WHERE id =
    (
```

```
        SELECT nameId FROM ThreadNames
        WHERE ThreadNames.globalTid = COMPOSITE_EVENTS.globalTid
      )
  ) as thread_name
FROM COMPOSITE_EVENTS
GROUP BY globalTid
ORDER BY CPU_utilization DESC
LIMIT 10;
```

Results:

```
PID       TID       CPU_utilization  thread_name
----------  ----------  ---------------  ---------------
19163     19163     98.4          radixSortThrust
19163     19168     1.35          CUPTI worker th
19163     19166     0.25          [NS]
```

Approach 2: Using Scheduling Events (when PMU data not available)

What this approach does: When CPU cycle counter data is not collected, this method calculates thread CPU time based on scheduling events (when threads are scheduled in/out), then calculates utilization percentages. This approach is less precise but still useful for understanding relative thread activity.

```
CREATE INDEX sched_start ON SCHED_EVENTS (start);

CREATE TABLE CPU_USAGE AS
SELECT
  first.globalTid as globalTid,
  (SELECT nameId FROM ThreadNames WHERE ThreadNames.globalTid = first.globalTid) as nameId,
  sum(second.start - first.start) as total_duration,
  count() as ranges_count
```

```
FROM SCHED_EVENTS as first
LEFT JOIN SCHED_EVENTS as second
ON second.rowid =
  (
    SELECT rowid
    FROM SCHED_EVENTS
    WHERE start > first.start AND globalTid = first.globalTid
    ORDER BY start ASC
    LIMIT 1
  )
WHERE first.isSchedIn != 0
GROUP BY first.globalTid
ORDER BY total_duration DESC;

SELECT
  globalTid / 0x1000000 % 0x1000000 AS PID,
  globalTid % 0x1000000 AS TID,
  (SELECT value FROM StringIds where nameId == id) as thread_name,
  ROUND(100.0 * total_duration / (SELECT SUM(total_duration) FROM CPU_USAGE), 2) as CPU_utilization
FROM CPU_USAGE
ORDER BY CPU_utilization DESC;

# Python equivalent for scheduling-based calculation:
def calculate_thread_cpu_time(sched_events):
  cpu_usage = {}
  for tid in unique_tids:
    tid_events = sched_events[sched_events['globalTid'] == tid]
    tid_events = tid_events[tid_events['isSchedIn'] == 1]  # Only sched-in events
```

```
    total_time = 0
    for i in range(len(tid_events) - 1):
        time_slice = tid_events.iloc[i+1]['start'] - tid_events.iloc[i]['start']
        total_time += time_slice

    cpu_usage[tid] = total_time
  return cpu_usage
```

Results:

```
PID       TID       thread_name     CPU_utilization
----------  ----------  ---------------  ---------------
19163     19163     radixSortThrust  93.74
19163     19169     radixSortThrust  3.22
19163     19168     CUPTI worker th  2.46
19163     19166     [NS]          0.44
19163     19172     radixSortThrust  0.07
19163     19167     [NS Comms]      0.05
19163     19176     radixSortThrust  0.02
19163     19170     radixSortThrust  0.0
```

Function Table

Goal: Create profiler-style function tables showing flat view (total time in each function across all call stacks) and bottom-up view (time spent directly in each function).

What the code does: Processes sampling callchain data to calculate time spent in functions, providing two views commonly used in profilers.

These examples demonstrate how to calculate Flat and BottomUp (for top level only) views statistics.

To set up:

```
ALTER TABLE SAMPLING_CALLCHAINS ADD COLUMN symbolName TEXT;
UPDATE SAMPLING_CALLCHAINS SET symbolName = (SELECT value FROM StringIds WHERE symbol =
StringIds.id);

ALTER TABLE SAMPLING_CALLCHAINS ADD COLUMN moduleName TEXT;
UPDATE SAMPLING_CALLCHAINS SET moduleName = (SELECT value FROM StringIds WHERE module =
StringIds.id);
```

To get flat view:

Flat view: Shows total time spent in each function across all call stacks (inclusive time).

```
SELECT symbolName, moduleName, ROUND(100.0 * sum(cpuCycles) /
  (SELECT SUM(cpuCycles) FROM COMPOSITE_EVENTS), 2) AS flatTimePercentage
FROM SAMPLING_CALLCHAINS
LEFT JOIN COMPOSITE_EVENTS ON SAMPLING_CALLCHAINS.id == COMPOSITE_EVENTS.id
GROUP BY symbol, module
ORDER BY flatTimePercentage DESC
LIMIT 5;
```

To get BottomUp view (top level only):

Bottom-up view: Shows time spent directly in each function (exclusive time, only leaf functions in call stacks).

```
SELECT symbolName, moduleName, ROUND(100.0 * sum(cpuCycles) /
  (SELECT SUM(cpuCycles) FROM COMPOSITE_EVENTS), 2) AS selfTimePercentage
FROM SAMPLING_CALLCHAINS
LEFT JOIN COMPOSITE_EVENTS ON SAMPLING_CALLCHAINS.id == COMPOSITE_EVENTS.id
WHERE stackDepth == 0
GROUP BY symbol, module
ORDER BY selfTimePercentage DESC
LIMIT 5;
```

```
# Python equivalent:
# Flat view - aggregate all occurrences of each function
flat_view = callchains.groupby(['symbol', 'module'])['cpuCycles'].sum()
flat_percentages = (flat_view / total_cycles * 100).sort_values(ascending=False)

# Bottom-up view - only leaf nodes (stackDepth == 0)
leaf_functions = callchains[callchains['stackDepth'] == 0]
bottomup_view = leaf_functions.groupby(['symbol', 'module'])['cpuCycles'].sum()
```

Results:

```
symbolName   moduleName   flatTimePercentage
-----------  -----------  ------------------
[Max depth]  [Max depth]  99.92
thrust::zip  /home/user_  24.17
thrust::zip  /home/user_  24.17
thrust::det  /home/user_  24.17
thrust::det  /home/user_  24.17
symbolName     moduleName                                  selfTimePercentage
--------------  ------------------------------------------  ------------------
0x7fbc984982b6  /usr/lib/x86_64-linux-gnu/libcuda.so.418.39  5.29
0x7fbc982d0010  /usr/lib/x86_64-linux-gnu/libcuda.so.418.39  2.81
thrust::iterat  /home/user_name/NVIDIA_CUDA-10.1_Samples/6_  2.23
thrust::iterat  /home/user_name/NVIDIA_CUDA-10.1_Samples/6_  1.55
void thrust::i  /home/user_name/NVIDIA_CUDA-10.1_Samples/6_  1.55
```

DX12 API Frame Duration Histogram

Goal: Analyze DirectX 12 application frame timing by measuring the duration between consecutive Present calls and creating a timing histogram.

What the code does: 1. Creates a view that pairs consecutive Present calls to calculate frame durations 2. Groups frame durations into millisecond buckets 3. Counts how many frames fall into each duration bucket

The example demonstrates how to calculate DX12 CPU frames durartion and construct a histogram out of it.

```
CREATE INDEX DX12_API_ENDTS ON DX12_API (end);

CREATE TEMP VIEW DX12_API_FPS AS SELECT end AS start,
  (SELECT end FROM DX12_API
    WHERE end > outer.end AND nameId == (SELECT id FROM StringIds
      WHERE value == "IDXGISwapChain::Present")
    ORDER BY end ASC LIMIT 1) AS end
FROM DX12_API AS outer
  WHERE nameId == (SELECT id FROM StringIds WHERE value == "IDXGISwapChain::Present")
ORDER BY end;
```

Number of frames with a duration of [X, X + 1] milliseconds.

```
SELECT
  CAST((end - start) / 1000000.0 AS INT) AS duration_ms,
  count(*)
FROM DX12_API_FPS
WHERE end IS NOT NULL
GROUP BY duration_ms
ORDER BY duration_ms;
```

```
# Python equivalent:
present_calls = dx12_api[dx12_api['function_name'] == 'IDXGISwapChain::Present']
present_calls = present_calls.sort_values('end')

frame_durations = []
```

```
for i in range(len(present_calls) - 1):
    duration = present_calls.iloc[i+1]['end'] - present_calls.iloc[i]['end']
    duration_ms = duration / 1000000.0  # Convert to milliseconds
    frame_durations.append(int(duration_ms))

# Create histogram
histogram = pd.Series(frame_durations).value_counts().sort_index()
```

Results:

```
duration_ms  count(*)
-----------  ----------
3            1
4            2
5            7
6            153
7            19
8            116
9            16
10           8
11           2
12           2
13           1
14           4
16           3
17           2
18           1
```

GPU Context Switch Events Enumeration

Goal: Track GPU context switches to understand GPU scheduling behavior and identify context switch patterns.

What the code does: Filters GPU context switch events to show only BEGIN (tag=8) and END (tag=7) events, which mark the boundaries of GPU context execution periods.

GPU Context Switch Event Tags: - **7**: END events (context execution ends) - **8**: BEGIN events (context execution begins)

GPU context duration is between first BEGIN and a matching END event.

```
SELECT (CASE tag WHEN 8 THEN "BEGIN" WHEN 7 THEN "END" END) AS tag,
  globalPid / 0x1000000 % 0x1000000 AS PID,
  vmId, seqNo, contextId, timestamp, gpuId FROM GPU_CONTEXT_SWITCH_EVENTS
WHERE tag in (7, 8) ORDER BY seqNo LIMIT 10;
```

Results:

tag	PID	vmId	seqNo	contextId	timestamp	gpuId
BEGIN	23371	0	0	1048578	56759171	0
BEGIN	23371	0	1	1048578	56927765	0
BEGIN	23371	0	3	1048578	63799379	0
END	23371	0	4	1048578	63918806	0
BEGIN	19397	0	5	1048577	64014692	0
BEGIN	19397	0	6	1048577	64250369	0
BEGIN	19397	0	8	1048577	1918310004	0
END	19397	0	9	1048577	1918521098	0
BEGIN	19397	0	10	1048577	2024164744	0
BEGIN	19397	0	11	1048577	2024358650	0

Resolve NVTX Category Name

Goal: Decode NVTX category names for NVTX markers and ranges to make the profiling data more human-readable.

What the code does: Joins NVTX events with their category definitions to resolve category IDs into meaningful category names, making it easier to understand the purpose of different NVTX annotations.

NVTX Event Types: - **33**: Category definition events (define new categories) - **34**: Mark events (instantaneous markers) - **59**: Push/Pop range events (nested ranges) - **60**: Start/End range events (paired ranges)

The example demonstrates how to resolve NVTX category name for NVTX marks and ranges.

```
WITH
 event AS (
  SELECT *
  FROM NVTX_EVENTS
  WHERE eventType IN (34, 59, 60) -- mark, push/pop, start/end
 ),
 category AS (
  SELECT
   category,
   domainId,
   text AS categoryName
  FROM NVTX_EVENTS
  WHERE eventType == 33 -- category definition events
 )
SELECT
 start,
 end,
 globalTid,
 eventType,
 domainId,
 category,
 categoryName,
```

```
 text
FROM event JOIN category USING (category, domainId)
ORDER BY start;
```

Results:

```
start     end       globalTid      eventType  domainId   category   categoryName           text
---------- ---------- --------------- ---------- ---------- ---------- ------------------------- ----------------
18281150   18311960   281534938484214  59        0         1         FirstCategoryUnderDefault  Push Pop Range A
18288187   18306674   281534938484214  59        0         2         SecondCategoryUnderDefaul  Push Pop Range
B
18294247            281534938484214  34        0         1         FirstCategoryUnderDefault  Mark A
18300034            281534938484214  34        0         2         SecondCategoryUnderDefaul  Mark B
18345546   18372595   281534938484214  60        1         1         FirstCategoryUnderMyDomai  Start End Range
18352924   18378342   281534938484214  60        1         2         SecondCategoryUnderMyDoma  Start End
Range
18359634            281534938484214  34        1         1         FirstCategoryUnderMyDomai  Mark A
18365448            281534938484214  34        1         2         SecondCategoryUnderMyDoma  Mark B
```

Rename CUDA Kernels with NVTX

Goal: Associate CUDA kernels with their surrounding NVTX ranges to provide more meaningful names and context for kernel analysis.

What the code does: 1. Finds the innermost NVTX push-pop range that encompasses each CUDA kernel launch 2. Maps the NVTX range text to the corresponding kernel execution 3. Enables analysis of kernels by their logical function rather than just their technical names

The example demonstrates how to map innermost NVTX push-pop range to a matching CUDA kernel run.

```
ALTER TABLE CUPTI_ACTIVITY_KIND_KERNEL ADD COLUMN nvtxRange TEXT;
CREATE INDEX nvtx_start ON NVTX_EVENTS (start);
```

```
UPDATE CUPTI_ACTIVITY_KIND_KERNEL SET nvtxRange = (
  SELECT NVTX_EVENTS.text
  FROM NVTX_EVENTS JOIN CUPTI_ACTIVITY_KIND_RUNTIME ON
    NVTX_EVENTS.eventType == 59 AND
    NVTX_EVENTS.globalTid == CUPTI_ACTIVITY_KIND_RUNTIME.globalTid AND
    NVTX_EVENTS.start <= CUPTI_ACTIVITY_KIND_RUNTIME.start AND
    NVTX_EVENTS.end >= CUPTI_ACTIVITY_KIND_RUNTIME.end
  WHERE
    CUPTI_ACTIVITY_KIND_KERNEL.correlationId == CUPTI_ACTIVITY_KIND_RUNTIME.correlationId
  ORDER BY NVTX_EVENTS.start DESC LIMIT 1
);

SELECT start, end, globalPid, StringIds.value as shortName, nvtxRange
FROM CUPTI_ACTIVITY_KIND_KERNEL JOIN StringIds ON shortName == id
ORDER BY start LIMIT 6;
```

```
# Python equivalent:
def find_innermost_nvtx_range(kernel_start, kernel_end, nvtx_ranges):
  # Find NVTX ranges that completely contain the kernel
  containing_ranges = []
  for nvtx in nvtx_ranges:
    if nvtx['start'] <= kernel_start and nvtx['end'] >= kernel_end:
      containing_ranges.append(nvtx)

  # Return the innermost (latest starting) range
  if containing_ranges:
    return max(containing_ranges, key=lambda x: x['start'])['text']
  return None
```

Results:

```
start     end       globalPid         shortName     nvtxRange
---------- ---------- ----------------- ------------- ----------
526545376  526676256  72057700439031808  MatrixMulCUDA
526899648  527030368  72057700439031808  MatrixMulCUDA  Add
527031648  527162272  72057700439031808  MatrixMulCUDA  Add
527163584  527294176  72057700439031808  MatrixMulCUDA  My Kernel
527296160  527426592  72057700439031808  MatrixMulCUDA  My Range
527428096  527558656  72057700439031808  MatrixMulCUDA
```

Select CUDA Calls With Backtraces

Goal: Analyze CUDA API calls along with their call stacks to understand the application code paths that lead to CUDA API usage.

What the code does: Joins CUDA runtime API calls with their associated call chains to show the complete stack trace for each CUDA call, helping identify where in the application CUDA calls originate.

```
ALTER TABLE CUPTI_ACTIVITY_KIND_RUNTIME ADD COLUMN name TEXT;
UPDATE CUPTI_ACTIVITY_KIND_RUNTIME SET name = (SELECT value FROM StringIds WHERE
CUPTI_ACTIVITY_KIND_RUNTIME.nameId = StringIds.id);

ALTER TABLE CUDA_CALLCHAINS ADD COLUMN symbolName TEXT;
UPDATE CUDA_CALLCHAINS SET symbolName = (SELECT value FROM StringIds WHERE symbol =
StringIds.id);

SELECT globalTid % 0x1000000 AS TID,
  start, end, name, callchainId, stackDepth, symbolName
FROM CUDA_CALLCHAINS JOIN CUPTI_ACTIVITY_KIND_RUNTIME ON callchainId ==
CUDA_CALLCHAINS.id
```

```
ORDER BY callchainId, stackDepth LIMIT 11;
```

Results:

TID	start	end	name	callchainId	stackDepth	symbolName
11928	168976467	169077826	cuMemAlloc_v2	1	0	0x7f13c44f02ab
11928	168976467	169077826	cuMemAlloc_v2	1	1	0x7f13c44f0b8f
11928	168976467	169077826	cuMemAlloc_v2	1	2	0x7f13c44f3719
11928	168976467	169077826	cuMemAlloc_v2	1	3	cuMemAlloc_v2
11928	168976467	169077826	cuMemAlloc_v2	1	4	cudart::driver
11928	168976467	169077826	cuMemAlloc_v2	1	5	cudart::cudaAp
11928	168976467	169077826	cuMemAlloc_v2	1	6	cudaMalloc
11928	168976467	169077826	cuMemAlloc_v2	1	7	cudaError cuda
11928	168976467	169077826	cuMemAlloc_v2	1	8	main
11928	168976467	169077826	cuMemAlloc_v2	1	9	__libc_start_m
11928	168976467	169077826	cuMemAlloc_v2	1	10	_start

SLI Peer-to-Peer Query

Goal: Filter and analyze SLI (Scalable Link Interface) peer-to-peer memory transfers between GPUs based on size, timing, and other criteria.

What the code does: Demonstrates how to query SLI P2P events with filtering conditions on resource size, time range, and sorting by transfer size to identify significant GPU-to-GPU transfers.

SLI P2P Event Classes: - **62**: Peer-to-peer transfer events between GPUs

The example demonstrates how to query SLI Peer-to-Peer events with resource size greater than value and within a time range sorted by resource size descending.

```
SELECT *
FROM SLI_P2P
```

```
WHERE resourceSize < 98304 AND start > 1568063100 AND end < 1579468901
ORDER BY resourceSize DESC;

# Python equivalent:
filtered_transfers = sli_p2p[
   (sli_p2p['resourceSize'] < 98304) &
   (sli_p2p['start'] > 1568063100) &
   (sli_p2p['end'] < 1579468901)
].sort_values('resourceSize', ascending=False)
```

Results:

```
start      end        eventClass globalTid       gpu        frameId    transferSkipped srcGpu     dstGpu
numSubResources resourceSize subResourceIdx smplWidth  smplHeight smplDepth  bytesPerElement
dxgiFormat logSurfaceNames transferInfo isEarlyPushManagedByNvApi useAsyncP2pForResolve
transferFuncName regimeName debugName  bindType
---------- ---------- ---------- ---------------- ---------- ---------- --------------- ---------- ---------- --------------- ------------
-------------- ---------- ---------- ---------- --------------- ---------- --------------- ------------ ------------------------- ---------------------
---------------- ---------- ---------- ----------
1570351100 1570351101 62       72057698056667136 0        771      0          256      512      1
1048576    0          256      256      1        16         2                3        0                0
1570379300 1570379301 62       72057698056667136 0        771      0          256      512      1
1048576    0          64       64       64       4          31               3        0                0
1572316400 1572316401 62       72057698056667136 0        773      0          256      512      1
1048576    0          256      256      1        16         2                3        0                0
1572345400 1572345401 62       72057698056667136 0        773      0          256      512      1
1048576    0          64       64       64       4          31               3        0                0
1574734300 1574734301 62       72057698056667136 0        775      0          256      512      1
1048576    0          256      256      1        16         2                3        0                0
1574767200 1574767201 62       72057698056667136 0        775      0          256      512      1
```

1048576	0	64	64	64	4	31	3	0	0

Generic Events

Goal: Analyze system-level events captured through generic event collection (like ftrace) to understand system behavior and syscall patterns.

What the code does: Demonstrates how to query generic events stored in JSON format, specifically showing how to create a histogram of syscall usage by process ID. The query uses a subquery to find the specific event type ID for "raw_syscalls:sys_enter" events, then counts occurrences by process ID.

Syscall usage histogram by PID:

```
SELECT json_extract(data, '$.common_pid') AS PID, count(*) AS total
FROM GENERIC_EVENTS WHERE PID IS NOT NULL AND typeId = (
  SELECT typeId FROM GENERIC_EVENT_TYPES
  WHERE json_extract(data, '$.Name') = "raw_syscalls:sys_enter")
GROUP BY PID
ORDER BY total DESC
LIMIT 10;
```

```
# Python equivalent:
import json

# Filter for syscall enter events
syscall_events = []
for event in generic_events:
    data = json.loads(event['data'])
    if 'common_pid' in data:
        syscall_events.append(data['common_pid'])

# Count syscalls by PID
```

```
pid_counts = pd.Series(syscall_events).value_counts().head(10)
```

Results:

```
PID       total
----------  ----------
5551      32811
9680      3988
4328      1477
9564      1246
4376      1204
4377      1167
4357      656
4355      655
4356      640
4354      633
```

Fetching Generic Events in JSON Format

Goal: Export generic events, types, and sources in JSON format for external analysis tools or custom processing pipelines.

What the code does: Constructs JSON objects from the database tables containing generic event data, enabling export to JSON Lines format for further processing with external tools.

Text and JSON export modes don't include generic events. Use the below queries (without the LIMIT clause) to extract JSON lines representation of generic events, types, and sources.

```
SELECT json_insert('{}',
   '$.sourceId', sourceId,
   '$.data', json(data)
)
FROM GENERIC_EVENT_SOURCES LIMIT 2;
```

```
SELECT json_insert('{}',
   '$.typeId', typeId,
   '$.sourceId', sourceId,
   '$.data', json(data)
)
FROM GENERIC_EVENT_TYPES LIMIT 2;

SELECT json_insert('{}',
   '$.rawTimestamp', rawTimestamp,
   '$.timestamp', timestamp,
   '$.typeId', typeId,
   '$.data', json(data)
)
FROM GENERIC_EVENTS LIMIT 2;
```

Results:

```
json_insert('{}',
   '$.sourceId', sourceId,
   '$.data', json(data)
)
-----------------------------------------------------------------------------------------------------------
{"sourceId":72057602627862528,"data":
{"Name":"FTrace","TimeSource":"ClockMonotonicRaw","SourceGroup":"FTrace"}}
json_insert('{}',
   '$.typeId', typeId,
   '$.sourceId', sourceId,
   '$.data', json(data)
)
```

```
----------------------------------------------------------------------------------------------------------------
{"typeId":72057602627862547,"sourceId":72057602627862528,"data":
{"Name":"raw_syscalls:sys_enter","Format":"\"NR %ld (%lx, %lx, %lx, %lx, %lx, %lx)\", REC->id, REC->args[0],
REC->args[1], REC->args[2], REC->args[3], REC->args[4], REC->args[5]","Fields":
[{"Name":"common_pid","Prefix":"int","Suffix":""},{"Name":"id","Prefix":"long","S
{"typeId":72057602627862670,"sourceId":72057602627862528,"data":
{"Name":"irq:irq_handler_entry","Format":"\"irq=%d name=%s\", REC->irq, __get_str(name)","Fields":
[{"Name":"common_pid","Prefix":"int","Suffix":""},{"Name":"irq","Prefix":"int","Suffix":""},
{"Name":"name","Prefix":"__data_loc char[]","Suffix":""},{"Name":"common_type",
json_insert('{}',
   '$.rawTimestamp', rawTimestamp,
   '$.timestamp', timestamp,
   '$.typeId', typeId,
   '$.data', json(data)
)
----------------------------------------------------------------------------------------------------------------
{"rawTimestamp":1183694330725221,"timestamp":6236683,"typeId":72057602627862670,"data":
{"common_pid":"0","irq":"66","name":"327696","common_type":"142","common_flags":"9","common_preempt_count":"0"}}
{"rawTimestamp":1183694333695687,"timestamp":9207149,"typeId":72057602627862670,"data":
{"common_pid":"0","irq":"66","name":"327696","common_type":"142","common_flags":"9","common_preempt_count":"0"}}
```

Arrow#

The Arrow type exported file, `.arrows`, uses the IPC stream format to store all tables in a file. The tables can be read by opening the file as an arrow stream. For example one can use the `open_stream` function from the arrow python package. For more information on the interfaces that can be used to read an IPC stream file, please refer to the Apache Arrow documentation [1, 2].

The name of each table is included in the schema metadata. Thus, while reading each table, the user can extract the

table title from the metadata. The table name metadata field has the key `table_name`. The titles of all the available tables can be found in section SQLite Schema Reference.

A sample function that reads all Arrow tables in a `.arrows` file is provided below in Python:

```
import pyarrow as pa

def read_tables(arrow_file):
  with pa.input_stream(arrow_file) as source:
    while source.tell() < source.size():
      try:
        yield pa.ipc.open_stream(arrow_file).read_all()
      except:
        continue
```

The Arrow directory exporter type, `_arwdir`, will create a directory with one arrow file per table/dataset.

JSON and Text#

JSON and TXT export formats are generated by serializing buffered messages, each on a new line. First, all collected events are processed. Then strings are serialized, followed by stdout, stderr streams if any, followed by thread names.

Output layout:

```
{Event #1}
{Event #2}
...
{Event #N}
{Strings}
{Streams}
{Threads}
```

For easier grepping of JSON output, the `--separate-strings` switch may be used to force manual splitting of strings, streams and thread names data.

Example line split: `nsys export --type=json --separate-strings sample.nsys-rep -- -`


```
{"type":"String","id":"3720","value":"Process 14944 was launched by the profiler"}
{"type":"String","id":"3721","value":"Profiling has started."}
{"type":"String","id":"3722","value":"Profiler attached to the process."}
{"type":"String","id":"3723","value":"Profiling has stopped."}
{"type":"ThreadName","globalTid":"72057844756653436","nameId":"14","priority":"10"}
{"type":"ThreadName","globalTid":"72057844756657940","nameId":"15","priority":"10"}
{"type":"ThreadName","globalTid":"72057844756654400","nameId":"24","priority":"10"}
```


Compare with: `nsys export --type=json sample.nsys-rep -- -`


```
{"data":["[Unknown]","[Unknown kernel module]","[Max depth]","[Broken backtraces]",
  "[Called from Java]","QnxKernelTrace","mm_","task_submit","class_id","syncpt_id",
  "syncpt_thresh","pid","tid","FTrace","[NSys]","[NSys Comms]", "..." ,"Process
  14944 was launched by the profiler","Profiling has started.","Profiler attached
  to the process.","Profiling has stopped."]}
{"data":[{"nameIdx":"14","priority":"10","globalTid":"72057844756653436"},
  {"nameIdx":"15","priority":"10","globalTid":"72057844756657940"},{"nameIdx":"24",
  "priority":"10","globalTid":"72057844756654400"}]}
```


Note that only last few lines are shown here for clarity, and that carriage returns and indents were added to avoid wrapping documentation.